

Bridging a Correlated-Noise Likelihood and Evidence-Based Inference onto the Next-Generation GIGA-LENS Scene API: a Certification, Benchmark, and Mechanism Campaign*

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Abstract

Our companion campaign ([Benson and Huang, 2026](#)) closed with a two-sided verdict on the real *HST*/WFC3-IR lens DESI-165.4754-06.0423: a drizzle-anchored correlated-noise likelihood is *necessary but not sufficient* — it flips the binned-product bimodality (a ~ 191 -nat per-basin evidence swing, exposing the steep basin as a noise-covariance artifact) yet over-corrects the slope to $\gamma = 1.103 \pm 0.008$, $\sim 17\sigma$ below the 1.433 native-diagonal anchor — and with the finding that real ill-conditioned and bimodal lens posteriors need a two-stage preconditioned recipe plus per-basin SMC evidence. Meanwhile the GIGA-LENS team’s next-generation rewrite (“the upstream development branches”) rebuilt the forward model around a unified multi-plane, multi-band *scene API* whose validation suite was, at the commit we vendored, not yet run, and which contains neither a correlated-noise likelihood nor an evidence layer. This report bridges the two lineages under a pre-registered, budget-fenced, proposer $\neq$ grader protocol (53.51 of a 100 A100h cap). The outcome is deliberately two-sided. *Delivered:* (i) the first external certification of the scene API forward model for the EPL+shear + Sérsic/shapelet configuration class — machine-precision cross-stack parity (forward image 5.7×10^{-15} ; gradients 1.5×10^{-11}) on both machines, with three documented convention reconciliations, including a real $\sim 2 \times 10^{-3}$ Sérsic- b_n approximant difference, and an fp128-referenced restatement of the Occam log-determinant gate after the original cross-algorithm reference proved noisier than the quantity it gated; (ii) a ported `CorrelatedImageData` likelihood term, exact in its analytic limits, that *reproduces the money number cross-stack*: $\gamma = 1.1005$ (offset 0.0027) with $\log Z$ agreement across stacks and machines to 0.11 nats in the low basin and 0.37 nats in the steep — $\gamma = 1.103$ now stands on two independent implementations; (iii) a sampler decision matrix from the first microcanonical-SMC benchmark on lens posteriors: prior-seeded MAMS-SMC is the only vehicle in the campaign that produced evidence at all (analytic targets, a double-source-plane ridge at $\lambda=1$, both basins of a 74-dim bimodal target with retention 1.000, and the correlated real-data refit), and where

*All work reported here — code, experiments, analysis, and text — was performed by Claude Code (Anthropic), guided by Greg Benson. This document is a technical report. Results obtained on the scene-API substrate are labeled *proposed (UNCERTIFIED — external)* throughout: we are the producer; the grader is the upstream GIGA-LENS development team (§5). Co-authorship on any publication whose results run through that substrate is offered to the team; their sign-off gates all such material.

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it fails it fails *by cost, never by health* — every non-completion is a wall-cap on a healthy sampler; on the 33-dim multi-plane real-data cell (MUSE cutouts provided by the team, used with permission) *neither* cold-start SMC nor warm MAMS converges at matched budget, while the classic single-stage MAP→SVI→HMC recipe produced no accepted posterior anywhere in the campaign, reproducing the known single-stage pathology on the new substrate. Unadjusted-MCLMC mutations are demoted for evidence work: a 20.1σ bias screen and a measured $\times 56$ minor-mode evidence inflation that SMC resampling does not launder at point estimate. Two cross-cutting negatives bind future evidence claims: bootstrap $\log Z$ errors understate true errors (measured $+3.06 \pm 0.35$ nats where the truth is ≈ 0), and a frozen MCMC-derived reference was itself indicted by an 11σ gate failure whose decomposition shows both basins with *more* evidence and γ -disjoint support — frozen references need coverage certificates. *Bounded, not decided*: the anchor arbitration (does 1.433 reproduce end-to-end on the new substrate?) closed PARTIAL-BY-VEHICLE-EXHAUSTION with no quotable γ . *Rejected*: stationarity of the noise-model class behind $\gamma = 1.103$ (calibrated $p = 0.010$; cross-block kernel spread $16.4\times$ the drizzle-registration envelope) — the standing working hypothesis for the over-correction, unconfirmed but unrefuted after the injection-recovery test was confounded by scene-subtraction residue. Every quantitative claim traces to a JSON artifact, a job id, and a gate row in `CAMPAIGN.md`.

1 Summary for Lens Modelers

This section collects, up front, the five things a lens modeler checks first. Every number is quoted from its on-disk artifact (the campaign spine, `research/synthesis_tables.md`, binds); results on the scene API substrate carry the report’s *proposed* (*UNCERTIFIED* — *external*) label. The system throughout is the real *HST*/WFC3-IR lens DESI-165.4754-06.0423. The summary diagnostics present the three *surviving* mass models: the E1 (PI-requested, 2026-07-22) **MCLMC** fit on the v2d native diagonal target, $\gamma = 1.4683$ [1.4343, 1.5048] — the first fully convergence-certified posterior on that target, consistent with the native-diagonal anchor ($\gamma = 1.433 \pm 0.034$) at $0.72\sigma_{\text{comb}}$ (item 4), and the lead model here; the **diagonal-low** solution ($\gamma = 1.293 \pm 0.012$, v3b binned; the anchor and diagonal-low fits are inherited from the companion campaign (Benson and Huang, 2026)); and the **steep** basin ($\gamma = 2.639$, evidence-disfavored at $\Delta \log Z \approx -29$). The correlated-likelihood binned-low fit ($\gamma = 1.1032 \pm 0.0086$) — the campaign’s money *number* — is deliberately excluded from the summary diagnostics: as a lens model of this system we can reject it, and it is discussed in this report only as the diagnosed artifact of a stationarity-violating noise-model class, in the mechanism sections (§7.4.1, §7.4.3).¹

(1) Data, model, residual, source — and the caustic structure. Figure 1 is *the* summary diagnostic — one model per row, and per row: (a) the observed cutout; (b) the best-fit model image, rendered at that model’s posterior-median parameters with its solved linear amplitudes, critical curve overlaid; (c) the reduced residual (data − model)/ σ ; and (d) the reconstructed (unlensed) source-plane surface brightness — the sampled- I_e Sérsic plus the 28 shapelet columns at their solved amplitudes — with the caustic overlaid. The rows are the three surviving models: the E1 v2d-native MCLMC fit ($\gamma = 1.468$; $\chi^2/\text{px} = 1.228$ on its home native pixels), diagonal-low ($\gamma = 1.293$; $\chi^2/\text{px} = 1.578$ — the field’s attainable floor, item 2), and the evidence-disfavored steep basin ($\gamma = 2.639$, $\Delta \log Z \approx -29$), whose real-space $\chi^2/\text{px} = 7.588$ (computed for this figure) makes the evidence verdict visible to the eye as coherent arc-scale residuals and an implausibly extended source reconstruction. The correlated optimum $\gamma = 1.103$ is dropped from the summary diagnostics deliberately — we can reject it as a lens model; its anatomy (the strong coherent lens-center residual its own whitened metric prices as “probably correlated noise”) is the subject of the mechanism

¹Source artifact: `data/e1_harvest.json`.

sections (§7.4.1, §7.4.3; same-pixel side-by-side renders in Fig. 12). Figure 2 remains as the compact same-axes comparison of the surviving models (MCLMC $\theta_E = 2.654''$; diag-low $\theta_E = 2.603''$), making the mass-model disagreement visible as caustic structure rather than only as a slope number.

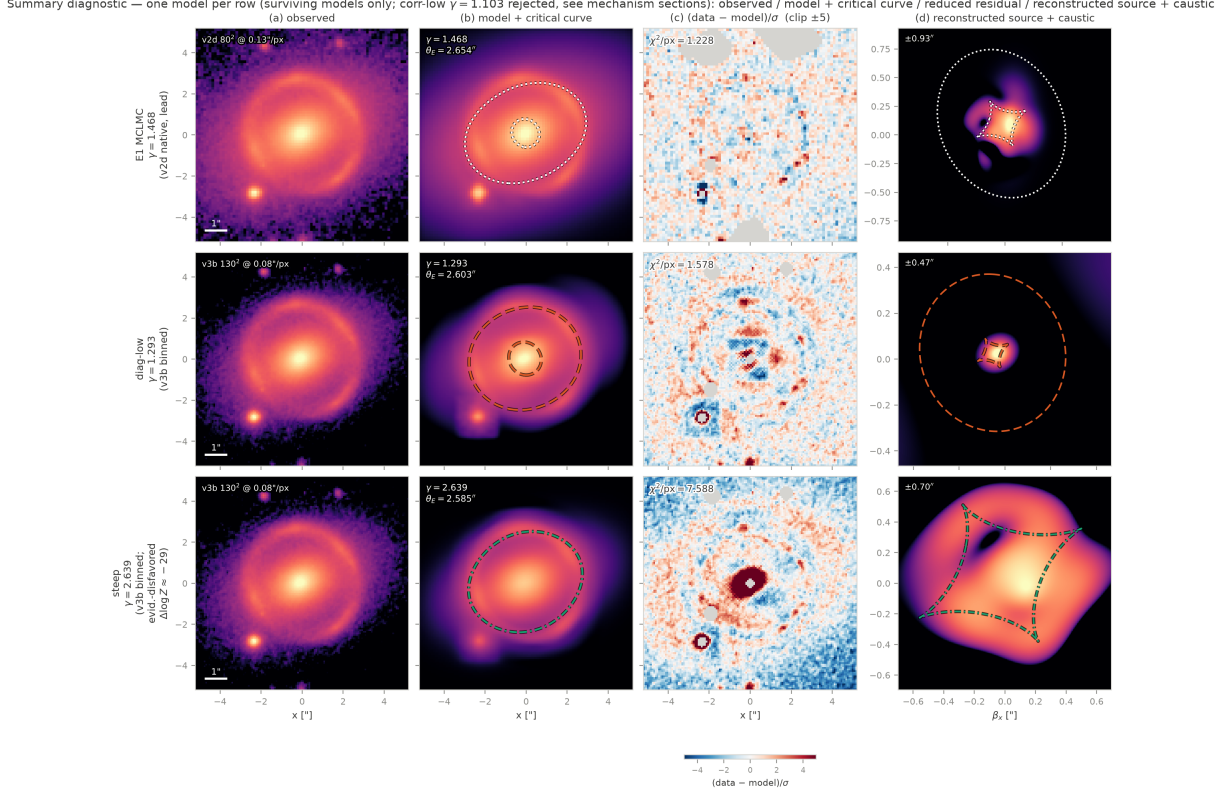


Figure 1: The summary diagnostic, one model per row — surviving models only (the corr-low $\gamma = 1.103$ model is excluded: rejected, and diagnosed as a noise-model artifact in §7.4.1, §7.4.3). Rows: E1 v2d-native MCLMC $\gamma = 1.468$ (lead; 80^2 @ $0.13''/\text{px}$), diagonal-low $\gamma = 1.293$ (v3b binned, 130^2 @ $0.08''/\text{px}$), steep $\gamma = 2.639$ (v3b binned; evidence-disfavored, $\Delta \log Z \approx -29$). Columns: (a) observed cutout (asinh stretch, normalized at the data’s 99.9th percentile; $1''$ scale bar); (b) model image at the per-dimension posterior-median parameters with the ridge-solved linear amplitudes (same stretch as the data panel), critical curve overlaid; (c) reduced residual $(\text{data} - \text{model})/\sigma$ over the keep mask, symmetric diverging scale *clipped at $\pm 5\sigma$* (masked pixels grey); (d) reconstructed unlensed source surface brightness (sampled- I_e Sérsic + 28 shapelets at their solved amplitudes) on a source-plane grid sized to $1.25\times$ the caustic extent (half-width labeled per panel), caustic overlaid. Display conventions follow the team’s GIGA-LENS discovery paper (Gu et al., 2022); renders use the parity-certified scene API stack (diagonal delta-bundle likelihood term). Assertions enforced at build time: each row’s render reproduces the report’s χ^2/px — MCLMC 1.228 (`data/e1_mclmc_home_chi2.json`) and diag-low 1.578 (`data/t04_realspace_headtohead.json`) to $< 1\%$ (measured $< 0.1\%$); steep’s 7.588 is computed and stated here — and each outer critical curve’s median radius matches its θ_E to $< 5\%$. Source artifacts: `data/rfig_modeler_params.npz`, `data/mclmc_diag_v2d.npz`, `data/rfig_modeler_checks.json`.

(2) Reduced χ^2 of the best-fit models. Table 1 gives both currencies. Under its own production metric the money fit is formally excellent: whitened $\chi^2/\text{dof} = 8946.8/9273 = 0.965$ (0.973 after subtracting the 46 nonlinear + 28 marginalized-amplitude parameters).² Two honest qualifiers frame

²Source artifact: `data/summary_rchi2_whitened_corrlow.json` — recomputed for this section at the identical posterior-median model point on the validated stack; it bit-matches the head-to-head artifact’s γ and whitened $\log L$ before being quoted.

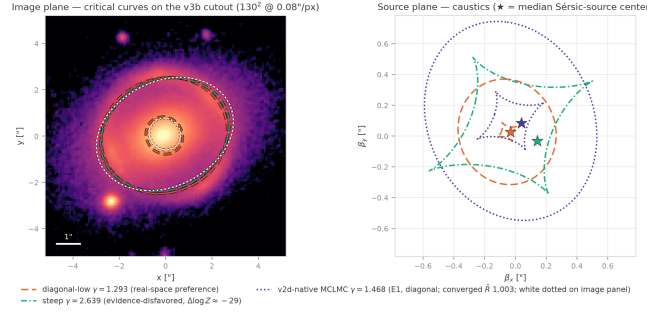


Figure 2: Compact comparison — critical curves (lens plane, over the **v3b** binned cutout, asinh stretch) and caustics (source plane; stars mark the median Sérsic-source centers) for the surviving posterior-median mass models on shared axes, following the critical-curve/caustic display conventions of the team’s GIGA-LENS discovery paper (Gu et al., 2022): diagonal-low $\gamma = 1.293$ (the real-space preference), steep $\gamma = 2.639$ (evidence-disfavored, $\Delta \log Z \approx -29$), and the E1 v2d-native MCLMC fit $\gamma = 1.468$ (violet on the source panel; neutral white dotted on the image panel). The corr-low $\gamma = 1.103$ curve was removed from this figure at PI direction — the model is rejected and appears only in the mechanism sections as the diagnosed artifact (§7.4.1, §7.4.3). Scale sanity: the outer critical curve’s median radius matches each model’s θ_E to $< 5\%$ for all three (diag-low $2.611''$ vs. $\theta_E = 2.603''$; MCLMC $2.652''$ vs. $2.654''$), and the $\gamma < 2$ models show the expected inner radial critical curve and caustic. Deflections from the vendored scene API EPL(50)+shear at the certified conventions; source artifacts `data/rfig_modeler_params.npz`, `data/mclmc_diag_v2d.npz`, `data/rfig_modeler_checks.json`.

the real-space column. First, *this field’s own-model floor is ≈ 1.6 , not 1.0*: the injection-build gate showed that even a ground-truth model of an injected scene scores $\chi^2/\text{px} = 1.598$ full-keep versus 0.952 on the sky subset — essentially all of the excess is bright-object scene-subtraction residue (bright-object subset 2.71, lens-center $r < 1.2''$ subset 5.46)³ — so the diagonal-low value 1.578 *is* the attainable field level, while 7.4–8.3 for the anchor and money models is genuine real-space misfit (the anchor row also carries a cross-product resolution handicap: the same parameters render $\chi^2/\text{px} = 1.251$ on their home native product). Second, the whitened 0.965 inherits the whitened metric’s known caveat: its stationary kernel class is *rejected* by the field at calibrated $p = 0.010$ (§7.4.1), and the same metric prefers the money model over diagonal-low by +63 nats in $\log L$ (+501 in log-posterior), inverting the real-space ordering — a reduced χ^2 of 0.965 here certifies internal consistency, not noise-model correctness. The E1 MCLMC row is evaluated by the same machinery on its *home* native product: $\chi^2/\text{px} = 1.228$ full-keep (1.484 arc band) at its posterior median — marginally better than the anchor point’s home 1.251 on the same pixels (the evaluator was cross-checked against the old-stack head-to-head value to 0.01% at the anchor point).⁴

(3) Sampling method. In HMC-vs-MCLMC vocabulary: every headline fit in this report was sampled by *adaptive tempered SMC* (Del Moral et al., 2006) ($\lambda : 0 \rightarrow 1$, per-stage adaptive step targeting a weighted ESS of $0.7N$, per-stage bootstrap $\log Z$) whose mutation kernel is *MAMS* — *the Metropolis-adjusted variant of MCLMC* (Robnik and Seljak, 2024a,b): microcanonical (MCLMC) dynamics made exact by a Metropolis accept/reject step, tuned to the MAMS paper’s target acceptance of 0.90. It is therefore neither plain HMC nor raw (unadjusted) MCLMC. The distinction is load-bearing: unadjusted-MCLMC mutations were demoted at the correctness gate (20.1σ

³Source artifact: `data/t11_injection_build_report.json`; `data/t11b_residue_mask_report.json`; `data/t04_realspace_headtohead.json`.

⁴Source artifact: `data/e1_mclmc_home_chi2.json`.

Table 1: Goodness of fit for the three campaign best-fit models on the same **v3b** binned pixels: real-space χ^2/px (diagonal-solved amplitudes; 16,653 keep px full field, 7,825 px arc band $r \in [1.2, 4.2]''$) and, for the production correlated fit, the whitened reduced χ^2 over the 9,273 whitened dof; plus the E1 MCLMC fit at its posterior median on its home **v2d** native product (marked ^a; same evaluator convention). Whitened $\log L$ for the three **v3b** models is in Table 8. Sources: `data/t04_realspace_headtohead.json`, `data/summary_rchi2_whitened_corrlow.json`, `data/t11_injection_build_report.json`, `data/e1_mclmc_home_chi2.json`.

model (γ)	χ^2/px full	χ^2/px arc band	whitened χ^2/dof
diag-low 1.293 (home reference)	1.578	1.929	—
anchor 1.433 (from v2d ; home 1.251)	7.436	3.622	—
corr-low 1.103 (money)	8.321	2.756	0.965
MCLMC 1.468 (E1; home v2d native px ^a)	1.228 ^a	1.484 ^a	—

^aE1 row only: evaluated on its home **v2d** *native* product (5,865 keep px; 2,937 arc-band px at $0.13''$), not the shared **v3b** binned pixels of the other rows — the direct analogue of the anchor’s home 1.251; `data/e1_mclmc_home_chi2.json`.

field’s own-model floor (truth model of injected scene): full-keep 1.598; sky subset 0.952 — the excess over 1.0 is bright-object scene-subtraction residue, faced by every fit on this product.

evidence bias on the ill-conditioned analytic target; a measured $\times 56$ minor-mode evidence inflation on the 74-d bimodal target), and everywhere the classic single-stage MAP→SVI→HMC recipe ran in this campaign it failed its own acceptance rule (Table 2; full decision matrix Table 9, §9.5, §9.6) — consistent with the companion finding that these real lens posteriors need the two-stage re-preconditioned recipe (Benson and Huang, 2026). The E1 epilogue adds **MCLMC proper** — the PI-requested vehicle; the team’s own vendored implementation (`full_mclmc_with_adapt_sharded`: windowed mass-matrix adaptation, run with their Stan-style mass-matrix regularization option enabled) — used as the direct posterior sampler on both diagonal targets from warm starts in the physical basin. On the native product it delivers the first fully convergence-certified posterior of the program (item 4); tempered SMC remains the evidence/model-comparison layer — the two are complementary, not competing.⁵

(4) Convergence metrics. The E1 MCLMC fits supply the campaign’s first $\hat{R}$ /ESS-certified direct-sampler rows, and the **v2d** native fit leads Table 2: $\hat{R}_{\text{worst}} = 1.0031$, $\text{ESS}_{\text{min}} = 30,334$ against its frozen gate $\hat{R} < 1.01 \wedge \text{ESS} \geq 1000$ over the pooled 64 chains (Fig. 3); its **v3b** twin fails both frozen gates and is reported as a NO-QUOTE migration finding (below). For the SMC fits, which have no $\hat{R}$, the honest analogues are independent-seed repeats, the per-stage weighted-ESS floor maintained by the adaptive λ -schedule, unique-particle retention after resampling, MAMS acceptance against its 0.90 target, and — strongest of all — the cross-stack, cross-machine reproduction. Table 2 is the consolidated convergence ledger for every fit quoted in this report, with $\hat{R}$ /ESS given where MCMC arms ran (the classic-recipe arms are both rejections, reported as such). For the two inherited diagonal fits (anchor 1.433, diag-low 1.293) the per-fit $\hat{R}$ /ESS live in the companion report’s campaign ledger (Benson and Huang, 2026) and are cited, not re-derived here. Table 2 covers the fits *quoted* in this report; Appendix A extends it to the complete per-fit ledger — one row for every fit the campaign ever ran, including the failed, rejected, collapsed, and budget-blocked arms.

⁵Source artifact: `data/mclmc_diag_v2d.json`; `data/e1_harvest.json`.

Table 2: Consolidated convergence ledger. wESS = final-stage weighted ESS; “unique” = unique particles after resampling; the adaptive λ -schedule holds per-stage wESS at $0.7N$ by construction. The two E1 MCLMC rows are gated at their own frozen thresholds ($\hat{R} < 1.01 \wedge \text{ESS} \geq 1000$, pooled 64 chains; containment band $\gamma \in [1.15, 2.0]$, report-never-drop). Sources: `data/mclmc_diag_{v2d,v3b}.json` + `data/e1_harvest.json`, `data/t02_t03_gate_eval.json` (+ per-seed run JSONs), `data/results-perlmutter/10g2_v3b_scene_seed2.json`, `data/b1r_decision_matrix.json` + `research/checkpoint_b1r_close.md`, `data/results-perlmutter/10_arbitration_classic.json`, `data/b3_run_ref_sys{0..3}_14-8.json`, [Benson and Huang \(2026\)](#).

fit / arm	vehicle, n	diagnostics	verdict
v2d native diag, MCLMC (E1, PI-requested) — $\gamma = 1.4683 [1.4343, 1.5048]$	vendored windowed-adaptation MCLMC; 2 seed groups $\times$ 32 chains $\times$ 4000 draws	$\hat{R}_{\text{worst}} = 1.0031$ vs gate < 1.01 ; $\text{ESS}_{\min} = 30,334$ vs ≥ 1000 ; γ : $\hat{R} 1.0017$, $\text{ESS} 39,794$; seed-group medians agree to 0.0002; 0 containment violations (all 256k draws in $[1.324, 1.637]$); 1.49 h on one L4	PASS both frozen E1 gates — γ quotable, not escalated; $+0.0353 = 0.72\sigma_{\text{comb}}$ from the 1.433 anchor (Fig. 3)
v3b binned diag, MCLMC (E1)	same; 4 seed groups $\times$ 16 chains	$\hat{R}_{\text{worst}} = 1.379$, $\text{ESS}_{\min} = 139$ (both lens-light I_e); 46/46 scene params over the $\hat{R}$ gate; 64/64 chains migrate below the $[1.15, 2.0]$ band during burn-in (Fig. 5)	NO-QUOTE + FIND-ING : structural migration, not sampling depth — the one allowed escalation declined
v3b corr low — $\gamma = 1.1032 \pm 0.0086$ (money)	tempered SMC + MAMS; 3 seeds $\times$ 128	$\lambda=1$ @28 stages every seed; wESS 118–128/128; unique 77–82/128; seeds $\{1.1032, 1.0967, 1.1005\} \Rightarrow \sigma_{\text{seed}}(\gamma) = 0.00325$, $\sigma_{\text{seed}}(\log Z) = 1.22$ nats	converged + seed-certified (T0.2, §7.1); $n=3$ caveat: $\pm 46\%$ on σ_{seed} itself
v3b corr steep ($\gamma \approx 2.64$)	same; 3 seeds $\times$ 96	$\lambda=1$ @21–22 stages; wESS 36–96/96 (seed-2 final resample low); unique 50–59/96; $\sigma_{\text{seed}}(\gamma) = 0.00664$	converged; $\Delta \log Z$ (steep – low) = $-28.9 / -32.3 / -31.2$ seed-stable
same fit, scene API port, Perlmutter A100 (L0-G2)	same; 1 seed	$\lambda=1$ @27/22 stages; wESS 123.5/128 (low), 47.6/96 (steep); unique 81/128, 54/96	PASS : $\Delta\gamma = 0.0027$; $\log Z$ agreement 0.11 (low) / 0.37 (steep) nats <i>across stacks and machines</i> (§7.7)
carousel 33-d multi-plane REAL (B1r; hardest cell)	same; 1 seed $\times$ 128	$\lambda = 0.151$ @36 stages; unique 92–104/128 every stage; MAMS accept 0.871–0.934 (mean 0.899 vs 0.90 target); stage wESS pinned at $0.7N$	no posterior : sampler healthy, killed by wall-cap — a cost verdict, not a convergence failure (§9.2)
classic MAP→SVI→HMC, warm, v2d (arbitration arm)	50 chains, 250 burn-in/750 draws; escalated 500/1500	$\hat{R}_{\text{worst}} = 44.3$ (escalated 11.7) vs rule < 1.05 ; $\text{ESS}_{\min} = 53.6$ (63.0 escalated); all chains railed $\gamma \approx 1.987 [1.983, 1.991]$ at the prior edge	NO-VERDICT : γ not quotable; reproduces the known T2 single-stage pathology on the scene API (§9.5)
classic reference chains, hs2 scenes $\times 4$ (B3)	frozen 50-chain recipe	$\hat{R}_{\text{worst}} = 4.36$ –7.87; $\text{ESS}_{\min} = 63.7$ –83.9 vs rule $\hat{R} < 1.05 \wedge \text{ESS} \geq 200$	all 4 rejected by the pre-registered acceptance rule (§9.6)
anchor 1.433 (v2d diag) and diag-low 1.293 (v3b diag)	companion campaign (two-stage re-preconditioned recipe)	provenance row: per-fit $\hat{R}$ /ESS ledgered in the companion report	converged there; cited, not re-derived (Benson and Huang, 2026)

E1 MCLMC diagonal fits — trace / rank diagnostics: v2d PASS ($\hat{R}_{\gamma}$ worst 1.0031, min-ESS 30,334) | v3b FAIL ($\hat{R}_{\gamma}$ worst 1.379, min-ESS 139; all 64 chains below the containment band)

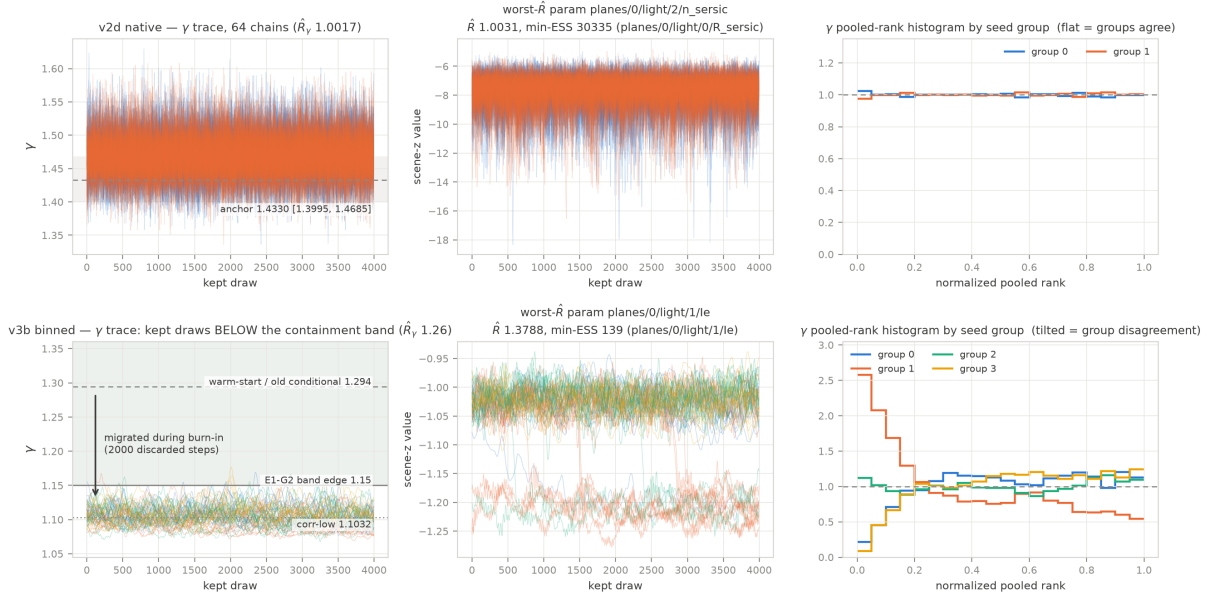


Figure 3: E1 MCLMC trace/rank diagnostics, both diagonal fits (plotted before any gate math per house rule). Top row, v2d native: the γ trace of all 64 chains sits on the anchor band 1.4330 [1.3995, 1.4685] ($\hat{R}_{\gamma}$ 1.0017), the worst- $\hat{R}$ parameter (a lens-light n_{sersic} , $\hat{R}$ 1.0031; the fit’s min-ESS, 30,334, sits on a lens-light R_{sersic}) mixes cleanly, and the per-seed-group pooled-rank histograms are flat (groups agree). Bottom row, v3b binned: the kept draws sit almost entirely *below* the E1-G2 containment-band edge (1.15; per-chain fraction of kept draws outside the band: min 0.94, median 1.00) after the chains migrated away from the 1.294 warm start during the 2000-step burn-in; the worst- $\hat{R}$ parameter (lens-light I_e , $\hat{R}$ 1.379, min-ESS 139) splits between shelves, and the tilted rank histograms localize the disagreement both within and between seed groups. Source artifacts: `data/mclmc_diag-{v2d,v3b}. {npz,json}`; `data/e1_harvest.json`.

(5) The posterior. Figure 4 shows the posterior corner for the production correlated v3b fit with the three seed repeats overlaid. The read-out: the Einstein radius is tight and stable ($\theta_E = 2.6239 \pm 0.0049$, a 0.2% constraint) while the two basins remain cleanly separated in γ (low 1.103 vs. steep 2.64) with a seed-stable evidence preference for the low basin of ≈ 29 –32 nats; the seed overlays coincide within $\sigma_{\text{seed}}(\gamma) = 0.00325$, which is what licenses quoting $\sigma_{\text{tot}} = 0.0086$ on the money number (§7.1).⁶ The regenerated corner now also overlays the E1 v2d-native MCLMC posterior (violet; 256,000 pooled draws, scene-native keyed extraction; its equal-weight draw median reproduces the run artifact’s pooled $q_{50} = 1.4683$ exactly), so the native-vs-binned displacement in γ is visible in the same axes as the seed-repeat spread.⁷ The number the correlated posterior constrains, and the standing caveat on interpreting it, are the subject of the rest of this report: $\gamma = 1.1032 \pm 0.0086$ is certified, cross-stack reproduced — and diagnosed as an artifact of a stationary noise-model class the field rejects.

The E1 caveat on the binned product — reported, not smoothed. The v3b arm of E1 is the honest counterweight to the clean v2d pass, and we state it plainly: *on the binned product, even MCLMC under the plain diagonal likelihood migrates toward the unphysical low- γ region.* Warm-started inside the physical basin (the $\gamma = 1.2941$ conditional cloud — itself only a conditional slice of an $\hat{R} = 843$ bimodal chain), all 64 chains in all four independent seed groups left the containment band during the 2000-step burn-in and spent essentially all of their kept draws at $\gamma \approx 1.08$ –1.15 (Fig. 5); the kept-draw pooled median, 1.1047, is descriptive only (the ensemble is unconverged, $\hat{R}_\gamma = 1.261$) but sits 0.0015 from the certified correlated-fit 1.1032. The one escalation the frozen protocol allowed was declined, with the reasoning on the ledger: the failure mode is structural migration out of the band, not sampling depth — doubling draws cannot un-migrate chains, and conditioning post hoc on the band would be a goalpost move — so the verdict is NO-QUOTE plus a finding. The finding is the sampling-side face of the campaign’s binned-product pathology (§7.4.1, §7.4.3): the binned pixels themselves prefer the low- γ shelf under likelihoods that discount what the native product resolves — seen with tempered SMC under both the correlated likelihood (the money fit) and the diagonal one (the B5 low basin at $\gamma = 1.092$, §9.3), and now with the PI’s own MCLMC under the plain diagonal likelihood. The finding’s honest scope is what this run tested: one product, one likelihood, one warm-start cloud, at a fixed 2000 + 4000-step budget — longer runs or other initializations could yet behave differently; what is established is the migration itself, not the shelf’s posterior weight. It further undermines the old 1.2941 conditional as a diagonal reference for this product — and it is why the v2d-native MCLMC fit, $\gamma = 1.4683$ [1.4343, 1.5048], is the quotable E1 deliverable.⁸

The E2 epilogue — the same sky in the collaborator’s own preparation. A second post-campaign epilogue (E2, 2026-07-23; §11.2) ran the identical E1 machinery, gates frozen and unchanged, on a collaborator’s own preparation of this lens (E. Odell’s cutout and PSF, which register at $0.064125''/\text{px}$ — exactly half our native scale, a *fourth* pixelization of this sky, with a parity flip relative to our raster). Warm-started from the healthy 1.433 anchor cloud, all 64 chains in all four seed groups migrated during burn-in to the same low- γ shelf (kept-draw pooled median 1.0999, 0.0033 from the corr-low 1.1032 — descriptive telemetry only: the fit fails both frozen gates, $\hat{R}_{\text{worst}} = 5.46$, NO-QUOTE), firing the pre-registered fork 2: the migration of Fig. 5 extends to a *second* resampled product and a second, fully independent preparation pipeline — matching

⁶Source artifacts: companion data/results/e2_v3b_low_smc_canary_fix.json; data/t02.t03_gate_eval.json.

⁷Source artifact: data/rfig_modeler_checks.json.

⁸Source artifact: data/e1_harvest.json; data/mclmc_diag_v3b.json.

Mass-parameter posteriors — v3b correlated SMC particles (128/seed low, 96/seed steep) + the E1 v2d-native MCLMC diagonal posterior (64 chains x 4000 draws)

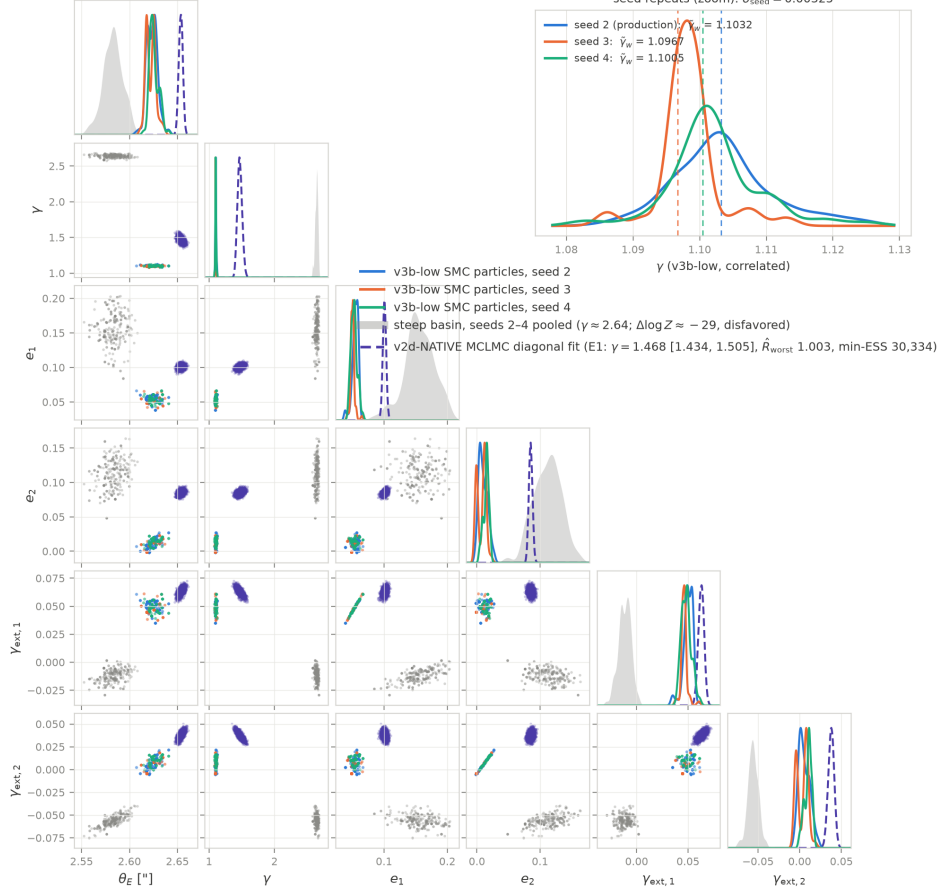


Figure 4: Posterior corner of the six mass parameters (θ_E , γ , e_1 , e_2 , $\gamma_{\text{ext},1}$, $\gamma_{\text{ext},2}$) from the stored equal-weight v3b-low correlated SMC particles, seeds $\{2, 3, 4\}$ overlaid — the SMC convergence visual (SMC has no $\hat{R}$; the seed-repeat spread $\sigma_{\text{seed}} = 0.00325$ plus the per-stage ESS/acceptance diagnostics of Table 2 are the analogues). Grey: the steep basin (seeds 2–4 pooled, $\gamma \approx 2.64$; $\Delta \log Z \approx -29$, disfavored by ≈ 29 – 32 nats). Inset: γ zoom with the per-seed weighted medians of record $\{1.1032, 1.0967, 1.1005\}$; the equal-weight particle medians reproduce them to ≤ 0.0027 , well inside $\sigma_{\text{stat}} = 0.008$. Violet: the E1 v2d-native MCLMC diagonal posterior (256,000 pooled draws; equal-weight draw median asserted equal to the run json’s pooled $\gamma_{50} = 1.4683$; it lies outside the γ zoom inset by construction). Source artifacts: companion `data/results/e2.v3b_low_smc_canary_fix.json` (+ seed-3/4 repeats), `data/t02.t03_gate_eval.json`, `data/rfig_modeler_params.npz`, `data/mclmc_diag_v2d.npz`, `data/rfig_modeler_checks.json`.

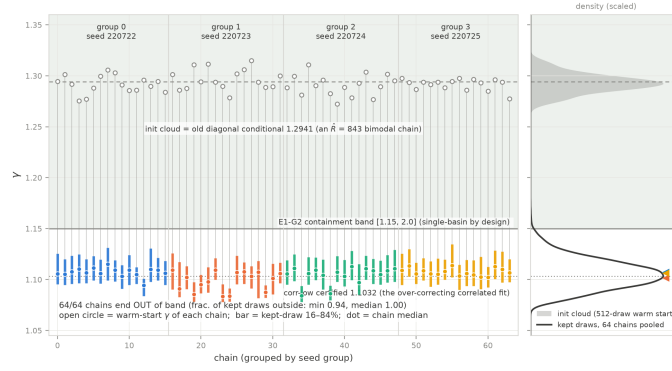


Figure 5: The E1 v3b migration finding (the run’s money visual, plotted before gate math). Left: for each of the 64 chains (grouped and colored by seed group), the warm-start γ (open circle, drawn from the 1.2941 diagonal-conditional init cloud) against its kept-draw 16–84% interval (bar) and median (dot): every chain exits the E1-G2 containment band $[1.15, 2.0]$ during burn-in and spends its kept draws at $\gamma \approx 1.08\text{--}1.15$ under the *diagonal* likelihood (fraction of kept draws outside the band: min 0.94, median 1.00; all 64 violations reported, none dropped, per the frozen report-never-drop rule). Right: init-cloud vs. pooled kept-draw marginals; the pooled kept-draw median 1.1047 sits 0.0015 from the corr-low certified 1.1032 — descriptive only, as the ensemble fails every convergence gate. Source artifacts: `data/mclmc_diag_v3b.json` (`gates.E1_G2`); `data/e1_harvest.json`.

the collaborator’s own unprompted “very migratory chains” report on his first-stage MCLMC. A companion demonstration on our own stored chains (O4, §11.2) returned an honest mixed result, reported as loudly as its intended pass: the collaborator-style stage-2 rescue (a fresh MCLMC preconditioned on the migrated stage-1 tail) improves the diagnostics most of the way ($\hat{R}_{\text{worst}} 1.379 \rightarrow 1.188$) while the chains stay at the shelf, but does *not* reach the frozen gates — so the strong “ $\hat{R}$ -blindness” claim is not supported at our settings, while the healthy v2d control converged exactly as pre-registered. The registration findings, the noise audit of his product (correlated like ours), and the updated product-dependence synthesis are collected in the epilogue section (§11.2).⁹

⁹Source artifact: `data/e2_harvest.json`.

What is new here, and what is deliberate confirmation

Confirmation (by design). The headline native-product MCLMC fit ($\gamma = 1.4683$) is, by design, the team’s own GIGA-LENS methodology — their vendored sampler and their standard likelihood class — executed on the new scene API substrate. Its value is the cross-implementation check: agreement with the old-stack anchor at $0.72\sigma_{\text{comb}}$ (item 4 above). A confirmation is only evidence if the two implementations are independently trustworthy — which is what item (i) below provides.

New (methodology and infrastructure). (i) The first external certification of the scene API forward model: the F1–F8 parity battery, on both campaign machines, including the Sérsic- b_n convention finding (§6, §6.2). This is what elevates cross-code agreement from coincidence to measurement. (ii) The correlated-noise (drizzle) likelihood ported as a scene API `LikelihoodTerm` with the Gaussian Occam term (§6.4), reproducing $\gamma = 1.1032$ across stacks and machines to $\Delta\gamma = 0.0027$ and 0.11 nats in $\log Z$ (§7.7). (iii) A per-basin evidence layer (tempered SMC, §9.3) and a measured scope rule for MCLMC: accurate within-basin location ($\Delta\gamma = 0.0016$ vs. the Metropolis-adjusted kernel on the 74-d bimodal target) but $\times 56$ minor-mode evidence inflation — use it for single-basin posterior fits (as in E1 here), not for evidence or basin weights.

New (findings). (iv) The binned data product is the pathology carrier, established from two independent directions: on the noise side, stationarity of the noise-model class is rejected at calibrated $p = 0.010$ (§7.4.1); on the sampling side, the E1 result that all 64 MCLMC chains migrate out of the physical band from the $\gamma = 1.2941$ warm start to ≈ 1.10 under the plain *diagonal* likelihood (Fig. 5) — so the old 1.29 diagonal value was a stuck-chain conditional, and the migration endpoint sits 0.0015 from the correlated-likelihood result. (v) The first formal SBC calibration exercise for this pipeline class (X2, §10.1), including the finding that the pre-existing frozen benchmark mock set cannot be used for SBC as generated: its generation prior is narrower than the fitting prior, under which SBC fails by construction. (vi) A cost-bounded sampler decision matrix on real lens posteriors (Table 9, §9).

Scope. Items (i)–(iii) are upstream-ready and ship in the handoff package (§10.2); items (iv)–(vi) are the campaign’s scientific claims; every verdict on the team’s substrate carries the *proposed* (*UNCERTIFIED* — *external*) label.

2 Introduction

2.1 Two lineages

The companion campaign (Benson and Huang, 2026) attacked the two ingredients that the code-comparison literature flags as under-addressed in fast GPU lens modeling (Galan et al., 2024): correlated image noise and honest posterior multimodality. It delivered a validated drizzle-anchored correlated-noise likelihood ($C = D^{1/2}KD^{1/2}$, convolutional whitening gated at operator-norm flatness $e_{\text{op}} \leq 0.02$, generalized ridge marginalization with the Gaussian Occam term) and the first systematic sampler benchmark on strong-lens posteriors. On the real system DESI–165.4754–06.0423 (Huang et al., 2025) the verdict was *necessary but not sufficient*: the correlated likelihood flips the binned-product bimodality — per-basin evidence swings ~ 191 nats, from $\Delta \log Z = +162$ (diagonal) to -29 (correlated), showing the upsampled steep basin to be a noise-covariance artifact — but over-corrects the slope, restoring a low basin at $\gamma = 1.103 \pm 0.008$ that sits $\sim 17\sigma$ below the 1.433 native-diagonal anchor, so that the correlated slopes across pixel scales *bracket* the anchor rather than unifying onto it. Explaining that bracket — and certifying the machinery on which any explanation would run — is unfinished business this campaign inherits. On the inference axis, the companion campaign found that the two hardest real posteriors (a 46-dim condition- 10^{14} marginalized target and a 74-dim bimodal target) defeat off-the-shelf samplers, and that per-basin tempered SMC evidence is the only trustworthy mode-weight instrument.

Independently, the GIGA-LENS team — authors of the original GPU recipe (Gu et al., 2022) and its multi-node successor (Huang et al., 2026) — has been rebuilding the library on what we refer to throughout as *the upstream development branches*: a JAX-only rewrite (Bradbury et al., 2018) around a unified “scene API” (multi-plane, multi-band scene composition; a documented Dataset/LikelihoodTerm seam; a physicality layer; experimental microcanonical sampler kernels). Three facts about the commit we vendored (§3) motivate the bridge. First, its validation suite was present but *unrun* — the forward model had no external certification. Second, it contains no correlated-noise likelihood, no analytic source marginalization with an Occam term, and no tempering/SMC evidence layer — precisely the assets the companion campaign validated on the old stack. Third, its likelihood seam is documented and stable enough to carry an upstream-shaped port. Per the campaign’s collision rules (§5) we do not characterize any unpublished results of those branches; where campaign cells ran on data or targets provided by the team (the 33-dim multi-plane “carousel” cell on real MUSE cutouts), the data were *provided by the team and used with permission, with any comparison against their own runs deferred pending their sign-off*.

2.2 Campaign goals

The campaign (branch `claude-giga-lens-linus`; plan frozen before any GPU spend) set five goals:

1. **Certify the substrate.** Cross-stack parity of the scene API forward model against the validated companion-campaign stack, at machine precision, on both campaign machines (gate battery F1–F8, §3).
2. **Port the payload.** An upstream-shaped `CorrelatedImageData` likelihood term on the documented likelihood seam, exact in its analytic limits, plus a mutation-kernel-generalized tempered-SMC evidence layer (MC-SMC, §4); then the acid test: does the new substrate *reproduce* the companion campaign’s money number $\gamma = 1.103$ and its evidence at posterior level?
3. **Benchmark the bet.** A pre-registered decision matrix for prior-seeded MAMS-SMC against warm MAMS-alone, MCLMC, and the classic MAP→SVI→HMC recipe, across analytic, synthetic-scene, double-source-plane, high-condition, bimodal, and real multi-plane target classes — with two-sided honest predictions written before the runs.
4. **Close mechanism suspects.** Why does the correlated likelihood over-correct? Free (0 GPU-h) and cheap discriminators for the profile-curvature, companion-galaxy, spectrum-flooring, and noise-stationarity suspects, then a confirmatory injection-recovery test on the real noise field.
5. **Arbitrate the anchor, and hand off.** A two-stack end-to-end check of the 1.433 premise itself; and a claims-register handoff to the team (certification evidence, port, a first formal simulation-based-calibration (Talts et al., 2018) run of the pipeline class, ops lessons), every scene-substrate verdict labeled *proposed* (*UNCERTIFIED* — *external*).

2.3 Findings in brief (two-sided by design)

House style holds honest negatives to be first-class results; we summarize both signs here and expand in the results sections that follow.

Certification delivered. F1–F8 pass at machine precision on both machines (forward image 5.7×10^{-15} relative; constrained-space gradients 1.5×10^{-11}), *given* three documented convention

reconciliations applied on our side of the fence with the vendor unpatched — the largest a real $\sim 2 \times 10^{-3}$ model-level difference between the two stacks’ Sérsic b_n approximants. One gate (F6, the Occam log-determinant) required a signed, ledgered restatement against fp128 truth after the original cross-algorithm reference was measured to be noisier than the tolerance it enforced (§6.3).

The port reproduces cross-stack. The ported correlated term is exact in its analytic limits (delta-kernel identity to 5.8×10^{-11} ; dense-Cholesky reference to 5.5×10^{-12} nats) and, run end-to-end on the new substrate on an independent machine, reproduces $\gamma_{\text{low}} = 1.1005$ (offset 0.0027 against the certified 1.1032 ± 0.0086), $\log Z$ to 0.11 nats in the low basin and 0.37 nats in the steep, and the low-basin evidence preference at seed-family magnitude ($\Delta \log Z = -29.36$ against $-28.9 / -32.3 / -31.2$). The companion campaign’s headline number now stands on two independent stacks.

The benchmark maps capability boundaries. Prior-seeded MAMS-SMC is the only vehicle that produced $\log Z$ at all in this campaign, and its failures are uniformly *cost walls on a healthy sampler*, never divergence or collapse. The classic single-stage recipe, at frozen settings, produced zero accepted posteriors campaign-wide — including a faithful reproduction, on the new substrate, of the known condition- 10^{14} single-stage pathology ($\hat{R}_{\text{worst}} = 44.3$, all chains prior-railed). The carousel-class real-data cell defeats *both* vehicles at matched budget (0 posterior draws each). MCLMC mutations are demoted for evidence: a 20.1σ bias screen and a $\times 56$ minor-mode inflation that resampling does not launder at point estimate. Two cross-cutting findings bind all future evidence gates: bootstrap $\log Z$ uncertainties understate true errors (measured $+3.06 \pm 0.35$ nats across an exact reparameterization where the truth is ≈ 0), and one of our own frozen references was indicted by the very gate it anchored — an 11σ failure decomposing into *more* evidence in both basins with γ -disjoint ensemble support, i.e. a within-basin coverage defect in the reference, not (only) the vehicle.

Stationarity rejected; mechanism standing but unconfirmed. The stationary noise-kernel class behind $\gamma = 1.103$ is provably violated by the real field (calibrated $p = 0.010$, replicated with the arc excluded; cross-block kernel spread $16.4\times$ the drizzle-registration envelope), while the cheap alternatives died: profile curvature is structurally dead at zero GPU cost, the companion galaxy is exonerated (-0.0021 shift), and spectrum flooring is rejected by thousands of nats. The confirmatory injection-recovery test returned NO-CONFIRM / NO-EXONERATE: positive-signed biases outside both pre-registered zones with a control that failed high *and* sick, implicating the injection construction (scene-subtraction residue) under the pre-registered honesty clause. Noise-model-class misspecification remains the working hypothesis — unconfirmed, unrefuted.

Arbitration bounded, not decided. The two-stack anchor check exhausted four vehicles without producing a quotable γ on the 46-dim diagonal target; it closed PARTIAL-BY-VEHICLE-EXHAUSTION, with the premise carried indirectly by machine-precision likelihood parity (F1–F6) plus the *converged* correlated-target reproduction above. “1.433 reproduces at posterior level on the new substrate” is therefore *untested*, and is flagged as such wherever the anchor is used.

The remainder of this report: §3 (substrate and certification battery), §4 (correlated likelihood port and the MC-SMC evidence layer), §5 (protocol, labeling, and budget fences), followed by the results sections (benchmark decision matrix; mechanism program; cross-stack reproduction and arbitration; gifts and handoff), the discussion of open items, and the reproducibility appendix.

3 Methods I: Substrate, Environment, and Cross-Stack Certification

3.1 Vendored substrate and pinned environment

The campaign runs against a frozen snapshot of the upstream development branch, vendored at commit 80916d2 (full SHA in `vendor/gigalens-linus/VENDORED_REF.txt`) via `git archive` — *unpatched*, installed `--no-deps`. Every campaign process asserts the vendor identity at import (`require_vendor_ref`), and a ledgered re-pin procedure (motivation → re-archive → diff review of the three subclassed surfaces → full gate-battery re-run green) guards against silent drift of the moving branch; it was never needed — the pin held for the whole campaign. All campaign-side adaptation lives in the shim package `cg12/` (parameter mapping, guards, scene factories, the correlated term, and the samplers); the vendor tree was never edited.

Table 3 lists the pinned environment. The single deviation from the upstream declared pins — `numpy` 2.4.6 versus their 2.1.3 — is recorded as a known deviation and covered by the gate battery. The old validated stack (commit 58ec9a7; Benson and Huang, 2026) is imported by path, never copied; whitener bundles and Foundry-I data products likewise. Production GPU cells ran single-GPU shared-QOS on NERSC Perlmutter A100s; certification and free checks ran on the local phoenix node (L4/A16).

Table 3: Pinned software environment (venv `cg12`, mirrored as `cg12-pm` on Perlmutter). Pins are seeded from the companion campaign’s `constraints.txt`; all installs are constrained.

Component	Version	Note
<code>python</code>	3.13	
<code>jax / jaxlib</code>	0.6.2	upstream’s declared pin; identical on both machines
<code>BLACKJAX</code>	1.3	(Cabezas et al., 2024)
<code>tensorflow-probability</code>	0.25.0	(Dillon et al., 2017)
<code>numpy</code>	2.4.6	<i>known deviation</i> vs. upstream 2.1.3; gate-covered
<code>tensorflow</code>	—	absent; the vendored dependency is verified inert
<code>vendor</code>	80916d2	UNPATCHED; <code>--no-deps</code> ; identity-guarded
<code>old stack</code>	58ec9a7	validated companion-campaign stack, by path

3.2 Reference-artifact parity design

Both stacks ship as a Python package named `gigalens`, so the originally planned single-process dual import is impossible (a name collision the upstream documentation itself warns about). The enacted design (ledgered decision D3) is *reference-artifact parity*: the old validated stack, running in its own certified virtual environment, dumps reference arrays (forward images, design-matrix columns, log-likelihoods, constrained-space gradients) to `data/parity_refs.npz` for the foundry v2d and v3b inputs at a reference point z_{ref} plus three seeded perturbations; the scene API stack then recomputes the identical quantities in the campaign environment and compares. Both sides run the same `jax` 0.6.2, which preserves 10^{-12} -class comparability across the process boundary. Comparison is *in constrained parameter space* through the campaign’s name-keyed parameter map (never positional ordering), so any flat-vector convention bug fails parity by construction.

3.3 The F1–F8 certification battery

Table 4 gives the battery and its frozen thresholds (fixed at P0, before any science ran); the measured results, on both machines, are reported with the certification results in §6 (Table 5). F1–F4 constitute, to our knowledge, the first external certification evidence for the scene API forward model in the EPL+shear + 4×Sérsic / Sérsic+shapelets ($n_{\max} = 6$) configuration class — the vendored validation suite was unrun at the pin. F8 repeats the entire battery under the Perlmutter-native pinned environment on an A100, so that the stack is certified on both machines before any production cell. Parity at these thresholds holds *given* three documented convention reconciliations, applied on our side of the seam with the vendor unpatched; they are themselves findings and are reported with the results (§6.2).

Table 4: Cross-stack certification battery design: old validated stack (58ec9a7) versus the scene API (80916d2), same foundry v2d/v3b inputs, $z_{\text{ref}} + 3$ seeded perturbations, constrained-space comparison, scored on the worst case over products and perturbations. Thresholds frozen at P0; F6 is quoted under its restated form (§6.3). Measured results: Table 5 and §6–§6.4.

Gate	Statement	Threshold
F1	forward image	$\leq 10^{-12}$ rel
F2	design-matrix columns	$\leq 10^{-12}$ rel
F3	diagonal masked loglik + χ^2	$\leq 10^{-8}$
F4	grad of marg. loglik (constrained)	$\leq 10^{-8}$ rel- L_2
F5	delta-kernel corr. term $\equiv$ stock	$\leq 10^{-10}$
F6	Occam $-\frac{1}{2} \log \det A$ vs fp128 truth	cond-scaled (§6.3)
F7	flat- z round-trip + name audit	exact (info)
F8	full battery, Perlmutter-native env	report-only
03-A	corr. term vs dense-Cholesky (32^2 toy)	≤ 0.1 nat
03-B	delta identities (fwd; conv $\equiv$ multiply)	$\leq 10^{-10}$
W	whitener-bundle revalidation	$e_{\text{op}} \leq 0.02$ strict

The whitener bundles imported from the companion campaign were revalidated in place (gate W): operator-norm flatness e_{op} reproduced exactly, erosion masks reproduced from first principles, kernel hashes pinned. The three production bundles (v2d, v3b, v3) are admissible under the strict $e_{\text{op}} \leq 0.02$ gate; the ledgered relaxed arm `v2d_relaxed` ($e_{\text{op}} = 0.0312$ against its own design target 0.05) is correctly *inadmissible-by-design* and fenced from production use.¹⁰

4 Methods II: The Correlated-Noise Likelihood Term and the MC-SMC Evidence Layer

4.1 CorrelatedImageData: an upstream-shaped port

The correlated-noise payload is ported as a dataset class `CorrelatedImageData` with its companion likelihood term `CorrelatedImageLikelihoodTerm` (`cgl2/correlated.py`), subclassing the scene API’s documented `Dataset` and `LikelihoodTerm` seam — shaped so that upstream adoption is a move, not a rewrite. Design points, each carrying a validation gate:

- **Frozen whitener bundles as data.** The constructor takes a frozen whitener bundle (convolution taps, $D^{-1/2}$ diagonal, eroded keep-mask, e_{op} , kernel hash) and validates it raise-never-default; `event_size` is the kept whitened degrees of freedom. The whitener seam is

¹⁰Source artifact: `data/whitener_manifest.json`.

deliberately pluggable: the stationarity rejection (§2) means a locally-stationary (per-region) kernel class is the anticipated successor, and the term is agnostic to which bundle it is handed.

- **Single-render contract.** One basis render per likelihood evaluation, via `lstsq.simulate` with `return_stacked=True`, honoring the upstream single-forward-eval contract and reusing the fused convolution/pooling path verbatim; residual and design columns are whitened by a grouped depthwise convolution wrapped in `jax.checkpoint` (the memory fix that keeps SMC particles under the ≤ 200 MB budget). Masked operations use `jnp.where`, never multiplication by a mask.
- **Ridge marginalization with the Occam term.** The linear (shapelet) amplitudes are analytically marginalized under a generalized ridge prior, *including* the $-\frac{1}{2} \log \det A$ Gaussian-evidence term that a plain linear solve omits — the term that makes downstream Bayes factors meaningful. Whitened χ^2 is exposed through the upstream `reports_chi2` channel.
- **Exactness in analytic limits.** With a delta kernel the term reduces exactly to the stock `ImageLikelihoodTerm` (F5: 5.8×10^{-11} ; `conv`≡multiply identity 0.0); on a 32^2 toy with a dense covariance it matches an exact Cholesky reference to 5.5×10^{-12} nats against a 0.1-nat gate (03-A).¹¹ Production wiring adds an old-vs-new whitened marginalized log-likelihood/gradient cross-stack check at z_{ref} plus three seeded perturbations (worst $|\Delta \log L_{\text{marg}}|$: 2.4×10^{-8} on `v2d`, 7.2×10^{-7} on `v3b`)¹² and a remat-exactness gate (checkpointed vs. un-checkpointed whitening, difference exactly 0.0).¹³ Separately, B0’s target-adaptor parity gate compares old and new log-densities at 64 prior draws (max $|\Delta|$ exactly 0.0).¹⁴

For SMC the likelihood is evaluated in `f64`: a `f32` forward path carries a ~ 0.3 -nat evaluation noise floor that would pollute Metropolis–Hastings acceptance in the mutation kernels.

4.2 MC-SMC: tempered SMC with microcanonical mutation kernels

The evidence layer (`cgl2/samplers/smc_micro.py`) generalizes the companion campaign’s validated adaptive tempered-SMC driver (Del Moral et al., 2006) — the scene API’s `ProbModel` log-prior / log-likelihood split maps exactly onto the BLACKJAX convention — with the mutation kernel abstracted over three choices:

- **HMC** — the proven baseline from the companion campaign.
- **MAMS** (Robnik and Seljak, 2024b) — the *primary* and the campaign’s bet: Metropolis-adjusted microcanonical dynamics, whose exact π_λ -invariance at every tempering stage is what makes the SMC evidence estimate unbiased.
- **MCLMC** (Robnik and Seljak, 2024a) — unadjusted microcanonical dynamics, admitted *diagnostic-only* behind a pre-registered bias gate (below), never as an evidence kernel.

Kernel logic is copied with attribution from the upstream experimental implementations, never imported (those modules reach into private BLACKJAX internals and the pre-rewrite simulator). Per- λ tuning was frozen at P0, before any science cell: mass matrix from the regularized weighted

¹¹Source artifact: `data/correlated_term_validation.json`.

¹²Source artifact: `data/parity_report_scene.json`.

¹³Source artifact: `data/10_remat_gate.json`.

¹⁴Source artifact: `data/smc.b0_report.json`.

particle covariance; step size from a short dual-averaging pilot per stage (10 iterations, acceptance target 0.90 for MAMS per the MAMS integrator recommendation, 0.80 for HMC; an energy-variance-per-dimension pilot for MCLMC); 4 kernel draws per stage; adaptive λ -schedule by bisection on the incremental-weight ESS at target $0.7N$ with systematic resampling; $\log Z$ accumulated per stage with a per-stage bootstrap error σ_{boot} . Gradient evaluations are ledgered exactly (pilot draws billed identically), which is what makes the budget-matched benchmark arms of the results sections well-defined.

B0 correctness gates (frozen at P0). On analytic targets the layer passes where it must and fails where the class fails, and both are recorded:¹⁵ target adapters build 11/11; on the 2-dim bimodal `mix2`, $|\log Z \text{ error}| = 0.073 \leq 3\sigma_{\text{boot}}$ with minor-mode weight 0.219 against truth 0.2; on the 46-dim ill-conditioned target the worst-parameter $|z| = 1.83$ passes while the $\log Z$ error is 2.40 nats at $\sigma_{\text{boot}} = 0.37$ — i.e. $6.6\sigma_{\text{boot}}$, the first measurement of the campaign-wide finding that *bootstrap $\log Z$ errors understate true errors on ill-conditioned targets*; on `funnel10` the layer *fails* (-0.66 -nat $\log Z$ deficit) — a pre-existing limitation of the adaptive tempered-SMC class, reproduced in the old validated stack and carried as a caveat, not a port defect. The pre-registered MCLMC bias screen fired: MCLMC-vs-MAMS $\Delta \log Z = +10.79 \pm 0.54$ on the ill-conditioned target, i.e. 20.1σ ,¹⁶ demoting MCLMC to diagnostic/cost-frontier-only for the entire campaign — a demotion whose independent confirmation ($\times 56$ minor-mode evidence inflation on the 74-dim bimodal target) appears in the results.

4.3 The checkpoint/resume retrofit

The benchmark’s first production wave lost ~ 17.6 of its 18.7 A100h to zero-artifact timeouts (only 1.12h produced readable science): the frozen SMC driver wrote results only at $\lambda = 1$ and printed no stage progress, so wall-capped jobs burned silently. The response — kept out of the frozen driver, which was never edited — is a checkpointing retrofit (`cgl2/samplers/ckpt.py`) with three properties:

1. **Per-stage full state.** A delegating `CheckpointKernel` (build/tune/mutate verbatim) writes a full-state checkpoint per λ -stage (post-mutation particles, λ , step size, trajectory length, acceptance, and the exact per-stage gradient/ $\log$ -probability ledger) plus one progress line; a pre-registered in-job wall cap raises *after* the checkpoint, with an exit-3 PARTIAL protocol — a wall hit preserves progress by construction, and no gate math is ever run on a non- $\lambda=1$ ensemble.
2. **Evidence-complete resume.** Weight-stage log-likelihoods are recorded through the driver’s existing `loglik_batch_fn` seam (delegation only), so a resumed run retains the *full* $\log Z$ accumulation and per-stage bootstrap — not a partial-from-resume approximation.
3. **Bit-identity.** On `--resume`, the RNG chain is fast-forwarded by replay of the recorded log-likelihoods; the retrofit gate proves a fresh run and an interrupted-then-resumed run *bit-identical* including $\log Z$ and its bootstrap, and that resume-after-completion rebuilds the identical result with zero new forward evaluations. (Documented caveat: under strict-f32 execution the checkpointed ensemble is an f64 reconstruction of the f32 state, so resume there is protocol-identical up to an f32-rounding-level perturbation; everywhere else it is bit-exact.)

¹⁵Source artifact: `data/smc.b0_report.json`.

¹⁶The ledger prose records “ 30σ ”; the artifact of record `data/smc.b0_report.json` gives $10.79/0.536 = 20.1$. Per the campaign’s discrepancy-flag rule (§5) the artifact value is quoted; the demotion ($> 3\sigma$ rule) is unaffected.

The retrofit ships with a 58-test suite, green on both machines. Its operational effect is visible throughout the results: the recovery lane turned the first wave’s zero-artifact failure mode into readable recoveries (3 of 4 cells), and the 12-hour, three-leg chained real-data carousel run — and every partial- λ cost readout quoted in the decision matrix — exists only because interrupted evidence runs became resumable and readable at all.

5 Methods III: Protocol

5.1 Pre-registration and frozen gates

Every experimental cell was preceded by a *design checkpoint* logged to the campaign record before submission: hypothesis, predicted direction and magnitude, falsifier, and derived threshold. Certification gates (F1–F8, B0) were frozen at P0 in the repository README before any science ran. Thresholds parameterized by then-unknown quantities carried declared placeholders that were *finalized, not moved*, when the parameterizing measurement landed: the seed-repeatability study fixed $\sigma_{\text{seed}}(\gamma) = 0.00325$ (low basin), 0.00664 (steep), and $\sigma_{\text{seed}}(\Delta \log Z) = 1.79$ nats (with the $n=3$ caveat — $\pm 46\%$ χ -distribution sampling error — quoted at every use), and the downstream gates inherited these by a ledgered finalization row.¹⁷ Where a pre-registered gate proved undecidable as written, the record shows the branch taken rather than a rescored verdict: outcomes in this report include FAIL-as-written, NOT-DECIDABLE-AS-REGISTERED, PARTIAL-BY-BUDGET, NOT-EVALUABLE-WITHIN-BUDGET, and AMBIGUOUS-AT-FROZEN-THRESHOLDS — honest negatives are first-class results, and no threshold was moved after data. Two further house rules: diagnostic figures are produced and inspected *before* gate text is written (a rule adopted from the upstream team’s research culture), and amendments (a re-registered gate, a descoped cell) enter the record as explicit ledgered decisions with the original always preserved.

5.2 Proposer $\neq$ grader: the UNCERTIFIED-external label

The campaign separates producing a result from certifying it. For every claim that runs through the scene API substrate, we are the *proposer*; the *grader* is the upstream GIGA-LENS team (and Greg Benson for our-side scope). Accordingly, every scene-substrate verdict in this report carries the label *proposed* (*UNCERTIFIED — external*), and the deliverable of record is a claims register in the team’s own lab-notebook format (`papers/handoff/CLAIMS.md`: fourteen claims, each with pre-registered criterion, measured numbers, named artifacts, and a mandatory doubt report). “Validated” is reserved, throughout this report, for the old certified stack; scene-API results are “reproduced/measured, UNCERTIFIED (external).” Claims derived from data or targets provided by the team are additionally marked SIGN-OFF GATED in the register: internal-memo material, publication-gated on the team’s explicit sign-off. In this report the carousel real-data cell appears under exactly that regime — *data provided by the team, used with permission; any comparison against their own runs is deferred pending sign-off* — and the upstream repositories are referenced only as the upstream development branches, without characterization of their unpublished results.

Four bright lines, frozen in the plan before the campaign began, bind the published text: (i) no result is presented so as to be readable as an MCLMC-vs-HMC efficiency comparison on a unimodal target — every sampler number in this report attaches to a multimodality, evidence, cold-start, or noise-model question; (ii) nothing derived from the upstream branches’ unpublished code or results appears externally without the team’s sign-off; (iii) the team’s ongoing staged campaigns

¹⁷Source artifact: `data/t02.t03_gate_eval.json`.

are untouched — our prototypes are offers, not applications; (iv) co-authorship is offered on any publication whose results run through their substrate.

5.3 Budget fences and operational discipline

The campaign ran under a 100 A100h hard stop with per-phase caps and a ledger-before-results rule: the estimated cost of every job is appended to the campaign ledger *before* any result is read, and actuals are harvested from `sacct`. Realized spend was 53.51 A100h — certification slice 2.21, injection-recovery 7.80, benchmark 22.45 of a 24-h cap, the user-funded real-data extension 15.85 of 18, and the payload/arbitration phase 5.21 of 17 (component sums differ from the total by 0.01h of per-row rounding).¹⁸ Sub-fences were binding, not advisory: chained runs carried per-leg wall caps with pre-declared stop conditions (the carousel chain stopped at its fence with $\lambda = 0.15$, and the cell was closed by an explicit ledgered decision, D9, rather than extended), budget-matched comparison arms were capped at their partner’s *realized* gradient ledger, and cells whose measured cost exceeded remaining headroom closed as NOT-EVALUABLE-WITHIN-BUDGET rather than overrun. Unused programmed items are closed explicitly in a final gate-ledger sweep (six programmed items proposed NOT-RUN with reasons — among them the S7 flow-preconditioned-MAMS arm, whose prerequisite B1 $\lambda=1$ ensemble never came to exist, and the X3 evidence-scored Vela-ladder prototype, never mobilized — plus B4 proposed NOT-EVALUABLE-WITHIN-BUDGET) rather than silently dropped.¹⁹

Operational rules that each cost real budget to learn once, then became protocol: production SMC jobs pin high-memory GPUs (`-C gpu&hbm80g`; memory sizings measured under rematerialization do not transfer to un-rematerialized runs); the submitted batch-script snapshot is verified by checksum via `scontrol write batch_script` (SLURM snapshots the script at submission — a “resubmitted fix” can silently run the old code); every campaign file is checksum-audited on the remote before production (single-writer manifest, union-merge on conflict); a dead-man watchdog with per-job expected artifacts and stall alerts armed every wave; and the lensing likelihood is never evaluated on CPU. A full ops-lessons table, with the job ids that motivated each rule, accompanies the reproducibility appendix.

5.4 Provenance rule

Every headline number in this report is traced to an on-disk artifact (JSON/NPZ path), and where it was produced on Perlmutter, to a SLURM job id; the campaign ledger `CAMPAIGN.md` records the gate row and the decision context. Where the ledger prose and the artifact of record disagree, the synthesis that feeds this report flags the discrepancy rather than smoothing it — seven such flags are on record, and in every case this report quotes the *artifact* value.²⁰ The experimental record — plan, ledger, design checkpoints, gate evaluations, and the claims register — ships with the repository.

6 Results I — Certification: the Scene API Reproduces the Validated Stack

The campaign’s foundation result: the next-generation scene API — the team’s upstream development branch, vendored unpatched at commit 80916d2 — evaluates the same forward model, likelihood, and

¹⁸Source artifact: `CAMPAIGN.md` (A100-hour ledger).

¹⁹Source artifact: `research/gate.ledger_final.md`.

²⁰Source artifact: `research/synthesis.tables.md` (§4, discrepancy flags).

gradients as the previous campaign’s validated stack to within machine precision, on both campaign machines, for the configuration class of the real lens. Everything downstream (the mechanism program of §7, the cross-stack reproduction of §7.7) stands on this section. All scene-substrate numbers are proposed (UNCERTIFIED—external): we are the producer; the upstream team is the grader.

The certification battery F1–F8 compares the scene API’s simulator and likelihood surfaces against the validated 58ec9a7 stack of the preceding campaign on the same DESI–165.4754–06.0423 inputs (v2d native and v3b binned products), at the reference point plus three seeded perturbations, compared in constrained parameter space. The battery ran at zero A100 cost (phoenix L4 free tier); its Perlmutter repetition (F8) took 58 s of A100 time.

6.1 The F1–F8 parity battery, both machines

All hard gates pass on both machines (Table 5).²¹ The forward image and design columns agree at the 10^{-15} level — the two stacks are evaluating the *same function*, not merely similar ones — and the constrained-space gradient, which compounds every convention in the chain (profiles, grids, PSF, masking, bijectors), agrees to 1.5×10^{-11} relative L_2 . The Perlmutter-native re-run under the production pinned environment (jax 0.6.2, x86 A100; job 55980038) passes every hard gate with the F1 statistic *bit-consistent* with the phoenix value (5.682×10^{-15}), certifying the stack on both machines and unblocking every scene-API production cell of the campaign.

Table 5: Cross-stack certification battery: scene API (vendored upstream development branch, unpatched) vs. the validated 58ec9a7 stack, same DESI–165.4754–06.0423 inputs, constrained space, $z_{\text{ref}} + 3$ seeded perturbations. F6 is the restated conditioning-scaled gate (§6.3); the F8 column’s F1 value is bit-consistent with phoenix. Phoenix column: `data/parity_report_scene.json`; Perlmutter column (F8, job 55980038, 58 s): `data/results-perlmutter/parity_report_scene.pmnative_55980038.json`.

gate (quantity)	tolerance	phoenix L4	Perlmutter A100 (F8)
F1 forward image (rel)	10^{-12}	5.7×10^{-15}	5.682×10^{-15}
F2 design columns (rel)	10^{-12}	3.9×10^{-15}	pass
F3 diagonal masked $\log L + \chi^2$	10^{-8}	7.5×10^{-9}	pass
F4 constrained gradient (rel L_2)	10^{-8}	1.5×10^{-11}	1.314×10^{-11}
F5/F5b delta-kernel $\equiv$ stock	10^{-10}	5.8×10^{-11} / 0.0	pass
F6 Occam vs fp128 truth (§6.3)	cond-scaled	v2d 1.19×10^{-11} ; v3b 4.48×10^{-10}	3.562×10^{-10}
F7 flat- z round-trip (46 names)	exact	0.0	pass

One scoping note, recorded for honesty: the pre-registered F1 statement (forward image, old vs. `SceneSimulator.simulate`) is gated as written; the *full* model image with each stack’s own solved source amplitudes is reported informationally (worst 1.1×10^{-12}), because it compounds F1×F2 through the $\text{cond}(A) \approx 7 \times 10^7$ amplitude solve — a path the likelihood itself never takes (the quadratic form that enters $\log L$ is gated via F3/F4).

6.2 Three convention reconciliations

Parity at the level of Table 5 holds *given three documented convention reconciliations*, each measured before being reconciled. They are findings in their own right — stock-vs-stock, the two stacks differ at exactly these scales:

²¹Source artifact: `data/parity_report_scene.json`; `data/results-perlmutter/parity_report_scene.pmnative_55980038.json`.

1. **Sérsic b_n approximant (a real $\sim 2 \times 10^{-3}$ model-level difference).** The old stack uses $b_n = 1.9992n - 0.3271$; the scene API uses $b_n = \exp(0.6950 + \ln n - 0.1789/n)$. Both are literature approximants to the exact b_n equation; we did not adjudicate which is closer to exact for the use range. Measured via an informational `native_profiles` arm: $\sim 2 \times 10^{-3}$ relative model-image difference for the tested class.²² Any cross-stack comparison at the 10^{-3} level needs this reconciliation.
2. **f32 shapelet Hermite prefactor** in the old stack ($\sim 10^{-8}$ basis delta).
3. **f32 coordinate grids + un-renormalized subgrid PSF kernel** in the old stack ($\sim 10^{-4} / 2 \times 10^{-6}$ image deltas at v2d/v3b).

The reconciliations are implemented on our side, as subclasses and instance overrides in `cgl2/scene_build.py` (docstring items 1–3); **the vendored branch is unpatched**. Declared environment deviation: our numpy is 2.4.6 vs. the branch’s 2.1.3 pin, covered by the gate battery. The certification covers one configuration class — EPL+Shear with $4 \times$ Sérsic lens light and Sérsic+shapelets ($n_{\max} = 6$) source — and makes no statement about other profile or multi-plane paths.

6.3 F6: an honest gate failure at the f64 noise floor, and its signed restatement

F6 — parity of the Occam term $-\frac{1}{2} \log \det A$ of the generalized ridge marginalization — **failed as originally written**, and the failure is worth more than the pass that replaced it. The original gate compared the scene-API jax-Cholesky value against a numpy `slogdet` reference at $\leq 10^{-10}$; on v3b it failed at 1.31×10^{-10} .²³ The noise-floor analysis showed why the gate, not the code, was wrong: at $\text{cond}(A) = 7.0 \times 10^7$, pure-numpy Cholesky and `slogdet` on the *identical* matrix already differ by 7.2×10^{-10} , and against an fp128 ground truth the jax-Cholesky value errs by 4.5×10^{-10} — *more accurate* than the numpy `slogdet` gate reference itself (7.1×10^{-10}) and than the old validated stack’s own stored value (1.3×10^{-9}). The defensible statement is that the gate *reference* errs at $\sim 7 \times 10^{-10}$ there, so demanding 10^{-10} cross-algorithm agreement is not meaningful at that conditioning (not that 10^{-10} is unreachable by any f64 implementation).

The gate was therefore restated — by signed exception (Benson sign-off, 2026-07-15), not silently moved — to compare against fp128 truth with a conditioning-scaled tolerance:

$$|\log \det A_{\text{jax-cho}} - \log \det A_{\text{fp128}}| \leq \max(10^{-10}, 5 \epsilon \text{cond}(A) \cdot 10^{-2}),$$

under which F6 passes: v2d 1.19×10^{-11} (tol 10^{-10} , floor governs at $\text{cond} = 7.9 \times 10^6$); v3b $4.48 \times 10^{-10} \leq 7.76 \times 10^{-10}$ ($\text{cond} = 7.0 \times 10^7$). The original recommendation and the full fp128 evidence are preserved in the artifact.²⁴ The transferable lesson, offered to anyone writing log det gates for marginalized likelihoods: at $\text{cond}(A) \gtrsim 10^7$ a cross-algorithm f64 tolerance below $\sim 10^{-9}$ tests the reference, not the code.

6.4 The correlated-noise likelihood port is exact in its analytic limits

The drizzle-anchored correlated-noise likelihood of the previous campaign was ported onto the scene API’s documented `Dataset/LikelihoodTerm` seam, as `CorrelatedImageData` plus its matching likelihood term: one `lstsq.simulate` render per evaluation, grouped-depthwise convolutional

²²Source artifact: `data/parity_report_scene.json` (`native_profiles` arm).

²³Source artifact: `data/parity_report_scene.json` (`products.v3b.F6.chol_minus_numpy`).

²⁴Source artifact: `data/parity_report_scene.json` (`products.*.F6`).

whitening, and generalized ridge marginalization with the Occam term. Its validation gates, all passed:²⁵

- **Delta-kernel identity (F5/F5b):** with a delta whitening kernel the correlated term reproduces the stock `ImageLikelihoodTerm` to 5.8×10^{-11} (χ^2 branch exact, 0.0).
- **Dense-Cholesky exact reference (03-A):** on a 32^2 toy with a full dense covariance, the ported term agrees with the exact reference to 5.5×10^{-12} nat (gate ≤ 0.1 nat); delta identities (03-B) hold to 9.1×10^{-13} / 0.0.
- **Whitener bundles re-validated on import:** operator-norm whiteness $e_{\text{op}} \leq 0.02$ (strict) holds for the three production bundles (v2d, v3b, v3); the `v2d_relaxed` bundle is correctly flagged inadmissible-by-design under the strict gate (it was built as the ledgered relaxed arm).

The $-\frac{1}{2} \log \det A$ term — which a plain least-squares implementation drops — is included, and matters for any downstream Bayes factor. This ported term is the engine of the cross-stack reproduction in §7.7.

7 Results II — The Bracket Mechanism on the Real Lens

The previous campaign ended with a verdict — the correlated likelihood is necessary but not sufficient — and a puzzle: on DESI-165.4754-06.0423 the three data products bracket the power-law slope (fine-correlated $\gamma = 1.816$, native-diagonal anchor 1.433, binned-correlated 1.103) instead of unifying. This campaign ran a pre-registered elimination program against that bracket. The outcome: two suspects are dead (radial profile curvature, the companion galaxy), one proposed cure is falsified (spectrum flooring), and the noise-model class — a single stationary convolution kernel per product — is directly rejected on the field that produced $\gamma = 1.103$, making noise-model-class misspecification the standing working hypothesis. The confirmatory injection experiment came back confounded, honestly reported below, so the mechanism is indicted, not convicted. Source/PSF representation remains alive and untested.

Table 6 is the final scoreboard; the subsections that follow give each row’s evidence. Except where noted, the mechanism program ran on the *validated old stack* (these are our-stack results, not scene-substrate results); X1-G0 and the T0.4 trilogy cost zero GPU hours.

7.1 The number under interrogation: γ certified at 1.1032 ± 0.0086 , with a seed-stable basin preference

Before interrogating the binned-correlated slope, the campaign certified it (T0.2: three matched-seed repeats of the production 128-particle tempered-SMC fit on v3b, seeds $\{2, 3, 4\}$; 2.21 A100-h with T0.3).²⁶ The low-basin medians are $\{1.1032, 1.0967, 1.1005\}$, giving $\sigma_{\text{seed}}(\gamma) = 0.00325$ — the pre-registered kill ($\sigma_{\text{seed}} > 0.024$) is not tripped — and

$$\begin{aligned} \gamma^{\text{binned}}(\text{corr, low}) &= 1.1032 \pm 0.0086, \\ \sigma_{\text{tot}} &= \sqrt{\sigma_{\text{stat}}^2 + \sigma_{\text{seed}}^2} = \sqrt{0.0080^2 + 0.00325^2} = 0.00864 \approx 1.08 \sigma_{\text{stat}}. \end{aligned}$$

²⁵Source artifact: `data/correlated_term_validation.json`; `data/parity_report_scene.json`; `data/whitener_manifest.json`.

²⁶Source artifact: `data/t02.t03_gate_eval.json`.

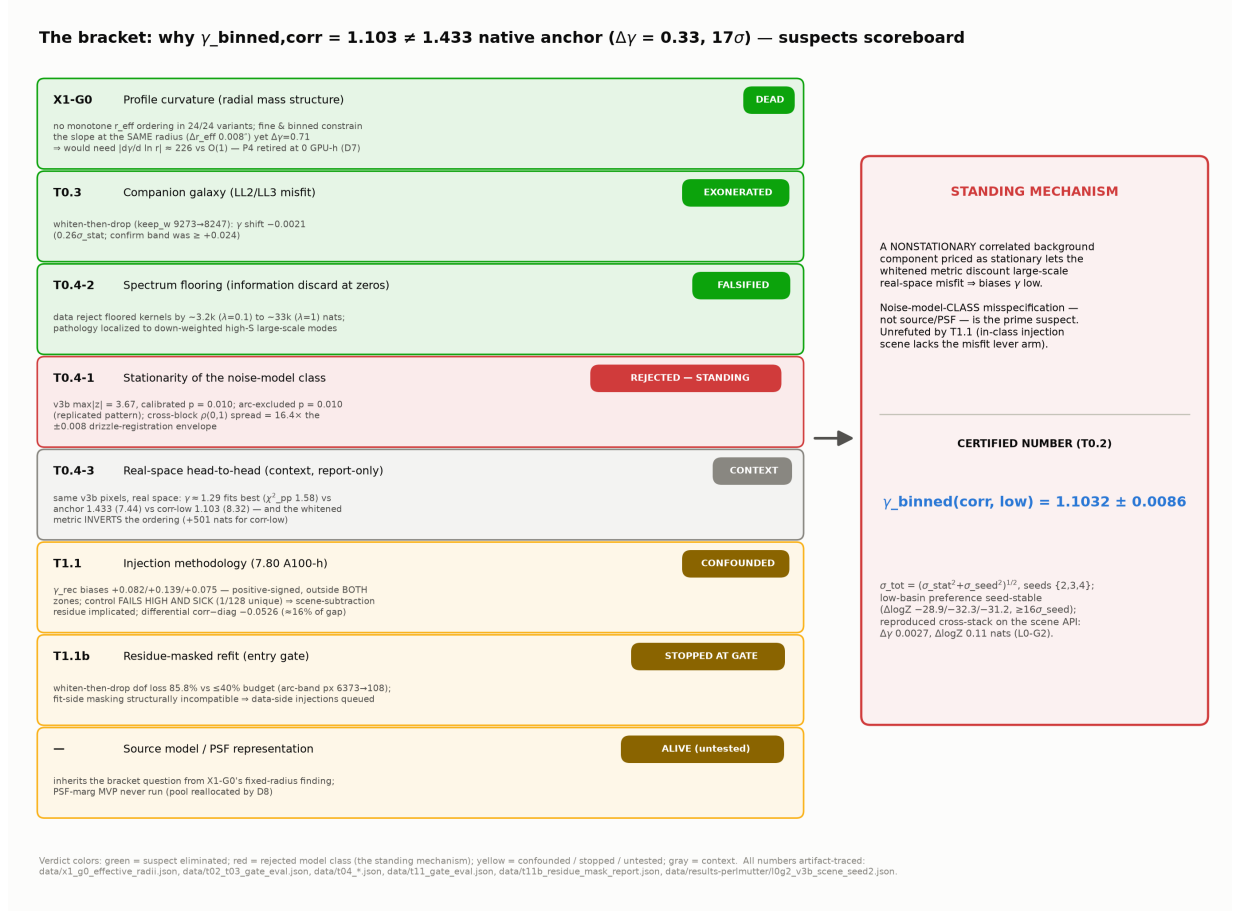


Figure 6: The bracket suspects scoreboard (the graphical form of Table 6): one card per pre-registered discriminating measurement, in program order. Profile curvature DEAD (X1-G0 entry-gate kill executed as written; P4 retired at zero GPU cost, decision D7); companion EXONERATED (T0.3); spectrum flooring FALSIFIED by 3.2k–33k nats (T0.4-2); stationarity of the noise-model class REJECTED at calibrated $p = 0.010$ — the standing mechanism (T0.4-1); the real-space head-to-head as context (T0.4-3; the anchor row carries a cross-product resolution handicap — honest caveat); injection–recovery CONFOUNDED by scene-subtraction residue under its pre-registered honesty clause (T1.1, $n=3$, no coverage claims); the residue-masked refit stopped at its entry gate (T1.1b, 0 A100-h); source/PSF representation ALIVE (untested). $\gamma = 1.1032 \pm 0.0086$ is certified (T0.2) and reproduced cross-stack (L0-G2); the E3 kernel-scan deferral remains the standing caveat on absolute- γ claims. Sources: the per-suspect artifacts cited in the subsections of this section.

Table 6: Final suspect scoreboard for the γ bracket (fine 1.816 / anchor 1.433 / binned 1.103). Every verdict is pre-registered; artifacts as cited in the subsections.

suspect	verdict	discriminating measurement
profile curvature (radial mass structure)	DEAD (X1-G0, §7.2)	no monotone r_{eff} ordering in 24/24 variants; would require $ d\gamma_{\text{loc}}/d\ln r \approx 226$ vs. $O(1)$ physical
companion galaxy (LL2/LL3 misfit)	EXONERATED (T0.3, §7.3)	whiten-then-drop shift -0.0021 ($0.26 \sigma_{\text{stat}}$; band was $\geq +0.024$)
spectrum flooring (information discard at spectral zeros)	FALSIFIED (T0.4-2, §7.4.2)	data reject floored kernels by $\sim 3.2\text{k}$ – 33k nats per point
stationarity of the noise-model class	REJECTED — standing mechanism (T0.4-1, §7.4.1)	calibrated $p = 0.010$; cross-block kernel spread $16.4\times$ the drizzle-registration envelope
injection methodology (T1.1 construction)	CONFOUNDED $\rightarrow$ data-side redesign (§7.5–7.6)	control fails high <i>and</i> sick; fit-side rescue loses 85.8% of whitened dof
source model / PSF representation	ALIVE (untested)	inherits the bracket question: the driver acts at fixed radius (X1-G0)

The steep basin repeats at $\{2.6393, 2.6522, 2.6485\}$ ($\sigma_{\text{seed}} = 0.00664$), and the per-matched-seed basin evidence gap $\Delta \log Z(\text{steep} - \text{low}) = -28.88 / -32.27 / -31.19$ nats is **seed-stable in sign and magnitude** — a low-basin preference $\geq 16 \sigma_{\text{seed}}$ from zero, with $\sigma_{\text{seed}}(\Delta \log Z) = 1.79$ nats (Fig. 7). That last number is load-bearing: the campaign’s cross-cutting finding that bootstrap errors understate evidence error makes this measured seed scatter the honest unit for every downstream $\Delta \log Z$ gate. Caveat, quoted with every use: $n = 3$ seeds, so the σ estimates themselves carry $\pm 46\%$ sampling error (χ -distribution). Downstream thresholds were re-parameterized from the measured σ_{seed} at this point — a ledgered finalization, not a goalpost move.

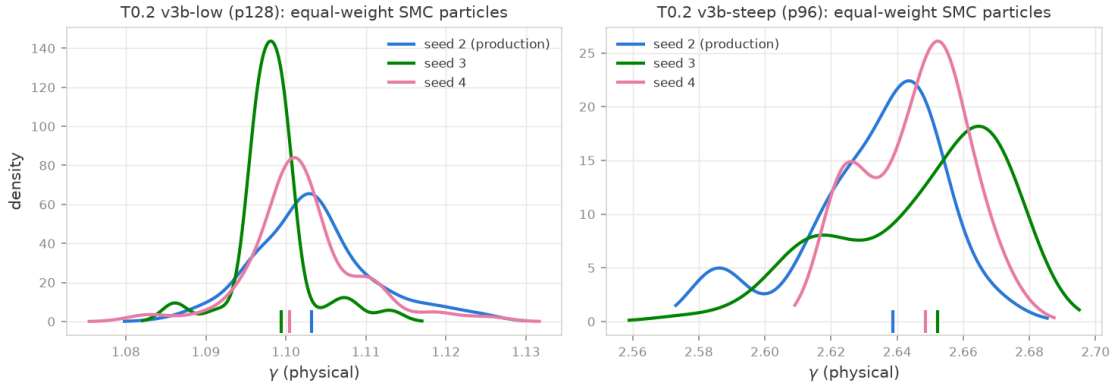


Figure 7: T0.2 seed-repeat certification: low- and steep-basin γ posteriors for seeds $\{2, 3, 4\}$ (production tempered SMC, v3b correlated likelihood). The money number is $\gamma = 1.1032 \pm 0.0086$ (σ_{tot} , seed scatter folded in); the basin preference $\Delta \log Z(\text{steep} - \text{low}) \approx -29$ to -32 nats is seed-stable. Source: data/t02.t03_gate_eval.json.

7.2 X1-G0: profile curvature is structurally dead

The P4 hypothesis held that the EPL single-power-law assumption drives the bracket: whitening and binning re-weight spatial frequencies, so each product constrains the slope at a different effective radius, and a curved $\kappa(r)$ then yields different local slopes. Its pre-registered entry gate (X1-G0, zero GPU cost) required the per-product effective radii to be monotone in the same sense as the γ ordering. **The gate failed in all 24 analysis variants** (3 statistics $\times$ primary weighting plus 7 robustness variants), and P4 was retired unspent on the strength of it (decision D7).²⁷

Table 7: X1-G0: slope-information effective radii by product (arc-brightness $\times (S/N)^2$ weighting; three statistics agree). The required ordering (γ : 1.816 $>$ 1.433 $>$ 1.103) would need r_{eff} monotone in the same sense; observed is $\mathbf{v3} > \mathbf{v3b} > \mathbf{v2d}$ — the middle- γ product has the *smallest* r_{eff} . Source: `data/x1_g0_effective_radii.json`.

product (likelihood)	γ	r_{eff} mean	log-mean	median
v3 fine 0.04'' (correlated)	1.816 ± 0.117	2.596''	2.578''	2.613''
v2d native 0.13'' (diagonal, anchor)	1.433 ± 0.035	2.501''	2.466''	2.543''
v3b binned 0.08'' (correlated)	1.103 ± 0.008	2.588''	2.571''	2.612''

The **fixed-radius magnitude argument** is stronger than the ordering kill. The fine and binned products are the same sky, and their slope information peaks at the same radius: $\Delta r_{\text{eff}} \approx 0.008''$ (r_{mean} 2.5964'' vs. 2.5882''), less than a quarter of a fine pixel — yet their slopes differ by $\Delta\gamma = 0.71$. For any curved $\kappa(r)$ to carry that gap across that lever arm would require a local-slope gradient $|d\gamma_{\text{loc}}/d\ln r| \approx 226$ per e -fold; physical profile curvature (NFW at r_s , dPIE break) delivers $O(1)$, and even the most charitable product pairing needs ≈ 10 (Fig. 8). The positive content of the kill: *the bracket driver differs between likelihoods at fixed radial information* — it must live in the noise/likelihood treatment or in source/PSF structure, not in the radial mass profile. What the gate does *not* kill: a BPL/dPIE fit could still win on evidence for other reasons; the pre-registered *mechanism* is what is excluded.

7.3 T0.3: the companion galaxy is exonerated

The companion galaxy (the LL2/LL3 light components, a known local misfit) was a candidate driver of the low slope. The pre-registered discriminator whitens first and then drops the companion's whitened pixels (keeping the production whitener fixed; 9273 $\rightarrow$ 8247 whitened dof), so the test isolates the companion's likelihood contribution without rebuilding the noise model. Result: $\gamma = 1.1011$ vs. production 1.1032 — a shift of -0.0021 ($0.26\sigma_{\text{stat}}$), statistically zero against the pre-registered detection band of $\geq +0.024$, with convergence diagnostics indistinguishable from production.²⁸ **The companion is exonerated**; the noise-model class keeps its place as prime suspect. (The companion run's $\log Z$ is not comparable to production — different data after the drop — and is not used.)

7.4 The T0.4 trilogy: the noise-model class is the standing suspect

Three zero-GPU checks on the real field, run on the validated stack with pre-registered forks, converge on one mechanism.

²⁷Source artifact: `data/x1_g0_effective_radii.json`; `research/x1_g0_mechanism_check.md`.

²⁸Source artifact: `data/t02.t03_gate_eval.json` (t03).

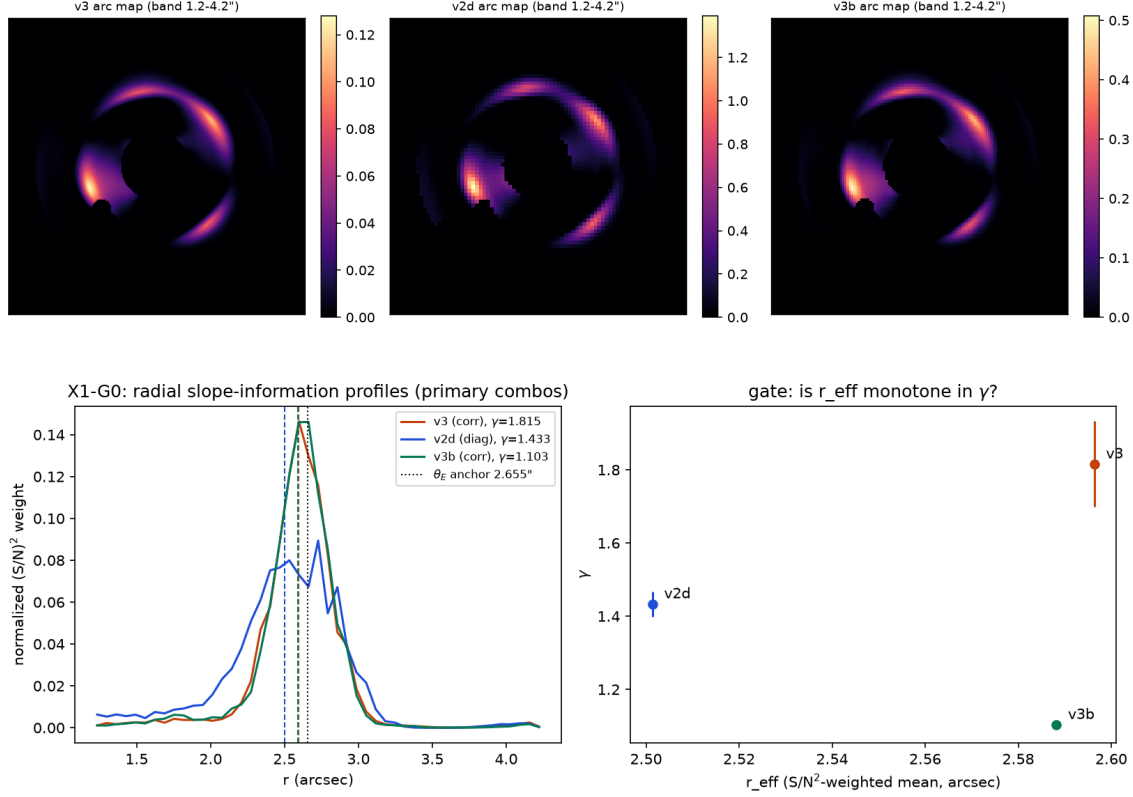


Figure 8: X1-G0 entry-gate kill. Top: arc-brightness $\times (S/N)^2$ slope-information weight maps for the three products. Bottom: radial $(S/N)^2$ slope-information profiles (left) and the γ -vs- r_{eff} gate panel (right): no monotone ordering exists (24/24 variants), and the fine/binmed pair constrains the slope at the same radius ($\Delta r_{\text{eff}} \approx 0.008''$) while disagreeing by $\Delta\gamma = 0.71$. Source: `data/x1_g0_effective_radii.json`.

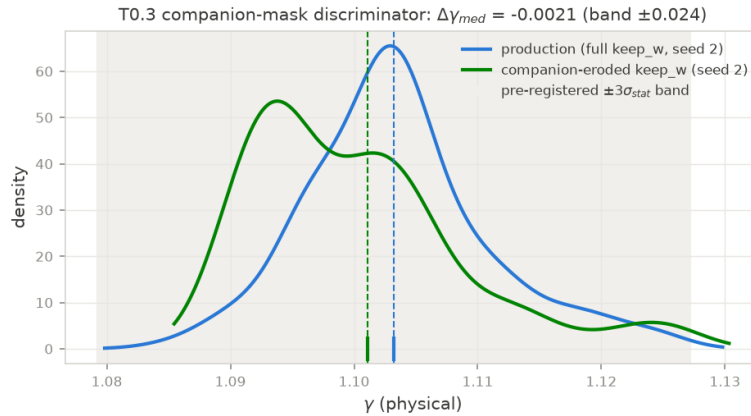


Figure 9: T0.3 companion-mask discriminator: the γ posterior under the production whitener with and without the companion's whitened pixels (9273 $\rightarrow$ 8247 dof). The shift is -0.0021 ($0.26\sigma_{\text{stat}}$) against a pre-registered detection band of $\geq +0.024$: the companion is exonerated. Source: `data/t02.t03_gate_eval.json`.

7.4.1 Stationarity of the noise-model class: REJECTED

The correlated likelihood behind $\gamma = 1.103$ prices noise with a *single stationary convolution kernel per product*. T0.4-1 tests that class directly: per-block kernel fits on a 3×3 grid over the field, with null bands calibrated by 200 exact FFT resamples of stationary fields under the global fitted kernel (mask geometry and finite-block scatter included), and a multiplicity-honest calibrated p from the same $\max|z|$ statistic on every null draw.²⁹ The verdict (Fig. 10):

- The money product **v3b** rejects the stationary class at $\max|z| = 3.67$, **calibrated** $p = 0.010$; the arc-excluded **v3** robustness arm also rejects at $p = 0.010$ ($\max|z| = 3.66$) — excluding the arc band *strengthens* the fine-product rejection (arc residuals were diluting the signal, not driving it).
- The rejection is a coherent, cross-product-replicated spatial pattern — top row of blocks over-correlated ($\rho(0, 1)$ up to 0.71 vs. global 0.60), middle-left block under-correlated (z down to -3.7) — the same sign pattern in **v3** and **v3b**, with and without the arc band. That is what a real shift-variant noise field predicts and a fit fluke does not.
- Drizzle sub-pixel registration cannot explain it: the enumerated registration envelope predicts a block-to-block $\rho(0, 1)$ spread of ± 0.008 ; the observed spread is ~ 0.28 peak-to-peak on **v3b** — $16.4\times$ **the envelope**. The nonstationarity must come from something else that varies across the skycell (coverage/weight-map structure and/or residual astrophysical background).

The noise-model class that produced $\gamma = 1.103$ is provably violated by the field it whitens. Honest caveats: the four test arms share pixels and are not independent — the case is the replicated pattern, not any single p -value.

7.4.2 Spectrum flooring: FALSIFIED as mechanism and as cure

A proposed benign reading of the low slope was information discard at spectral zeros: the near-singular fine whitener amplifies low-power modes, and flooring the whitener spectrum (raising $s_{\text{floor}} = \lambda$) should then cure the known “fine-low gaming” pathology, in which optimization drives $\gamma \rightarrow 1.02$ while whitened log-probability improves and real-space χ^2 collapses. T0.4-2 rebuilt the whitener across $\lambda \in \{0.05 (\equiv \text{production, unfloored}), 0.1, 0.3, 1.0\}$ — the production build is reproduced to $\max|\Delta h| = 1.0 \times 10^{-9}$ in the taps, and the production floor is $s_{\text{floor}} = 0.05$, at which the floor does not engage (the plan’s “0.1” label is corrected in the artifact) — and scored three frozen points (the **v3b**-low start A, the railed fine-low MAP B, the fine-steep MAP C) under exact cross- λ normalization ($\log|C(\lambda)|$ via Szegő, dense-Cholesky anchored to $\leq 7 \times 10^{-4}$ nat/px).³⁰

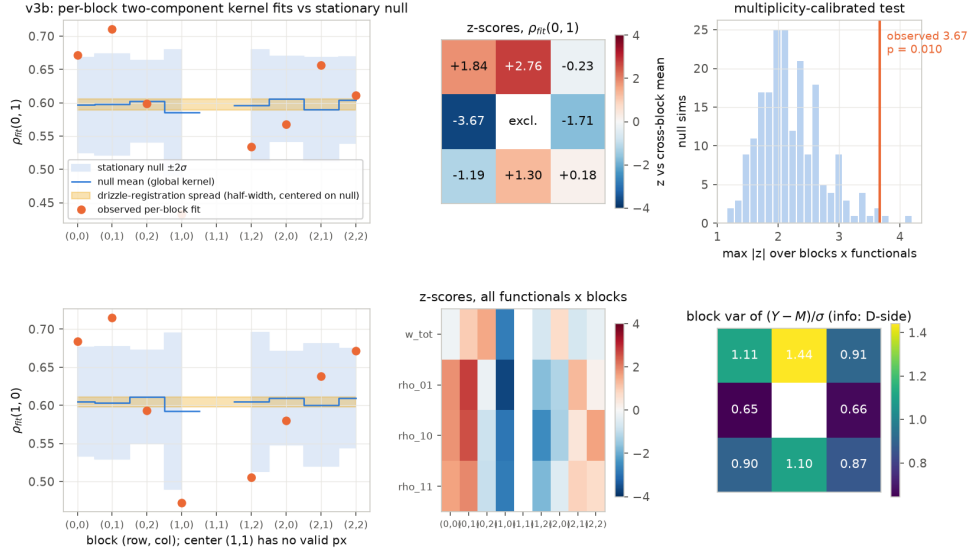
Two results (Fig. 11):

- **Flooring never re-aligns the orderings.** The whitened-vs-real-space inversion (B beats A by ~ 4200 whitened nats at production while real space prefers A by $\Delta\chi_{\text{pp}}^2 = 37$) persists essentially unchanged from the unfloored whitener through $\lambda = 1.0$ (85% of modes floored): still +3832 nats. If the pathology lived at the spectral zeros, flooring them would have fixed it.
- **The data reject the floored kernels outright:** the normalized likelihood drops by $\sim 3.2\text{k}$ nats per point at $\lambda = 0.1$ ($-3187/ -3328/ -3304$ for A/B/C) and by $\sim 33\text{k}$ nats at $\lambda = 1.0$ ($-33,060/ -33,450/ -33,370$). “Cure by flooring” costs thousands of nats of evidence per step — more than any plausible model could repay.

²⁹Source artifact: `data/t04.stationarity.json`; `data/t04.stationarity_noarc.json`.

³⁰Source artifact: `data/t04.lambda.arm.json`.

T0.4 stationarity test — v3b: $\max|z|=3.67$, naive 2σ gate FAIL, calibrated $p=0.010$ (REJECTED)



T0.4 stationarity test — v3: $\max|z|=3.66$, naive 2σ gate FAIL, calibrated $p=0.010$ (REJECTED)

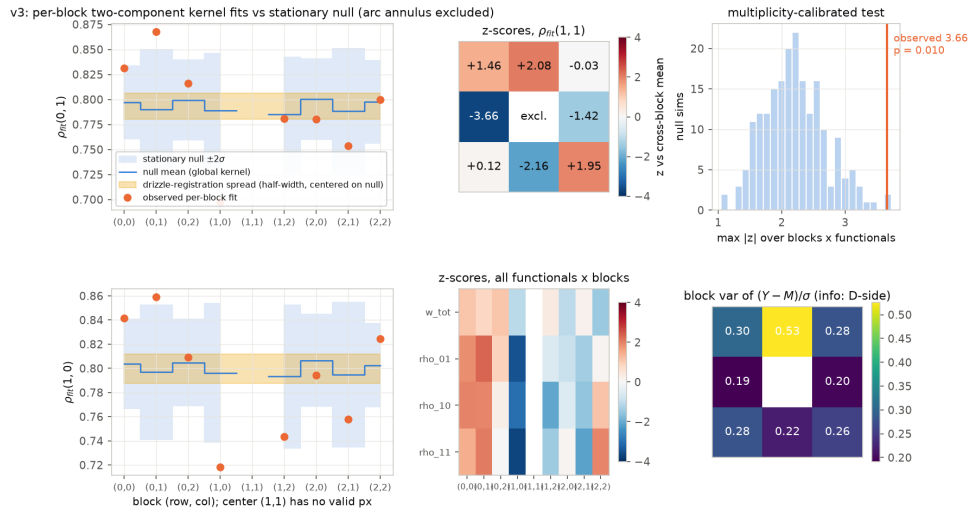


Figure 10: T0.4-1 stationarity rejection. Top: the money product v3b — per-block kernel functionals vs. null-calibrated bands ($\max|z| = 3.67$, calibrated $p = 0.010$). Bottom: the arc-excluded fine product v3 ($\max|z| = 3.66$, $p = 0.010$) — excluding the arc band *strengthens* the fine-product rejection (with the arc included, v3 gives $p = 0.17$). The spatial pattern (top row over-correlated, middle-left under-correlated) replicates across products and pixel-set arms; its amplitude is $16.4\times$ the drizzle-registration envelope. Sources: `data/t04_stationarity.json`; `data/t04_stationarity_noarc.json`.

The pathology is localized instead in what flooring never touches: the *down-weighting of high-power large-scale modes* ($S_{\max}/\text{mean} \approx 99$ — the fitted correlated-background component, $w_b \approx 0.27$), which prices large-scale real-space residual structure at $\sim 1/10$ amplitude weight. Those are exactly the modes T0.4-1 shows are nonstationary: the two checks indict the same component.

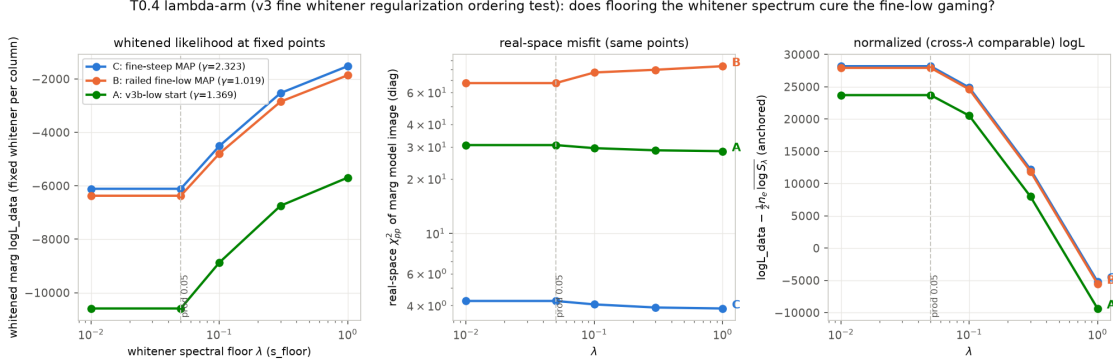


Figure 11: T0.4-2 λ -arm: whitened and normalized likelihoods of the three frozen points vs. spectrum floor λ . The gaming inversion is flat in λ (flooring does not cure it) while the normalized likelihood rejects the floored kernels by 3.2k–33k nats per point. Source: `data/t04_lambda_arm.json`.

7.4.3 Real space vs. the whitened metric: the orderings invert

T0.4-3 (report-only by design, a free ordering check rather than a calibrated test) asks which solution the binned pixels themselves prefer in plain real space. Forward renders of three on-disk posterior medians on the *same* v3b pixels, each given its ridge-solved (real-space optimal) source amplitudes:³¹

Table 8: T0.4-3 real-space head-to-head on the same v3b binned pixels (diag-solved source amplitudes; 16,653 keep px full field, 7,825 px arc band $r \in [1.2, 4.2]''$). The production whitened metric inverts the real-space ordering. Source: `data/t04_realspace_headtohead.json`.

model (γ)	χ_{pp}^2 full	χ_{pp}^2 arc band	whitened $\log L_{\text{data}}$	whitened log-posterior
anchor 1.433 (from v2d)	7.44	3.62	−5442.1	−5615.7
corr-low 1.103 (money)	8.32	2.76	− 4599.2	− 4683.6
diag-low 1.293 (home reference)	1.58	1.93	−4662.3	−5184.3

In real space the binned data’s preference is the home-product diagonal-low solution at $\gamma \approx 1.29$ (χ_{pp}^2 1.58 full / 1.93 arc band); the 1.103 corr-low model is real-space poor (8.32 / 2.76), its residual dominated by smooth lens-center misfit — the same currency as the fine-low gaming. **The production whitened metric inverts this ordering**, preferring corr-low over diag-low by +63 nats in $\log L_{\text{data}}$ and +501 **nats in log-posterior**. The 1.103 solution is a whitened-metric optimum, not a real-space one, *on its own product*. Honest caveat on the anchor row: its full-field number carries a cross-product resolution handicap (parameters fit at $0.13''$ rendered at $0.08''$; the same point renders $\chi_{\text{pp}}^2 = 1.25$ on its home product), so the defensible summary is that *neither* extreme- γ model is the binned data’s real-space preference — the $\gamma \approx 1.29$ diagonal solution is (Fig. 12).

Mechanism synthesis (working hypothesis, not a conviction). One mechanism spans all of T0.4 and the X1-G0 constraint: the fitted stationary kernels contain a large correlated-background

³¹Source artifact: `data/t04_realspace_headtohead.json`.

T0.4 real-space head-to-head on the SAME v3b binned pixels (forward renders at posterior medians; amps ridge-solved, diagonal metric)

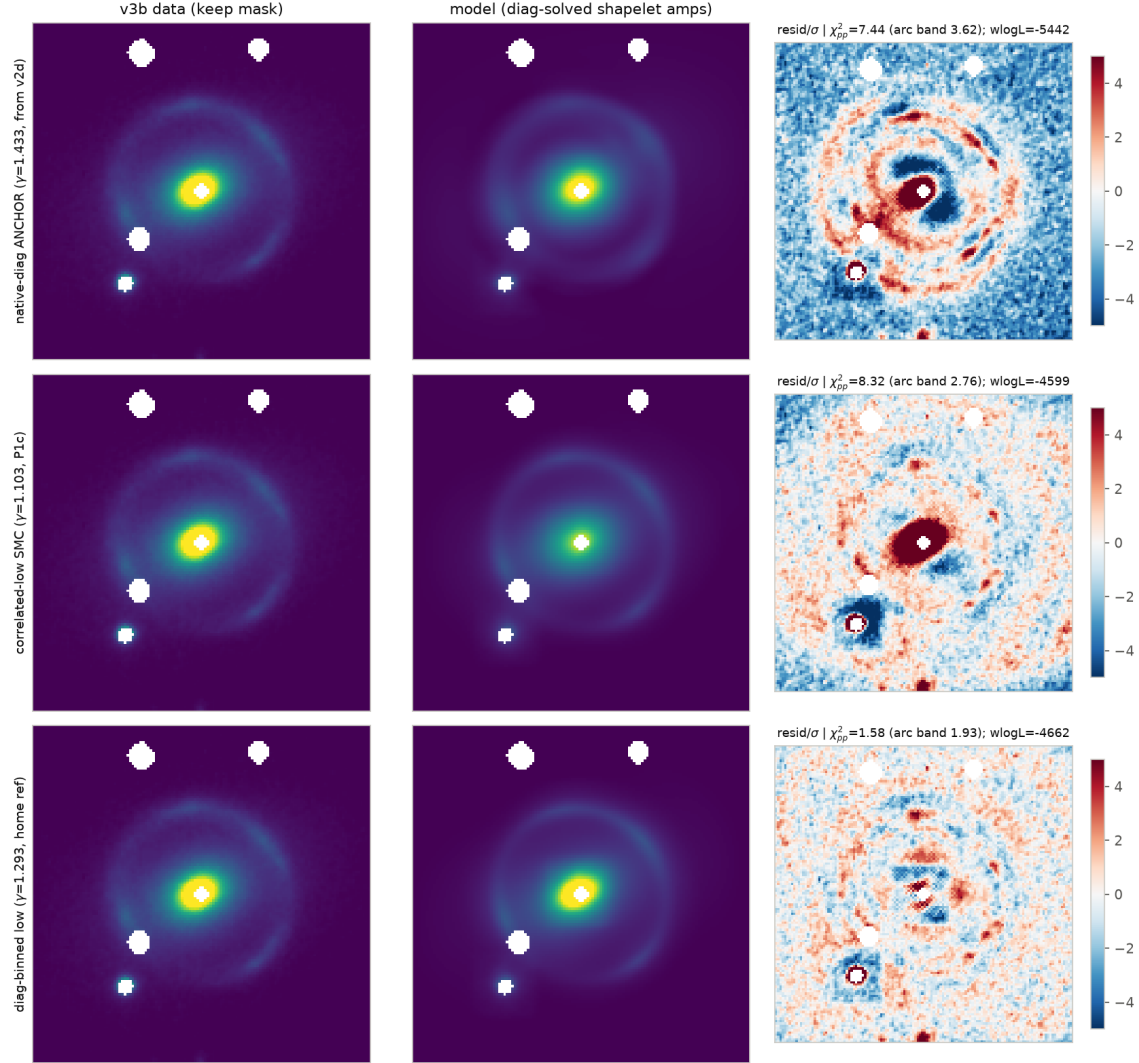


Figure 12: T0.4-3 residual maps on the same v3b pixels for the three posterior medians. The corr-low ($\gamma = 1.103$) render leaves strong coherent lens-center residuals that the whitened metric prices as “probably correlated noise”; real space prefers the $\gamma \approx 1.29$ diagonal solution. Source: data/t04_realspace_headtohead.json.

component ($w_b \approx 0.25\text{--}0.27$) estimated from sky that is provably *not* stationary across the field; the whitener therefore prices large-scale real-space misfit at $\sim 1/10$ weight everywhere, including regions where that pricing is wrong; solutions that spend exactly this currency (fine-low gaming, the 1.103 optimum) become whitened-optimal while real space disagrees — a coherent route to a low-biased γ at fixed radial information. Noise-model-*class* misspecification is thereby the prime suspect for the 1.103 over-correction; a locally-stationary (per-region kernel) class is the indicated next likelihood iteration, and the ported `CorrelatedImageData` deliberately keeps its whitener seam pluggable for it. Standing caveat: the kernel-hyperparameter scan (E3) was deferred by plan and never funded, and the alternative class was never fitted — the mechanism rests on a class-level rejection plus consistent circumstantial evidence, and the confirmatory injection test came back confounded, as the next subsection reports.

7.5 T1.1: injection–recovery is confounded — an honest negative

T1.1 was the mechanism’s confirmatory experiment (7.80 A100-h; jobs 55952480/55952481/55958518 + diagonal control 55952483): inject arcs of known slope $\gamma_{\text{truth}} = 1.4330$ into the *real v3b* drizzle residual field at three sub-pixel shifts, refit with the production correlated pipeline, and read the recovered slope against pre-registered zones — CONFIRM-low if $\text{median}(\gamma_{\text{rec}} - 1.433) < -0.078$; EXONERATE if $|\text{median}| < 0.026$; a diagonal-likelihood control on the same data predicted in $[1.29, 1.43]$.³²

The result lands outside both zones, on the positive side (Fig. 13): recovered medians 1.5151/1.5719/1.5076, biases +0.0822/ +0.1389/ +0.0747 (own- σ z +2.25/ +3.23/ +1.82); median bias +0.0822 (+0.0784 dropping the duplicate-flagged inj2 — same conclusion). The mechanism’s directional prediction failed at face value. **And the control failed high and sick:** $\gamma_{\text{rec}} = 1.5677 \notin [1.29, 1.43]$, with total resample collapse (1 unique particle of 128, $\sigma_\gamma = 4 \times 10^{-16}$) and the injected source’s amplitude railed at $\approx 58\times$ truth. Per the pre-registered honesty clause, a control failing its own window implicates the *injection construction*, not the whitener: both likelihood classes on the same injected field recover γ high by +0.07–0.14, the only shared element is the data, and the real residual field carries the real lens’s imperfect bright-object scene subtraction. The nuisance readout shows exactly the absorption signature — recovered sources $\times 1.8\text{--}3$ larger and $\times 2.4$ brighter than injected truth in all three correlated runs. $n = 3$; no coverage claims.

Two salvage data survive, quoted at their honest strength:

- **The whitener-isolating differential:** on identical data (inj1), $\gamma(\text{corr}) - \gamma(\text{diag}) = -0.0526$ — sign consistent with the mechanism, magnitude $\approx 16\%$ of the real-data 0.330 gap, and *no defensible error bar* (the diagonal leg is a degenerate point mass).
- **What T1.1 could not test by construction:** the injected scene is exactly in-class (truth source = fitted shapelet ridge solve), so the mechanism’s lever arm — discounted large-scale structural *misfit* — is largely absent by design. Even a clean exonerate here would not have refuted T0.4-1’s direct stationarity rejection, which stands unrefuted.

Verdict: NO-CONFIRM / NO-EXONERATE. The mechanism diagnosis continues to rest on T0.4’s direct evidence; T1.1 neither strengthens nor weakens it decisively.

³²Source artifact: `data/t11_gate_eval.json`; `research/t11_injection_recovery.md`.

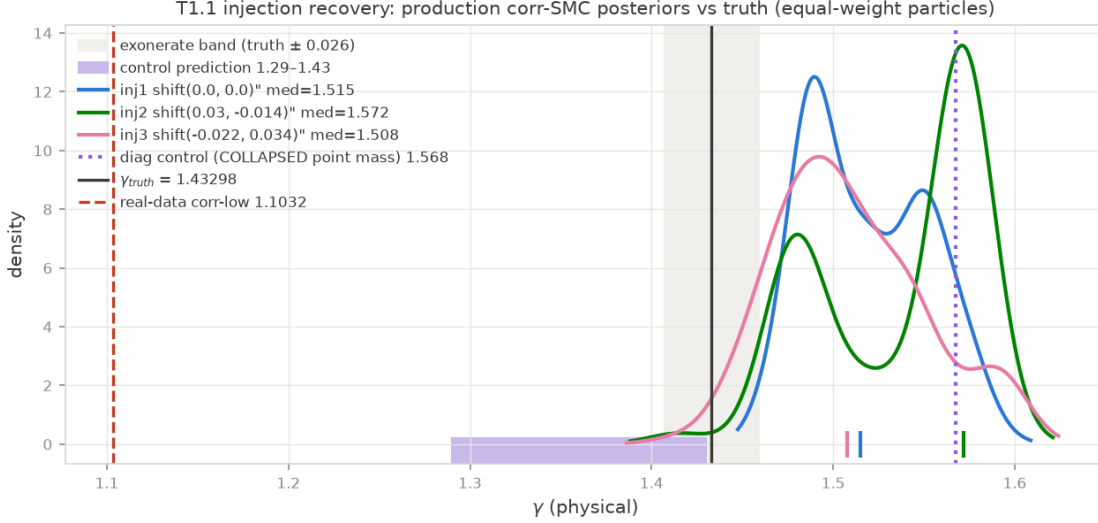


Figure 13: T1.1 injection–recovery: recovered γ posteriors for the three correlated injections and the diagonal control vs. the injected truth 1.4330 and the pre-registered zones. All arms land high (scene residue absorption); the control is both out-of-window and degenerate. Source: `data/t11_gate_eval.json`.

7.6 T1.1b: the fit-side rescue fails structurally — injections must be rebuilt data-side

The obvious rescue — mask the scene-subtraction residue on the fit side (whiten with the production kernel, then drop residue-dominated whitened pixels) — was given a pre-declared entry gate before any GPU spend: the kept-dof loss had to stay $\leq 40\%$. **It failed structurally, and T1.1b stopped at zero cost:** whiten-then-drop loses 85.8% of the whitened dof (9273 $\rightarrow$ 1320; the region drop alone loses 47.6%), and the arc band — where the slope information lives — retains 108 of 6373 whitened pixels, all outer sky.³³ The 21×21 whitening kernel support smears the excluded regions across the field: fit-side residue masking is structurally incompatible with this injection design, not merely expensive.

The redesign conclusion is data-side: residue-free injection substrates must be built before whiten-bias numbers are quotable — kernel-sampled noise-only fields, sky-set bootstrap, or a deeper multi-start scene subtraction. This was queued as the follow-on experiment and remains unfunded at program close; it is the sharpest single item on the handoff list, because it is what stands between “indicted” and “convicted” for the noise-model mechanism.

7.7 L0-G2: $\gamma = 1.103$ reproduces across stacks and machines

The final mechanism-program result is a positive control on everything above (0.53 A100-h; job 55985451, 31 min). The ported correlated likelihood (§6.4) refit the v3b low and steep basins on the scene-API substrate — an independent implementation stack, on a different machine, mirroring the production recipe at seed 2. Against the pre-registered gate ($|\gamma - 1.1032| \leq 0.017$ and $\Delta \log Z(\text{steep} - \text{low}) < 0$), the refit **passes**:³⁴

- $\gamma_{\text{low}} = 1.1005 [1.0992, 1.1065]$, i.e. $\Delta = 0.0027$ from the certified 1.1032 — well inside both the gate and σ_{seed} ;

³³Source artifact: `data/t11b_residue_mask_report.json`.

³⁴Source artifact: `data/results-perlmutter/10g2_v3b_scene_seed2.json`.

- $\log Z_{\text{low}} = -4770.97$ vs. the old-stack production -4771.08 : **0.11 nats of agreement across stacks *and* machines**;
- $\Delta \log Z(\text{steep} - \text{low}) = -29.36$, in sign and magnitude inside the old-stack seed family ($-28.9 / -32.3 / -31.2$).

The `CorrelatedImageData` port is thereby licensed at posterior level, and $\gamma = 1.103$ now stands on two independent stacks — the over-correction of §7.4 is a property of the *likelihood class*, not of either implementation. Scope notes (UNCERTIFIED—external; proposed to the upstream team as CGL2-C5): the same frozen whitener bundle feeds both stacks, so this is a stack-implementation cross-check, not an independent noise model; single seed on the new stack, with the old-stack σ_{seed} ($n = 3, \pm 46\%$) as context.

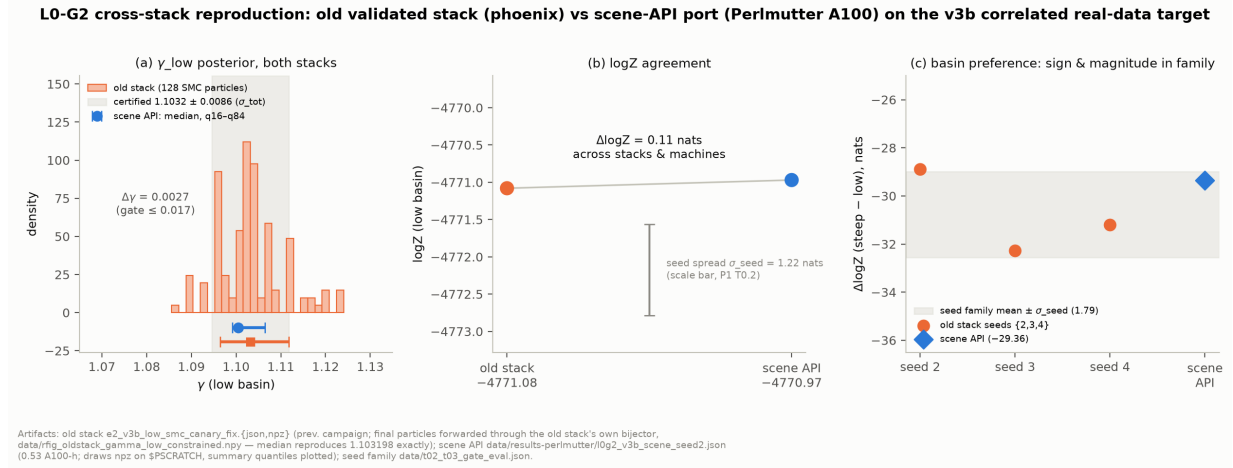


Figure 14: L0-G2 cross-stack reproduction of the correlated v3b refit. (a) γ_{low} : the previous campaign’s production SMC ensemble (128 final particles, phoenix, old validated stack; particles forwarded through that stack’s own bijector — the median reproduces the ledgered 1.103198 exactly) against the scene-API port’s summary quantiles (Perlmutter A100, seed 2): $\Delta\gamma(\text{median}) = 0.0027$, inside both the pre-registered 0.017 gate and the certified 1.1032 ± 0.0086 band. (b) $\log Z$ agreement to 0.11 nats across stacks *and* machines (-4771.08 vs. -4770.97), with the old-stack seed spread of $\log Z_{\text{low}}$ ($\sigma_{\text{seed}} = 1.22$ nats) as the honest scale bar (per-run bootstrap σ is not recorded in these summary artifacts). (c) $\Delta \log Z(\text{steep} - \text{low}) = -29.36$ on the scene API, inside the old-stack seed family ($-28.88 / -32.27 / -31.19$). The scene side is plotted from summary quantiles (the posterior-draws NPZ resides on Perlmutter scratch). Sources: `data/results-perlmutter/l0g2_v3b_scene_seed2.json`; `data/rfig_oldstack_gamma_low_constrained.npy`; `data/t02_t03_gate_eval.json`.

8 Results III — Fermat-Potential Sensitivity: a Triple-Disclaimed Teaser

Read the disclaimers first; they are the load-bearing part of this section. (1) DESI-165.4754–06.0423 is **not a time-delay lens** — no source variability, placeholder redshifts; nothing here is a Δt prediction for a real system. (2) The image positions are **synthetic**: four fixed points on a $\theta_E = 2.655''$ circle about the anchor mass center, the same points for every posterior. (3) The correlated binned-low posterior is the product **known to over-correct** ($\gamma = 1.103$, 17σ below the 1.433 anchor; §7) — so this teaser demonstrates *sensitivity* of Fermat observables to the

coadd-covariance modeling choice; it does *not* calibrate which answer is right. Additionally, the primary comparison pair differs by data product (native vs. binned), not only by noise model; the same-product secondary arm isolates the noise model.

With that stated, the number (0 GPU-h; per-sample EPL+shear Fermat potentials through a numpy port of the EPL deflection — Tessore–Metcalf angular series, validated against the vendored jax implementation to $\max |\Delta| = 1.3 \times 10^{-15}$; fractional shifts are distance/redshift-free):³⁵ swapping the noise model moves the median Fermat-potential differences $\Delta\phi$ between image pairs by

$$88\% (\sim 10.7\sigma) \text{ anchor} \rightarrow \text{corr}, \quad 61\% (\sim 17\sigma) \text{ diag} \rightarrow \text{corr, same binned product},$$

against each posterior’s own $\Delta\phi$ width of 4–8% — versus the $\sim 1\%$ scale at which such shifts become TDCOSMO-relevant. Representative opposite-pair values (arcsec²): diagonal-native -0.344 ± 0.028 ; diagonal-binned-low -0.107 ± 0.004 ; correlated-binned-low -0.040 ± 0.004 . The physical reading is simple: $\Delta\phi$ between images tracks $(\gamma - 1)$, so the correlated likelihood’s slope deflation $1.433/1.293 \rightarrow 1.103$ propagates near one-to-one into a $\sim 4\text{--}9\times$ collapse of $\Delta\phi$ (Fig. 15).

The takeaway is deliberately modest: on a real drizzled *HST* product, the noise-covariance modeling choice moves Fermat observables by tens of percent while typical time-delay error budgets carry no line item for coadd/drizzle noise covariance. The teaser’s only claim is that such a line item is worth pricing on an actual time-delay lens — with a likelihood class that survives §7.4.1’s stationarity test, which the present correlated class does not.

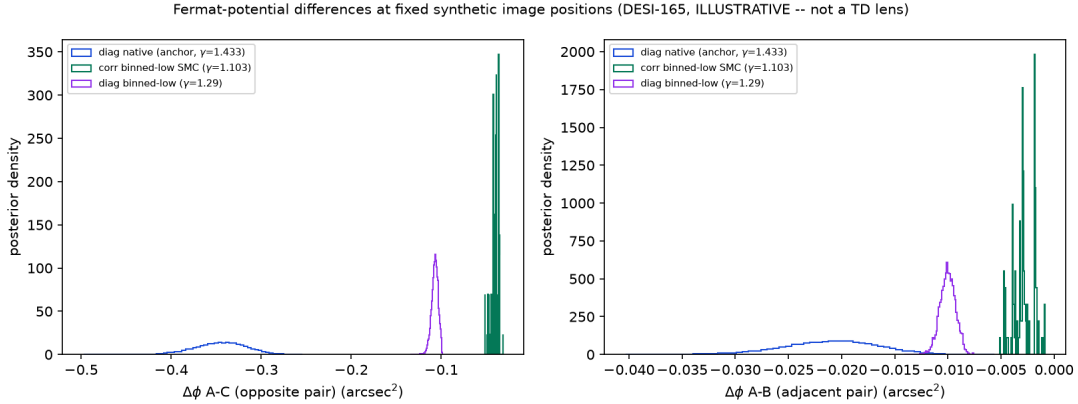


Figure 15: Fermat-potential difference posteriors for an opposite and an adjacent synthetic image pair under the three noise-model posteriors (diagonal-native anchor, diagonal-binned-low, correlated-binned-low). Illustrative only (see the triple disclaimer): the noise-model choice moves $\Delta\phi$ by 61–88% where $\sim 1\%$ is the TDCOSMO-relevant scale. Source: `data/fermat_dt_teaser.json`.

9 The Sampler Decision Matrix

The benchmark’s deliverable is not a ranking but a *decision matrix*: for each target class that a lens modeler actually faces, which inference vehicle converges, which yields trustworthy evidence, and at what cost. Four vehicles were exercised under the frozen P0 protocol: **S1**, cold-start prior-seeded MAMS-SMC (tempered SMC with Metropolis-adjusted microcanonical mutations, adaptive λ -schedule, per-stage bootstrap $\log Z$); **warm MAMS**, a MAP-warm-started MAMS-alone arm (no

³⁵Source artifact: `data/fermat_dt_teaser.json`; `research/fermat_dt_teaser.md`.

The sampler decision matrix — target class × vehicle (P2 deliverable)

	S1 prior-seeded MAMS-SMC (the campaign bet)	Warm MAMS (MAP-warm, no tempering)	MCLMC-mutation SMC (DEMOTED at B0)	Classic MAP→SVI→HMC (frozen B3 settings)
T0 analytic mix2 / funnel10 / illicond46 (B0) 0 A100-h (local GPU)	PASS 2/3 mix2 PASS (log2 err 0.073); illicond46 p-PASS, log2 err 2.40 nats = 6.6x basic; funnel10 FAIL -0.66 nats	not run (HMC-mutation SMC arm ran instead)	DEMOTED illicond46 $\Delta \log Z = +10.79 \pm 0.54$ = 20.1σ vs MAMS → never an evidence kernel	not run
hs2 easy scene 22-d unimodal x4 systems (B3) 12.4 L4-h free tier; 0 A100-h	BLOCKED-BY-BUDGET Q14 arms inside the 3 h/arm L4 fence at 1h512; the COST row is the result	not run	not run	BLOCKED-BY-REFERENCE A14 complete, Q14 accepted ($R = 36 \pm 7.87$; ESS 64-84 vs $R = 1.05 \pm 555a200$)
DSPL thin-ridge 21-d, orig vs ratio coords (B2/B2) 2.9 A100-h	MIXED $\lambda = 1$, @64 stages, sanity CLEAN; failure NOT-DECIDABLE-A5-REGISTERED; B2 PASS-UNDERPOWERED ($p = 0.062$)	not run	not run	not run
T2 46-d cond=10^{-14} v2d marg46 / diagonal scene (B4 → arbitration) B4 3.5 h; arb 3.8 h = 0.9 h	BLOCKED-BY-COST healthy sampler, 3x wall-capped ($\lambda = 0.587$ @ 3 h A100) → vehicle budget-infeasible	not run (classic arm was warm-started instead)	not run	NO-VERDICT unconverged; $R = 44.3$, chains rallied $y = 1.99$ at prior edge → reproduces the known T2 single-stage pathology
T3 74-d bimodal foundry_v3b/74, diagonal (B5) 1.87 A100-h (+1.66 infra-fail)	CONVERGED Q1 PASS both basins @$\lambda = 1$, retention 1.000; Q2 FAIL-as-written 11x → indicates frozen reference coverage	not run	AMBIGUOUS (G3) x58 minor-mode inflation (22-148) within-basin y intact (1.0935 vs 1.0919)	not run
Carousel 33-d multi-plane REAL MUSE cutouts (B1r, team-provided data) 11.3 + 4.5 A100-h (+11.0 wave-1)	BLOCKED-BY-COST $\lambda = 0.151$, @36 stages, 0 posterior samples — healthy sampler, killed by cost	BLOCKED-BY-COST wall fence at burn 11:50; 0 draws, ESS=0 (41.5% of B⁰ grade)	not run	not run (candidate vehicle; est 17-20 h)
v3b correlated REAL 46-d whitened likelihood (B0-G2) 0.53 A100-h	CONVERGED PASS: $y = 0.0027$, log2 Δ 0.11 nats cross-check, best evidence rev of the campaign	not run	not run	not run

■ CONVERGED (evidence delivered)
 ■ PARTIAL / MIXED / AMBIGUOUS
 ■ BLOCKED (budget / cost / reference)
 ■ NO-VERDICT / FAIL / DEMOTED
 ■ not run

S1 is the only vehicle that produced log Z anywhere; every S1 non-completion is a wall-cap on a healthy sampler (cost, never health). Classic at frozen single-stage settings produced no accepted posterior in the campaign. Sources: research/synthesis_tables.md §1; research/gate_ledger_final.md; per-cell artifacts in data.

Figure 16: The sampler decision matrix as a capability grid (the graphical form of Table 9): 7 target classes × 4 vehicles. Cell status is the gate-ledger final disposition (green = converged with evidence delivered; yellow = partial/mixed/ambiguous; salmon = blocked by budget, cost, or reference; red = no-verdict/fail/demoted; gray = not run), with the discriminating measurement and the realized cost in each run cell. The honest reading is baked in: S1 prior-seeded MAMS-SMC is the only vehicle that produced log Z anywhere; every S1 non-completion is a wall-cap on a healthy sampler; the classic recipe at frozen settings produced no accepted posterior; MCLMC was demoted at B0 ($\Delta \log Z = +10.79 \pm 0.54 = 20.1\sigma$, artifact value) and stayed diagnostic-only. Sources: `research/synthesis_tables.md §1`; `research/gate_ledger_final.md`; per-cell artifacts as cited in Table 9 and the subsections.

tempering); **MCLMC-mutation SMC**, formally demoted to a diagnostic at the B0 correctness gate (below) and never used as an evidence kernel; and the **classic** MAP→SVI→HMC baseline recipe at frozen single-stage settings. Table 9 is the full matrix; the subsections that follow give each row’s evidence and the two cross-cutting findings. Every cell is pre-registered (design checkpoint with derived thresholds logged before submission) and closed in the final gate ledger (§12). The per-fit convergence diagnostics behind every cell — one row for each fit the campaign ever ran, including the failed, rejected, and budget-blocked arms — are tabulated in Appendix A.

9.1 Reading the matrix: measured capability boundaries

Three boundaries are measured, not conjectured. First, **prior-seeded MAMS-SMC works** — **converges and yields evidence** — on analytic targets, the DSPL ridge, the 74-d bimodal target (both basins, retention 1.000), and the correlated real-data v3b refit; it is the only vehicle in the campaign that produced log Z at all. Second, **when it fails, it fails by cost, never by health**: every non-completion (B1 mocks, B1r real carousel, B3 at $N=512$, the three arbitration wall-caps) is a wall-clock cap on a sampler whose per-stage diagnostics were clean at the kill point — λ -progress per A100-hour is the binding constraint on scene-substrate real targets. Third, **the classic single-stage recipe produced zero accepted posteriors in this campaign** (B3 0/4, the arbitration arm’s NO-VERDICT, and 30/32 unhealthy fits in the X2 SBC arm at

Table 9: The sampler decision matrix. Rows are target classes; columns are vehicles. “—” = not run (pre-registered scope, budget close, or prerequisite unmet; dispositions in `research/gate.ledger.final.md`). All A100 costs are actual ledgered hours. The carousel row uses real MUSE cutouts provided by the team, used with permission; comparison against the team’s own runs is deferred pending sign-off. The DSPL and easy-scene target builders come from the upstream development branches (used with permission). Scene-substrate cells are proposed (UNCERTIFIED — external): reproduced/measured on the vendored scene API, with certification belonging to the receiving team.

target class	S1 prior-seeded MAMS-SMC	warm MAMS	MCLMC (demoted)	(de- classic MAP→SVI→HMC	evidence yield
T0 analytic (2-d bimodal; funnel-10; illcond-46) — B0, 0 A100-h	mix2 PASS (log Z err 0.073); — funnel10 FAIL −0.66 nat (class limitation, reproduced in the old validated stack); illcond46 PASS worst- z 1.83 but log Z err 2.40 nat = $6.6\sigma_{\text{boot}}$	— (an HMC-mutation SMC arm ran as comparison)	DEMOTED: $\Delta \log Z$ vs MAMS $+10.79 \pm 0.54 = 20.1\sigma$ on illcond46	n/a	analytic refs; σ_{boot} provably understates log Z error (finding A)
easy scene 22-d unimodal $\times 4$ — B3, free tier, 0 A100-h	BLOCKED-BY-BUDGET 0/4 inside the 5h/arm L4 fence at $N=512$; the readable result is the cost row	—	—	4/4 complete but 0/4 pass the frozen acceptance rule ($\hat{R}_{\text{worst}} 4.36\text{--}7.87$) $\Rightarrow$ BLOCKED-BY-REFERENCE	none (no accepted posterior on either arm)
DSPL thin ridge 21-d — B2/B2', 2.9 A100-h	$\lambda=1$ @64 stages, sanity clean; — $\hat{m}$ 0/512 vs. imported band $\Rightarrow$ falsifier control shows the band is realization-specific; B2' PASS-UNDERPOWERED	—	—	—	$\Delta \log Z(\text{orig-ratio}) = +3.06 \pm 0.35_{\text{boot}}$ where true $\Delta \approx 0$ (finding A, measured)
T2 46-d cond- 10^{14} — B4 + arbitration, 4.4 A100-h	healthy but wall-capped $\times 3$ (A100 $\lambda=0.587$ @3.5h; L4 $\lambda=0.418$ at a 12.3-h resume leg) $\Rightarrow$ budget-infeasible ; B4 itself not evaluable within budget	(the classic — arm was warm-started)	—	NO-VERDICT: $\hat{R}_{\text{worst}} 44.3$, all chains railed $\gamma \approx 1.99$ — the known T2 single-stage pathology, reproduced cross-stack	none; premise carried by likelihood parity + the L0-G2 row
T3 74-d bimodal — B5, 1.9 A100-h (+1.7 infra)	G1 PASS both basins — ($\lambda=1$, retention 1.000); G2 FAIL-as-written 11.0 σ — decomposition indicts our own frozen reference’s within-basin coverage (finding B)	—	G3 AMBIGUOUS: minor-mode evidence inflated $\times 56$; within-basin γ untouched	—	occupancy robust across all estimators; absolute log Z NOT robust at the ± 5 -nat gate scale
carousel 33-d multi-plane, real — B1r, 15.85 A100-h	$\lambda=0.1506$ @36 stages after 11.33h, 0 posterior samples (sampler healthy; killed by cost)	budget-matched S6br: wall fence in burn round 11/30, 0 draws , ESS=0 at 41.5% of the grad budget	—	—	none — NEITHER CONVERGES ; the cost row is the deliverable
v3b correlated , real, 46-d — L0-G2, 0.53 A100-h	PASS: $\gamma \Delta 0.0027$, log $Z \Delta 0.11$ nat cross-stack; $\Delta \log Z(\text{steep-low})$ in the seed family	—	—	—	best evidence row of the campaign; correlated port licensed

reduced budget) — consistent with the prior campaign’s finding that this posterior family needs the two-stage re-preconditioned recipe; its unique value here was diagnostic (§9.5).

9.2 Carousel (B1r): neither vehicle converges at campaign budget

The carousel cell is the campaign’s designated hard real-data row: a 33-dim multi-plane model on real MUSE cutouts (data provided by the team, used with permission; any comparison against the team’s own runs is deferred pending their sign-off). The cold-start arm S1r ($N=128$, three chained checkpointed legs) spent 11.33 A100-h to reach $\lambda = 0.1506$ at 36 stages — *zero* posterior samples, with the sampler demonstrably healthy at the fence (92–104 unique particles of 128, acceptance 0.871–0.934), so $\lambda = 1$ is infeasible under any campaign-scale budget.³⁶ The pre-registered close-out framing was written *before* the baseline ran, including the branch that would have scored against our own bet (“any nonzero baseline ESS $\Rightarrow$ decisive cold-start LOSS”). That branch did not fire: the warm MAMS-alone baseline S6br, budget-matched at S1r’s realized 3,078,912 gradient evaluations, hit its 4.5-h wall fence in burn round 11 of 30 — **sampling never started** (0 draws, ESS = 0, $\hat{R}$ not computable) at 1,277,632 grads = 41.5% of the budget, with burn itself healthy (acceptance 0.873–0.902).³⁷ Per-gradient throughput agrees across arms to $\sim 5\%$ ($\sim 284\text{k}$ vs. 272k grads/h), so the truncation is a *target-cost* fact, not arm inefficiency. The row’s honest verdict: **carousel-class real targets are out of reach for both vehicles at these budgets** (an until-converged warm arm is estimated at 17–20 h from the wave-1 anchors, untested). The descoped gates (2-seed self-consistency, log Z repeatability) remain descoped and are stated as such in the amendment of record.³⁸

9.3 T3 bimodal (B5): both basins pass; the gate that fails indicts our own frozen reference

On the 74-d bimodal target, S1 delivers the multimodality certificate **G1** in full: the low basin reaches $\lambda = 1$ in 31 stages (retention 1.000, $\gamma_{\text{eqw}} = 1.0919$, $\log Z = 38376.33 \pm 0.33_{\text{boot}}$) and the steep basin in 22 stages ($\gamma_{\text{eqw}} = 2.5552$, $\log Z = 38518.23 \pm 0.31_{\text{boot}}$).³⁹

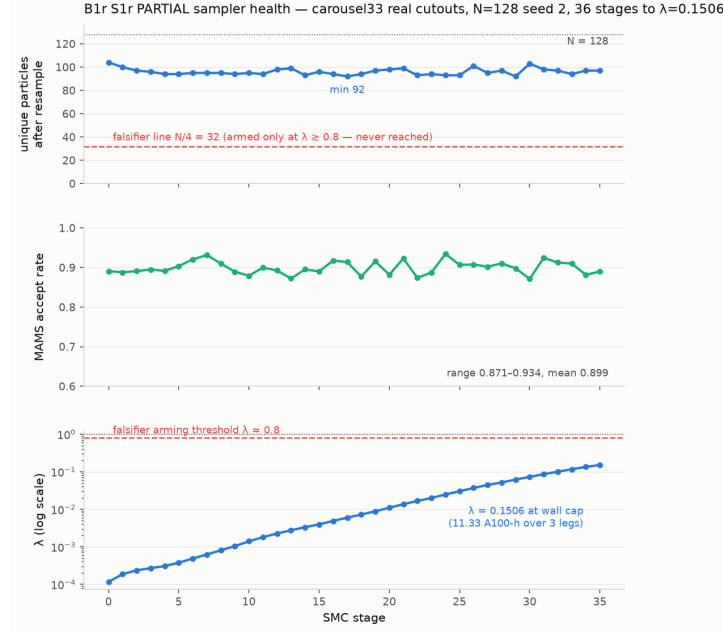
G2 — basin $\Delta \log Z$ against the frozen per-basin evidence reference from *our own previous campaign* (P2c) — **fails as written**, by a lot: $\Delta(\text{steep} - \text{low}) = +141.90 \pm 0.45$ vs. the reference $+162.2 \pm 1.8$, a -20.30 -nat miss against a 5.54-nat band ($11.0\times$ the gate σ -scale). We frame this carefully, because the honest decomposition cuts *toward the reference*: the per-basin offsets are $+25.16$ (low) and $+4.86$ (steep) — *both* basins carry *more* evidence than the reference, in the same direction — and both $\lambda = 1$ ensembles are γ -*disjoint* from the fixed-step HMC chains the reference was seeded and mutated from (low basin: ours [1.071, 1.126] vs. chains [1.265, 1.381]), with two *independent* mutation kernels agreeing on the location (MAMS 1.0919, MCLMC 1.0935). The adaptive anneals reach higher-log-probability shelves that the reference’s short fixed-step chains never visited. Comparability was verified line by line before scoring (same target constructor, identical evidence construction, the importance weights cancel; the reference’s own on-disk smoke reproduces its $\Delta = 162.7$ under its kernel), so the discrepancy is real — and with $n=1$ seed on the measuring side, final attribution between reference and vehicle is explicitly undecidable within the closed budget. What the fail *does* establish is cross-cutting finding B (§9.7): frozen MCMC-derived

³⁶Source artifact: `data/results-perlmutter/b1r128-carousel133.s1.seed2.PARTIAL.json`.

³⁷Source artifact: `data/b1r_decision.matrix.json`.

³⁸Source artifact: `research/checkpoint_b1r_close.md`.

³⁹Source artifact: `data/b5_gate_eval.json`.



S6br (56252401): warm MAMS-alone, Track-A budget-matched at $B^* = 3,078,912$ grads (D9) — wall fence fired IN BURN, zero sampling rounds

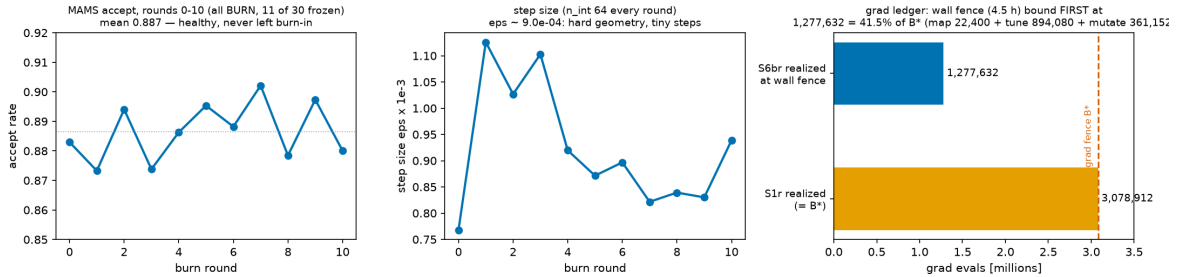


Figure 17: The carousel cell B1r (real MUSE cutouts provided by the team, used with permission; comparison with the team’s own runs deferred pending sign-off); both panels plotted before gate text per house rule. Top: the S1r cold-start MC-SMC arm over three chained checkpointed legs to $\lambda = 0.1506$ at 36 stages (11.33 A100-h) — unique-particle counts 92–104 of 128 and acceptance 0.871–0.934 at the fence: a healthy sampler killed by cost. Bottom: the budget-matched warm-MAMS arm S6br (job 56252401) hits its 4.5-h wall fence in burn round 11 of 30 — sampling never started (ESS = 0 at 1,277,632 grads = 41.5% of the matched 3,078,912-gradient budget), with burn itself healthy. Sources: `data/results-perlmutter/b1r128_carousel33_s1_seed2.PARTIAL.json`; `data/b1r_decision_matrix.json`.

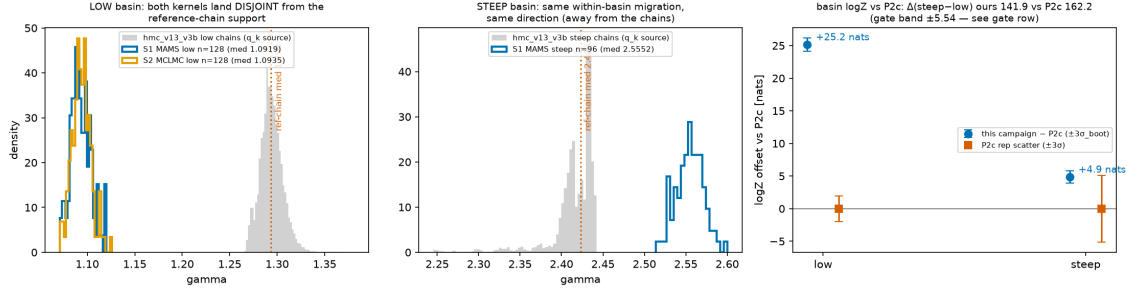


Figure 18: B5 both-basin $\lambda = 1$ posterior overlay on the 74-d bimodal target **foundry_v3b74**, drawn *before* any gate math per house rule (jobs 56006049 / 56251555 / 56006052). The low basin’s two independent mutation kernels agree on location (MAMS $\gamma_{\text{eqw}} = 1.0919$, MCLMC 1.0935); the steep basin sits at 2.5552. Source: `data/b5_gate_eval.json`.

evidence references can carry within-basin coverage error far exceeding the gate scale. This is a statement about our own prior artifact, not about any external one.

G3, the pre-registered MCLMC-laundering probe, lands **ambiguous at the frozen thresholds** because its two frozen clauses conflict on the measured value: the diagnostic MCLMC arm inflates minor-mode evidence by $+4.03 \pm 0.49$ nats = $\times 56$ [$\text{CI}_{95, \text{boot}}$ 22–146] — $8.3\sigma_{\text{boot}}$ from zero and $6.0\sigma_{\text{boot}}$ above the pre-registered $1.5\text{--}3\times$ ceiling, so SMC resampling does *not* launder the unadjusted-MCLMC bias at point estimate — yet the shift sits inside the imported 5.56-nat σ_{seed} repeatability band, so the laundering falsifier technically fires. Within-basin location is untouched (γ 1.0935 vs. 1.0919): the bias lands in the evidence, not the posterior. Either way the physical conclusion is robust across every estimator: the minor mode is negligible ($w_{\text{low}} \leq 1.3 \times 10^{-60}$ in all constructions), while *absolute* evidence calibration at the ± 5 -nat gate scale is not.

9.4 DSPL (B2/B2’): the realization-calibration lesson

The DSPL thin-ridge cell was scored twice, and the sequence is the lesson. As written, B2 asked S1 (in the original, deliberately pathological coordinates) to recover a pre-registered minor-arm mass band 0.103 ± 0.045 , imported at design time. The measured $\hat{m} = 0/512$ fails the band — but the pre-registered falsifier control, an *exact* reparameterization of the same posterior on the same data (which cannot suffer coordinate mode-death), measures this realization’s true arm mass at $5/512 = 0.0098 \pm 0.0043$, itself far outside the band.⁴⁰ The band does not transfer across data realizations; the gate was **not decidable as registered**, and the pre-registered branch mandated re-registration rather than silent re-scoring. The re-registered B2’ (rule and attainable-significance clause declared before scoring) compares the arms directly: two-sided Fisher $p = 0.0619 \geq \alpha = 0.01$ — **PASS-UNDERPOWERED (consistent)**, with the honest caveat that with only five minor-arm particles in total no outcome could have failed at α ; minor-arm under-coverage up to $\sim \times 3$ is not excluded (the one-sided signal, $p = 0.031$, is $\sim 2\sigma$ and suggestive at most). The dominant arm is statistically identical across coordinates ($\Omega_{m,0}$ median 0.4701 vs. 0.4702). The headline stands at “no detected coordinate pathology” strength — and the cell’s most valuable by-product is an evidence-error datum: $\Delta \log Z(\text{orig} - \text{ratio}) = +3.06 \pm 0.35_{\text{boot}}$ nats on identical data where the exact reparameterization forces the true $\Delta \approx 0$ (§9.7).

⁴⁰Source artifact: `data/b2_gate_eval.json`.

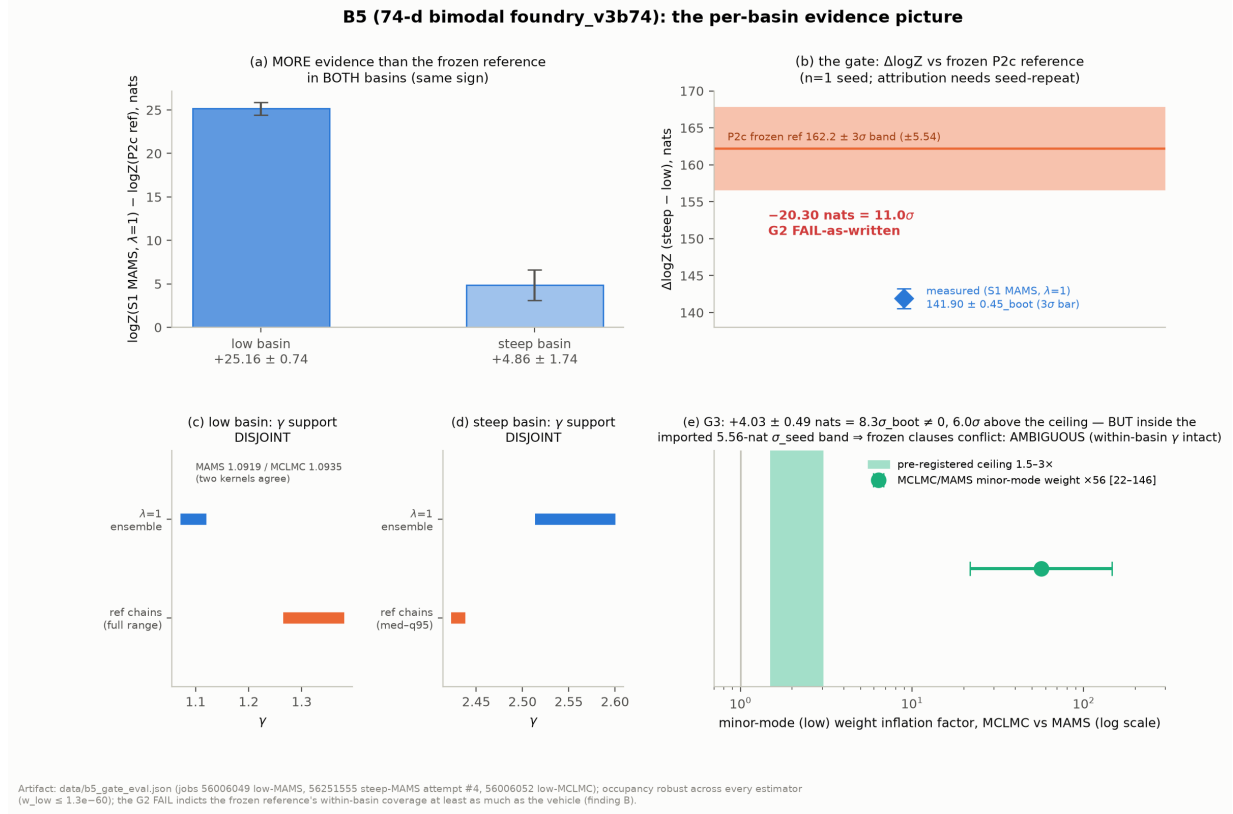


Figure 19: B5 per-basin evidence picture on the 74-d bimodal target. (a) The S1 $\lambda = 1$ ensembles find *more* evidence than the frozen P2c reference in *both* basins ($+25.16 \pm 0.74$ low, $+4.86 \pm 1.74$ steep — same sign). (b) The G2 gate as written therefore fails at 11.0σ (measured $\Delta = 141.90 \pm 0.45_{\text{boot}}$ vs. reference 162.2 ± 1.8 , band ± 5.54 nats) — but the decomposition in (a) plus (c,d) indicates the frozen reference's within-basin coverage at least as much as the vehicle: both $\lambda = 1$ ensembles are γ -disjoint from the reference-chain support that the reference was seeded *and* mutated from, while two independent kernels (MAMS 1.0919, MCLMC 1.0935) agree on the low-basin location (cross-cutting finding B; $n=1$ seed, attribution undecidable within the closed budget). (e) The MCLMC diagnostic arm inflates minor-mode evidence $\times 56$ [$CI_{95, \text{boot}}$ 22–146] = $+4.03 \pm 0.49$ nats — $6.0\sigma_{\text{boot}}$ above the pre-registered 1.5–3 $\times$ ceiling yet inside the imported $5.56\text{-nat } \sigma_{\text{seed}}$ repeatability band, so the two frozen G3 clauses conflict: AMBIGUOUS, and the B0 demotion stands. Minor-mode occupancy is robust across every estimator ($w_{\text{low}} \leq 1.3 \times 10^{-60}$). Source: data/b5_gate_eval.json (jobs 56006049, 56251555 attempt #4 chunk-32, 56006052; frozen P2c reference constants therein).

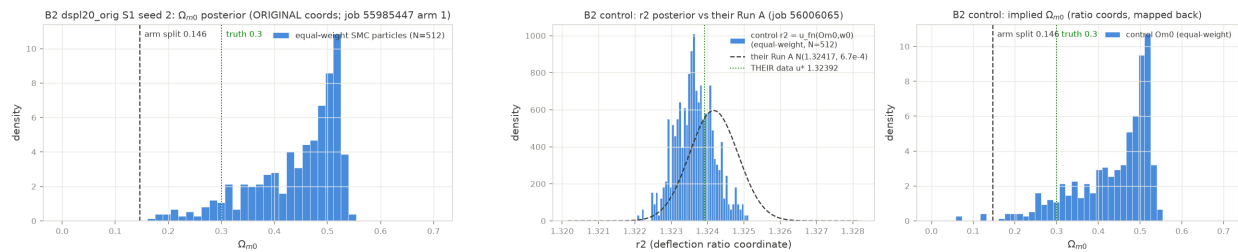


Figure 20: DSPL thin-ridge cell (B2/B2'). Left: the dominant arm is statistically identical across coordinates ($\Omega_{m,0}$ median 0.4701 vs. 0.4702). Right: the exact-reparameterization ratio-coordinates falsifier control that proved the imported minor-arm band realization-specific (this realization's true arm mass $5/512 = 0.0098 \pm 0.0043$ vs. the imported band 0.103 ± 0.045), forcing the pre-registered re-registration to B2' (PASS-UNDERPOWERED). Source: data/b2_gate_eval.json.

9.5 The classic arm: a NO-VERDICT that reproduces the T2 pathology cross-stack

The anchor-arbitration classic arm — warm-started MAP→SVI→HMC at the frozen B3 reference settings on the parity-certified 46-d cond- 10^{14} v2d target — is a **NO-VERDICT (unconverged)** under the pre-registered worst-parameter rule: $\hat{R}_{\text{worst}} = 44.3$ (11.7 at the escalated stage, vs. the 1.05 gate), $\text{ESS}_{\text{min}} = 63$ at the escalated stage (53.6 at the initial 50-chain stage — both quoted, per the flagged-record rule), with every chain railed at the prior edge, $\gamma \approx 1.987$ [1.983, 1.991] — so γ is not quotable.⁴¹ The scientific content is what this reproduces: the same single-stage-HMC pathology this posterior exhibits on the old validated stack (single-stage $\hat{R} = 3.1$; only the two-stage re-preconditioned recipe converges), now demonstrated on the scene substrate at 0.9 A100-h. The difficulty is target-intrinsic, not stack-specific — exactly the premise-check the arbitration needed once every SMC vehicle had wall-capped, and the cheapest useful thing a non-converging run can prove.

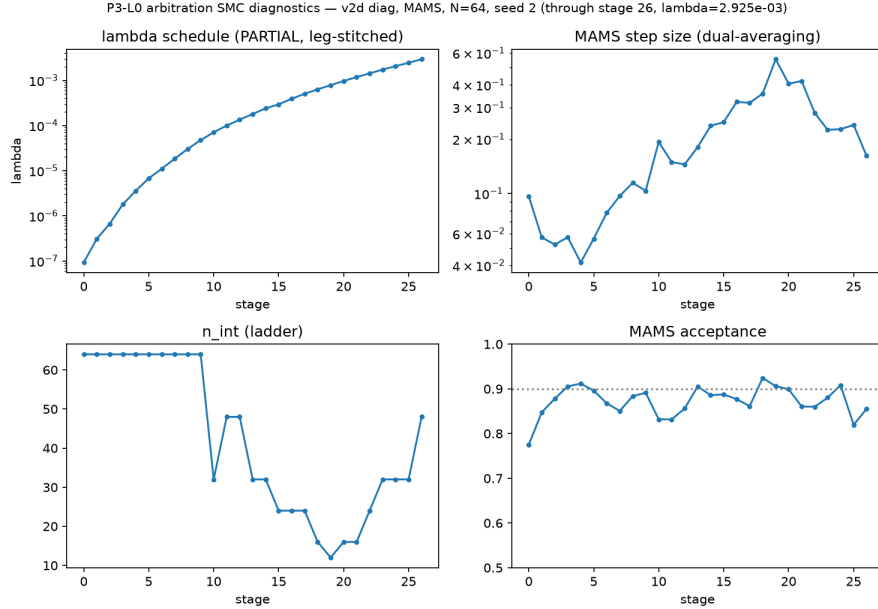


Figure 21: The arbitration MC-SMC arm on the 46-d cond- 10^{14} v2d target (job 56168446): healthy per-stage diagnostics to $\lambda = 0.587$ at the 3.5-h A100 fence — budget-infeasible, not unhealthy. Together with the L4 resume leg ($\lambda = 0.4183$ at a 12.26-h leg, on-disk record) and the classic arm’s NO-VERDICT, this is the vehicle-exhaustion evidence behind the anchor arbitration’s PARTIAL-BY-VEHICLE-EXHAUSTION close. Disposition: `research/gate.ledger_final.md`.

9.6 Easy scene (B3): double-blocked, and honest about it

The nominally easiest row — four 22-d unimodal systems from the hundred-systems class — returned no evaluable accuracy cell, for two independent reasons that we report as such. The SMC arms are **blocked by budget**: no arm finished inside the pre-registered 5-h/arm free-tier L4 fence at $N=512$ (12.38 L4 GPU-h consumed; the readable result is the cost row: scene-substrate MC-SMC at $N=512$ costs > 5 h/target on an L4).⁴² The classic reference arms are **blocked by reference**:

⁴¹Source artifact: `data/results-perlmutter/10_arbitration_classic.json`.

⁴²Source artifact: `data/b3_cells.json`.

all four posteriors completed but all four fail the frozen reference-acceptance rule ($\hat{R}_{\text{worst}}$ 4.36–7.87 vs. < 1.05 ; ESS_{min} 63.7–84 vs. ≥ 200), so “4/4 references complete” must not be read as “references usable” — each per-cell artifact records `reference_accepted: false`. The double block is itself a datum: it is additional, independent evidence for the single-stage-recipe convergence finding above, on targets designed to be easy.

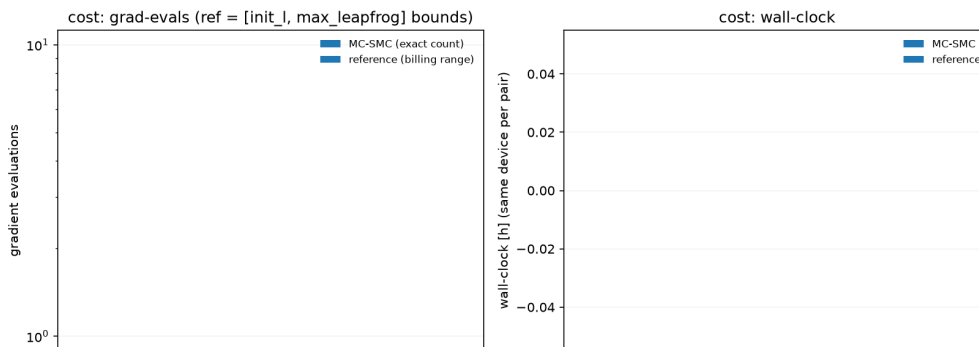


Figure 22: B3’s readable result is its cost row: scene-substrate MC-SMC at $N=512$ costs > 5 h per 22-d target on the free-tier L4 (12.38 L4 GPU-h consumed, 0/4 arms complete inside the pre-registered 5-h/arm fence). Source: `data/b3_cells.json`.

9.7 Two cross-cutting findings

(A) Bootstrap error bars understate evidence error. Measured twice, in different ways. At B0, MAMS-SMC on the analytic ill-conditioned target returns $\log Z -145.80$ vs. the analytic -148.20 : a 2.40-nat error at $\sigma_{\text{boot}} = 0.37$, i.e. $6.6\sigma_{\text{boot}}$ (the HMC-mutation arm errs by 21.6 nats at $\sigma_{\text{boot}} = 0.41$).⁴³ On DSPL, on *identical data* where an exact reparameterization forces the true difference to zero, $\Delta \log Z(\text{orig} - \text{ratio}) = +3.06 \pm 0.35_{\text{boot}}$. The practical rule the campaign adopted, and that we recommend: never gate a $\Delta \log Z$ claim on σ_{boot} alone; the seed-repeat scale ($\sigma_{\text{seed}} = 1.79$ nats here, $n=3$, $\pm 46\%$ sampling error on that estimate) is the load-bearing error bar for every evidence gate.

(B) Frozen-reference coverage risk. The B5-G2 fail decomposes into both-basin positive evidence offsets with both ensembles γ -disjoint from the reference chains and two independent kernels agreeing on the new location (§9.3): frozen references built from short fixed-step MCMC chains can carry within-basin coverage error far exceeding the σ -scale of any gate built on them. The X2 frozen-set validity question (§10.1) is the same genus on the SBC side. The lesson we apply to ourselves first: an evidence reference needs a coverage certificate before it is frozen into a gate.

Finally, the demotion that held all campaign: **unadjusted-MCLMC mutations are not an evidence kernel**. The B0 bias screen measured $\Delta \log Z = +10.79 \pm 0.54$ vs. MAMS on the ill-conditioned target — 20.1σ by the artifact of record (the ledger prose says “ 30σ ”; the flagged, binding artifact value is 20.1)⁴⁴ — and B5-G3’s $\times 56$ minor-mode inflation confirms the channel on a real-lens-derived target. Within-basin locations stay correct in both measurements: MCLMC remains useful precisely as the cheap *diagnostic* it was demoted to.

⁴³Source artifact: `data/smc.b0_report.json`.

⁴⁴Source artifact: `data/smc.b0_report.json`.

10 Calibration Gifts and the Handoff Package

10.1 X2: the first formal SBC of the pipeline class

The X2 cross-pollination arm ran the first formal simulation-based calibration (SBC; rank-statistic) test of this pipeline class: $N=32$ prior-matched mock systems through the certified port of the classic MAP→SVI→HMC pipeline at reduced free-tier budgets (~ 29 A16-h; $N=32$ of the assigned ≥ 64 — an under-delivery selected by the pre-stated budget rule and reported as such). Every verdict is *proposed* (*UNCERTIFIED* — *external*): the ranks are the deliverable; interpretation and certification belong to the team.⁴⁵

Three results, in decreasing order of confidence. **(i) A validity question about the frozen mock set**, posed as a question and not a finding of fault: the pre-existing frozen SBC set carries, per its own provenance in the upstream development branches, a generation prior narrower than the current fitting prior — under which SBC fails by construction (truth draws must follow the fitting prior). Such a mismatch is a legitimate robustness-testing design if intentional; the question for the team is whether a matched-prior set should exist. Our arm therefore ran on prior-matched *regenerated* mocks. (Detail is internal-memo material, publication-gated on the team’s sign-off.) **(ii) Severe lens-light rank miscalibration**: the lens-light photometric block shows one-sided, sign-coherent rank pile-ups — I_e rank-location $z = -5.27$ ($p = 7.6 \times 10^{-11}$; 18/32 systems in the bottom bin), $R_{\text{sersic}} z = +5.76$ ($p = 1.5 \times 10^{-10}$), $n_{\text{sersic}} z = +4.75$ — which is not an under-mixing shape; three candidate channels (genuine pipeline miscalibration, reduced MAP/SVI budget, plug-in error map) are not separable from this run and are said so in the claims register. The mass block shows no location bias (worst $|z| = 2.25$) but 6 of 8 parameters fail uniformity via U-shapes, the classic under-mixing signature, with only 2 of 32 fits passing the health rule at this budget. **(iii) The glass-house disclosure**, included in the same harness by pre-registered rule: our own previous campaign’s E1c SBC *failed* the same γ -rank gate ($p = 6.5 \times 10^{-5}$, 68% coverage 0.34) and *flipped to PASS* ($p = 0.53$) when unhealthy fits were rerun deep and re-read healthy-only — sampler pathology can fake rank failures, and that amendment path is exactly what $n_{\text{healthy}} = 2$ blocks here. The turnkey harness ships in the handoff package for an $N \geq 64$ rerun at reference settings.⁴⁶

10.2 The handoff package

The handoff directory (`papers/handoff/`) contains the engagement memo (with its close-of-campaign addendum) and `CLAIMS.md`: fourteen claims in the receiving team’s own claims-register format, every verdict *proposed* (*UNCERTIFIED* — *external*), with the sign-off-gated claims (those touching data or targets from the upstream development branches) explicitly marked, and a mandatory doubt report attached to each. That file — not this report — is the interface of record for the team: it names the artifact behind every number and reserves “validated” for the old certified stack throughout.

The gap list `research/handoff_gap_list.md` inventories the package honestly against what the plan promised: delivered in substance are the claims register, the certified correlated-noise likelihood port (`cgl2/correlated.py`, `cgl2/marg.py`, with its exact-limit validation record), the tempered-SMC implementation with checkpoint/resume (`cgl2/samplers/`, 58-test suite), and the turnkey SBC harness (`30_sbc_gift.py`). Promised but *not* packaged at close: the SMC-stage adapter specification, the correlated-likelihood handoff document and upstream-shaped patch files, the DSPL note, and the Hessian-stage restore (never started). These remain owed offers, listed as

⁴⁵Source artifact: `data/x2_sbc.json`.

⁴⁶Source artifact: `research/x2_sbc_gift.md`.

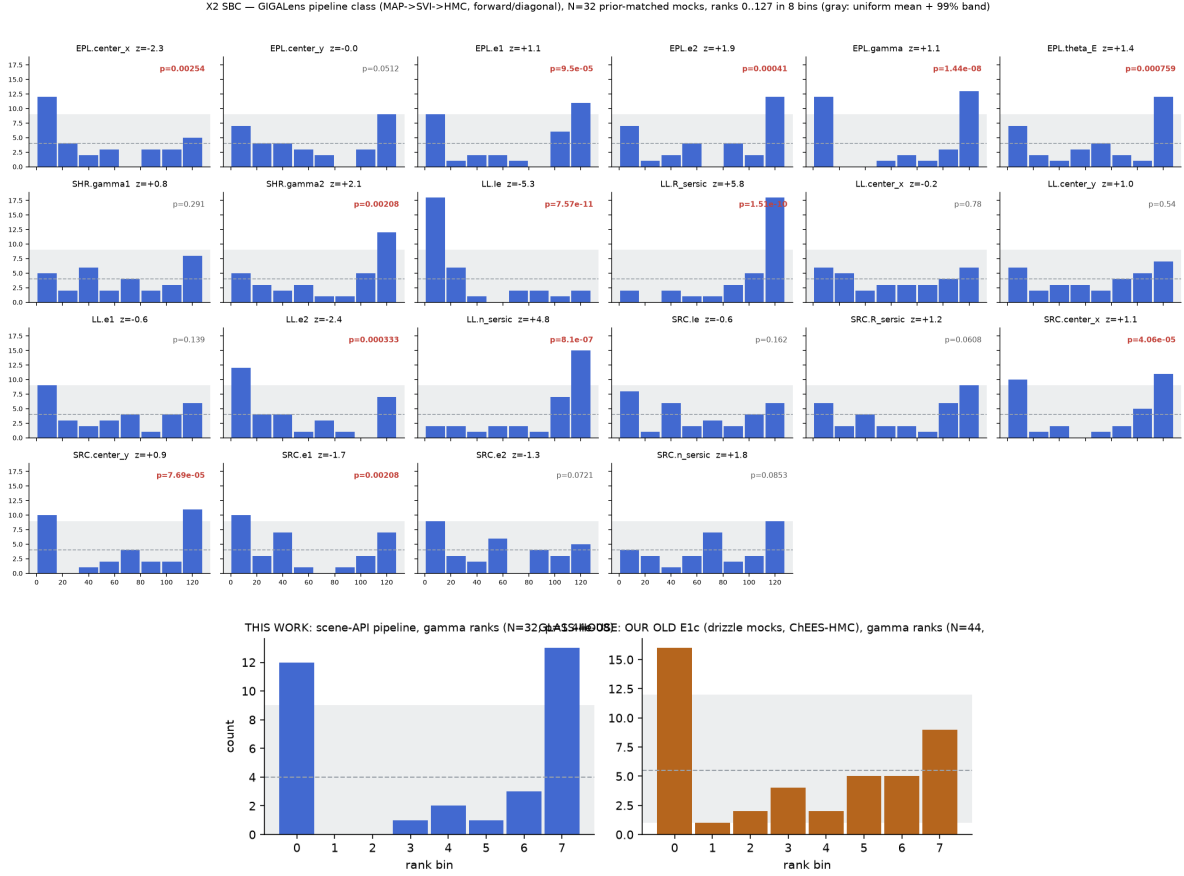


Figure 23: X2: the first formal SBC of the pipeline class ($N=32$ prior-matched mocks, reduced free-tier budget; all verdicts proposed, UNCERTIFIED — external; the pipeline runs on the upstream development branches, whose unpublished results are not characterized here). Top: rank histograms — the lens-light photometric block shows one-sided, sign-coherent pile-ups ($|z|$ up to 5.76), which is not an under-mixing shape; the mass block shows under-mixing U-shapes without location bias (worst $|z| = 2.25$). Bottom: the glass-house disclosure — our own previous campaign’s E1c γ -rank gate failed ($p = 6.5 \times 10^{-5}$) and flipped to PASS ($p = 0.53$) when unhealthy fits were rerun deep and re-read healthy-only: sampler pathology can fake rank failures, and $n_{\text{healthy}} = 2$ blocks that amendment path here. Source: `data/x2_sbc.json`.

such in the memo addendum, not silently dropped.

11 Discussion

11.1 What the campaign settles

Five things are now measured that were conjectures at the start. (1) **The bridge holds:** the scene-substrate forward model is certified against the validated old stack at machine precision for our config class, and the correlated-noise likelihood ported onto it reproduces the certified real-lens result on an independent stack and machine — γ to $\Delta = 0.0027$ and $\log Z$ to 0.11 nats (L0-G2). The $\gamma = 1.103$ result now stands on two stacks. (2) **The mechanism narrows:** stationarity of the noise-model class behind that number is rejected on the real field ($p = 0.010$, replicated spatial pattern), the companion and spectrum-flooring suspects are eliminated, and the profile-curvature fork died at its zero-cost entry gate — noise-model *class* misspecification is the standing working hypothesis for the over-correction, with source/PSF representation the surviving alternative. (3) **prior-seeded MAMS-SMC is the campaign’s only evidence-producing vehicle**, and its failure mode is purely economic: cost walls on healthy samplers, never divergence or mode loss. (4) **Unadjusted MCLMC is not an evidence kernel** (20.1 σ screen; $\times 56$ minor-mode inflation; locations intact) — but is a cheap, useful diagnostic. (5) **The classic single-stage recipe does not converge on this posterior family**, now shown on both stacks and three independent cells (B3, arbitration, X2 health rates); the two-stage re-preconditioned recipe remains the only known converging direct-sampler path.

A post-campaign epilogue (E1, 2026-07-22; PI-requested MCLMC fits of both diagonal targets, §1) sharpened two of these after close. On (5), the direct-sampler gap is narrower than the campaign body suggests: the team’s own vendored MCLMC, warm-started and run with its windowed mass-matrix adaptation and regularization option, delivered the first fully convergence-certified posterior on the v2d native diagonal target ($\hat{R}_{\text{worst}} = 1.0031$, $\text{ESS}_{\text{min}} = 30,334$; $\gamma = 1.4683 [1.4343, 1.5048]$, $0.72\sigma_{\text{comb}}$ from the anchor) at 1.49 free-tier L4-hours — so the convergence failures above are specific to the classic single-stage recipe, not to direct samplers per se. On (2), the mechanism gains its sampling-side face: on the binned product the same vehicle under the same *diagonal* likelihood migrated every one of 64 chains, in all four seed groups, out of the physical basin it was warm-started in and down to $\gamma \approx 1.08\text{--}1.15$ (kept-draw pooled median 1.1047, 0.0015 from the correlated fit’s 1.1032; NO-QUOTE — the ensemble fails every gate, and the one allowed escalation was declined because migration is structural, not a depth problem). The binned data’s preference for the unphysical low- γ shelf is therefore a property of the *product*, not of any single likelihood or sampler: it is reached by tempered SMC under the correlated likelihood and, from a physical-basin warm start, by MCLMC under the plain diagonal one — which is the T0.4 mechanism narrative restated from the sampling side, and one more reason the mechanism’s confirmatory data-side injection test (above) is worth running.⁴⁷

11.2 Epilogue E2: the collaborator’s own data — registration, the apples-to-apples fit, and two service findings

After E1, a collaborator on the team’s sampling effort (E. Odell) shared his own working cutout and PSF of DESI-165.4754-06.0423 — 140² cutout, 29² PSF, with no error map, mask, pixel scale, or exposure metadata attached — together with a workflow report (priv. comm.): his own first-stage MCLMC had “very migratory chains,” and a second stage preconditioned on the first

⁴⁷Source artifact: `data/e1_harvest.json`.

stage’s tail converged $\hat{R} \ll 1.01$ at $\sim 8\text{--}16$ chains $\times \sim 5000+5000$ draws. E2 is the pre-registered epilogue on that material: registration and packaging (O1), the apples-to-apples MCLMC fit under the E1-frozen gates (O2), a noise audit of his preparation with the T0.4-1 machinery (O3), and a demonstration of his stage-2 rescue on our own stored E1 chains (O4, run on *our* data only). All gates were frozen before any GPU work; everything ran on the free tier (6.8 L4-h total; no A100-h), and his data stays outside the repository per the campaign’s data bright line.⁴⁸

Registration (O1): a fourth pixelization, a parity flip, and a stack-dependent companion.

Registered against our calibrated $0.04''$ fine product (arcsinh-stretched, high-pass-filtered ZNCC over a scale grid and all eight dihedral orientations), his cutout sits at $0.064125''/\text{px}$ — *exactly half our native* $0.12825''$, a fourth pixelization of this sky beyond our $0.128''/0.08''/0.04''$ trio, and measurably *not* the $0.065''$ paper-drizzle scale — with a *parity flip* (a left-right mirror) relative to our raster; registration residual 0.300px against a 0.5px gate. His PSF is sampled at the cutout’s own scale (decisive: the half-scale hypothesis drops the aligned cosine from 0.885 to 0.705) and is *sharper and more symmetric* than our empirical PSF (moment-FWHM $0.449'' \times 0.456''$ vs. $0.555'' \times 0.475''$). Two preparation-level findings fell out of the registration census. First, the bright compact companion at $(-2.34'', -2.86'')$ — masked by our pipeline and a named suspect earlier in the campaign (T0.3, §1 item 2 sources) — *exists only in our stack*: his frame shows nothing at that position, while carrying nine compact sources ours lacks (single-visit-class artifacts; all masked) — a drizzle epoch/frame-selection question for the collaborator, not a science claim. Second, his files carry no noise model, so the packaged error map is a *documented v1 assumption* (background RMS from his own empty sky plus our archival exposure metadata; packaged sky $\chi^2/\text{px} = 0.9987$), to be replaced by his exact numbers.⁴⁹

The apples-to-apples fit (O2): pre-registered fork 2 fires. The E1 vehicle — constants, warm-start path, and both frozen gates byte-identical; the registered flip and offset absorbed grid-side so priors and warm starts remain in the campaign frame — was warm-started from the transported healthy anchor cloud (median 1.4334). Every one of the 64 chains in all four seed groups migrated during the 2000-step burn-in to the same low- γ shelf (Fig. 24): kept-draw pooled median $1.0999 [1.0727, 1.1103]$ — *descriptive telemetry, not a quotable posterior*: the fit fails both frozen gates ($\hat{R}_{\text{worst}} = 5.46$, $\text{ESS}_{\text{min}} = 66$; all 64 chains spend *every* kept draw outside the $[1.15, 2.0]$ band), and the one allowed escalation was declined for the E1 reason (structural migration, not sampling depth). Per-chain migration $\Delta\gamma = 0.27\text{--}0.45$; the endpoint sits 0.0033 from the certified corr-low 1.1032 and 0.0048 from the E1 v3b shelf. Of the three pre-registered interpretation forks, fork 1 — the native answer robust across four pixelizations, the strongest closure available — is dead for the resampled family; **fork 2 fires**: the migration extends to a second resampled product *and* a second, fully independent preparation pipeline, on the collaborator’s own data — matching his independent “very migratory chains” report, so the pathology is not an artifact of our preparation. A goodness-of-fit finding rode along: even the fit’s best kept draw leaves $\chi^2/\text{px} = 26.8$ under the v1 noise model (the shelf fit is a grossly bad model of his data, not merely an unconverged one), and the pooled per-dimension median is off-manifold at $\chi^2/\text{px} = 72.7$ — medians of unconverged multi-basin ensembles are not models.⁵⁰

⁴⁸Source artifact: `data/e2_harvest.json`; CAMPAIGN.md E2 design checkpoint; `research/checkpoint_e2_o4.md`.

⁴⁹Source artifact: `data/odell_registration.json`; `data/odell_cutout.npz` (meta).

⁵⁰Source artifact: `data/mclmc_diag_odell.{npz,json}`; `data/e2_harvest.json`; `data/e2_odell_noise.json` (model.render).

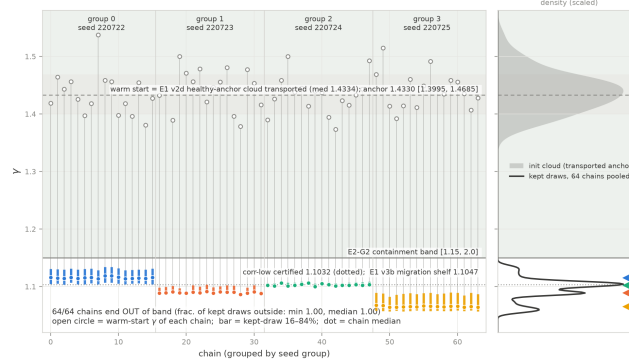


Figure 24: The E2 apples-to-apples fit on the collaborator's own registered 0.064125" product (his preparation, his PSF; plotted before gate math). Left: for each of the 64 chains (colored by seed group), the warm-start γ (open circle, drawn from the transported healthy 1.433 anchor cloud) against its kept-draw 16–84% interval and median: every chain exits the E2-G2 containment band [1.15, 2.0] during burn-in and equilibrates at $\gamma \approx 1.06\text{--}1.14$ (fraction of kept draws outside the band: min 1.00, median 1.00 — every kept draw; all 64 violations reported, none dropped). Right: init-cloud vs. pooled kept-draw marginals; the pooled kept-draw median 1.0999 sits 0.0033 from the corr-low certified 1.1032 and 0.0048 from the E1 v3b migration shelf 1.1047 — descriptive only (the ensemble fails both frozen gates, $\hat{R}_{\text{worst}} = 5.46$, NO-QUOTE). Source artifacts: `data/mclmc_diag_odell.{npz,json}` (gates.E1.G2.violations); `data/e2_harvest.json`.

The stage-2 rescue demonstration (O4): falsifier fired, control held. This experiment ran entirely on our own stored E1 chains, and is offered as a service finding about the shared two-stage recipe (the collaborator's rescue is empirically the same move as the two-stage re-preconditioned recipe used throughout this program — ours included). The pre-registered prediction: a stage-2 MCLMC preconditioned on the migrated v3b stage-1 tail (init draws, tail covariance as inverse mass, tail mean; 16 chains $\times$ 5000+5000 at the collaborator's stated scale) would converge pristinely at the shelf — demonstrating that stage-2 convergence metrics cannot detect stage-1 basin selection. **The falsifier fired instead**, and per the frozen checkpoint it is reported as loudly as a pass: stage-2 $\hat{R}_{\text{worst}} = 1.1875$, $\text{ESS}_{\text{min}} = 81$ ($\hat{R}_{\gamma}$ 1.128) — no pristine- $\hat{R}$ -at-the-shelf at these settings. What survives is a weaker but still useful statement (Fig. 25): the rescue moves the diagnostics most of the way toward the pass region ($\hat{R}_{\text{worst}}$ 1.379 $\rightarrow$ 1.188, $\hat{R}_{\gamma}$ 1.261 $\rightarrow$ 1.128) while the chains stay at the shelf (γ median 1.0849, telemetry; -0.019 vs. the stage-1 tail — falsifier F2, a return toward the band, did *not* fire, so the E1/E2 migration interpretation stands); a looser-but-common threshold ($\hat{R} < 1.2$) *would* have certified the unphysical solution, and the split-half readout ($\hat{R}_{\text{worst}}$ 1.033 on chains 0–7 vs. 1.321 on 8–15) shows an 8-chain practitioner could land on either side — at the rigorous frozen gates ($\hat{R} < 1.01$, $\text{ESS} \geq 1000$) the diagnostics still flag stage-1 basin selection. The control held exactly as pre-registered: the identical rescue on the healthy v2d stage-1 converges cleanly at the E1 posterior ($\hat{R}_{\text{worst}} = 1.0029$, $\text{ESS}_{\text{min}} = 8305$; $\gamma = 1.4683$, $0.0008\sigma_{\text{comb}}$ from the E1 quotable value; 0/16 chains out of band) — the property probed is stage-1-conditional, as designed. The collaborator's reported $\hat{R} \ll 1.01$ second stage is *not* reproduced on our stored chains at his stated scale; understanding that difference needs his exact configuration (chain count and length, preconditioning details, thresholds) — a question for him, not an assumption in this report.⁵¹

⁵¹Source artifact: `data/o4-{v3b,v2d}_stage2.{npz,json}`; `research/checkpoint_e2_o4.md`; `data/e2_harvest.json`.

E2-O4 — Evan-style stage-2 rescue on OUR stored E1 chains ($16 \times 5000+5000$, his stated scale): v3b improves ($\hat{R}_{\text{worst}} 1.379 \rightarrow 1.188$) but does NOT certify (F1 fired: ' $\hat{R}$ -blindness' NOT supported at these settings); v2d control clean

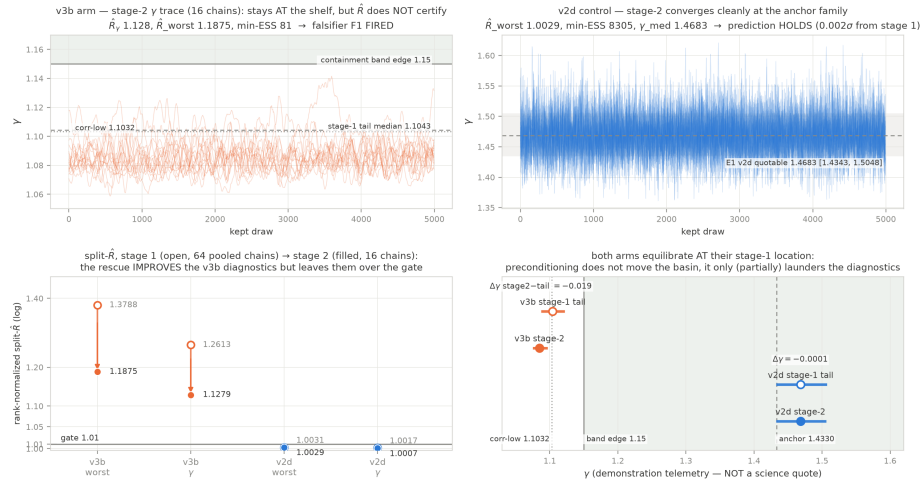


Figure 25: The O4 stage-2 rescue demonstration on our stored E1 chains ($16 \times 5000+5000$, the collaborator's stated scale; plotted before gate math). Top left: the v3b arm's stage-2 γ trace stays at the migrated shelf but does not certify ($\hat{R}_\gamma 1.128, \hat{R}_{\text{worst}} 1.1875$, min-ESS 81 — falsifier F1 fired). Top right: the v2d control converges cleanly at the anchor family ($\hat{R}_{\text{worst}} 1.0029$, min-ESS 8305), matching the E1 quotable posterior to $0.0008\sigma_{\text{comb}}$. Bottom left: stage-1 (open) $\rightarrow$ stage-2 (filled) split- $\hat{R}$ dumbbells against the 1.01 gate — the rescue improves the v3b diagnostics but leaves them over the gate. Bottom right: both arms equilibrate *at* their stage-1 location (all γ values demonstration telemetry, not science quotes): preconditioning does not move the basin. Source artifacts: `data/o4.{v3b,v2d}_stage2.{npz,json}`; `research/checkpoint_e2.o4.md`; `data/e2_harvest.json`.

The noise audit (O3): his preparation is correlated like ours; the stationarity verdict is model-quality-limited. The T0.4-1 machinery (masked ACF, two-component drizzle-family fit, 3×3 block stationarity against a 200-draw calibrated null) was run on his product in two arms — the pre-declared model-subtracted arm (residuals of the best available model on his grid: the failed fit’s maximum-log-density kept draw, a residual-minimizing reference point, not a certified posterior) and a no-model robustness arm added because that model is poor; only verdicts on which *both* arms agree are claimed. **Correlated like ours — confirmed on both arms** (Fig. 26): nearest-lag $\rho_{01}/\rho_{10} = 0.844/0.807$ (no-model) and $0.762/0.847$ (model-subtracted) against the pure-drizzle prediction 0.600 at his scale ratio — excess correlation *above* the entire drizzle-registration envelope $[0.51, 0.66]$ — with the same drizzle+background kernel family as our products (fitted correlated weight 0.92 on both arms; the two-component family fits the clean arm at $\max|\text{resid}| 0.030 \leq 0.05$, the production gate). **Stationarity: not established either way.** The clean no-model arm does *not* reject ($\max|z| = 2.20$, calibrated $p = 0.305$ — the same non-rejection class as our fine product, $p = 0.17$), while the model-subtracted arm rejects hard ($\max|z| = 6.34$, $p < 0.005$) but is confounded: the best available model still leaves sky $\chi^2/\text{px} = 18.8$, so its rejection measures model error, not a noise property. The supported memo line is “your product carries the same correlated-noise structure ours does”; the sharper nonstationarity question needs his exact noise metadata and a converged fit.⁵²

E2-O3 noise audit — Even’s own product is correlated like ours (both arms: ρ_{01} 0.84 no-model / 0.76 model-sub vs pure-drizzle 0.60; corr. weight 0.92; drizzle-family fit max|resid| 0.030 $\leq$ 0.05) | stationarity NOT REJECTED on the clean no-model arm (max|z| 2.20, cal. p 0.305); model-sub arm rejects (max|z| 6.34, p 0.005) but is CONFOUNDED by the unconverged model

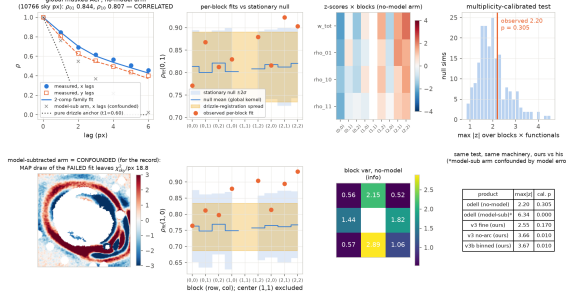


Figure 26: The O3 noise audit of the collaborator’s own product (T0.4-1 machinery; plotted before gate math). Top row, no-model arm: global masked ACF over 10,766 sky pixels against the two-component drizzle-family fit and the pure-drizzle anchor ($t_1 = 0.60$); per-block ρ_{01} against the stationary null and the drizzle-registration spread; per-block z -scores; and the multiplicity-calibrated test (observed $\max|z| 2.20$, $p = 0.305$ — no rejection on the clean arm). Bottom row: the model-subtracted arm is kept for the record but *confounded* — the best available model (the failed E2 fit’s max-log-density draw) leaves sky $\chi^2/\text{px} = 18.8$, so its hard rejection ($\max|z| 6.34$) measures model error; the summary table runs the same test with the same machinery on our own products (fine $p = 0.17$; binned $p = 0.010$, rejected). Source artifact: `data/e2_odell_noise.json`.

The product-dependence synthesis, updated. E2 sharpens the E1 synthesis into a clean split by product family. *The native scale is well-behaved:* the `v2d` native diagonal target has now passed the strictest convergence gates in this program twice (the E1 fit; the O4 control), both times at the anchor family. *The resampled products migrate:* both resampled products of this sky — our $0.08''$ binned drizzle and his $0.064125''$ half-native preparation, produced by two fully independent pipelines — migrate every chain from a physical-basin warm start to the same $\gamma \approx 1.10$ shelf under the plain diagonal likelihood, in every seed group (128 of 128 chains across the two products), and the correlated-likelihood money fit sits on the same shelf. Whatever the mechanism’s final form

⁵²Source artifact: `data/e2_odell_noise.json`.

(the T0.4 stationarity rejection is its noise-side face, §7.4.1), it travels with *resampling of this data*, not with any one pipeline, likelihood class, or sampler — and the native-scale anchor family (1.4330 old stack; 1.4683 scene-API MCLMC) remains the only convergence-certified answer on this sky.

11.3 What it opens

The honest open list is long, and most of it is deliberate non-spend recorded in the gate ledger rather than silent omission. **Data-side injections:** the T1.1 injection-recovery test came back confounded — the control arm failed high *and* sick, implicating the injection construction (scene-subtraction residue) under the pre-registered honesty clause — and the fit-side rescue (T1.1b) failed its entry gate with an 85.8% whitened-dof loss. Residue-free injections must be built on the data side (kernel-sampled noise-only fields, sky-patch bootstrap, or deeper multi-start subtraction); the mechanism’s confirmatory test is therefore still unrun, with one directional datum in its favor (same-data corr–diag differential -0.0526 , no defensible error bar). **The two-stage port:** a converged scene-substrate posterior on the $\text{cond-}10^{14}$ target needs the two-stage re-preconditioning recipe ported (~ 0.5 day + 2–4 A100-h, estimated, not spent); until then “the 1.433 anchor reproduces at posterior level on the new substrate” is untested. **The locally-stationary whitener class:** the rejected stationary class needs a successor; the ported likelihood deliberately keeps a pluggable whitener seam for it, and the kernel-scan arm (E3) that would inform it was deferred by plan and never funded — it lapses at program close as the standing caveat on absolute- γ claims. **L1/L2 never ran:** there is *no* production-SMC coverage certificate (L1) and no supersampling/sub-pixel-PSF decomposition (L2); nothing in this report should be read as implying either exists. **PSF marginalization was promoted and then never built** (its pool was reallocated by a later decision); the D7 promotion language should not be read as delivered work. The B4 preconditioning question (does per- λ ensemble preconditioning replace the two-stage recipe at $\text{cond-}10^{14}$?) closed not-evaluable-within-budget and is genuinely open.

11.4 Guidance for practitioners

The decision matrix compresses to four rules. (1) If you need **evidence or mode structure** — $\log Z$, basin weights, cold-start robustness — on targets up to the 74-d bimodal class, prior-seeded MAMS-SMC is the vehicle, at single-digit A100-hours per basin; take the evidence error from seed repeats, not from the per-run bootstrap (finding A). (2) If you need a **posterior on a carousel-class real multi-plane system**, budget an order of magnitude more than we did or use a warm-started method and accept the cold-start blind spot: at $\lesssim 16$ A100-h *neither* family converges (B1r). (3) Use **MCLMC as a diagnostic**, freely — and never as the evidence kernel without Metropolis adjustment or a measured bias budget. (4) **Certify your references:** before freezing an MCMC-derived posterior or evidence reference into an acceptance gate, demand a coverage certificate at the gate’s own σ -scale (findings B; B3’s blocked-by-reference row; the X2 frozen-set question). Our own frozen reference failed this standard at 11σ -equivalent scale — the most expensive single lesson of the campaign, and one we incurred against ourselves.

11.5 Operational lessons

Four ops findings each cost real budget once and are now protocol. (1) **Pin the high-memory GPUs and re-measure after remat changes:** two independent OOM failures (an injection arm on a 40-GB node, 1.49 h burned; the arbitration’s 21-GiB single mutate allocation) traced to memory sizings measured *with* rematerialization active and applied to runs without it — memory figures do not transfer across remat settings; the fix is a hardware constraint pin plus fresh sizing.

(2) **The scheduler snapshots scripts at submission:** a fix “resubmitted” after editing the batch script re-ran the old code (two failed attempts); protocol now verifies the *submitted* snapshot’s checksum against the intended file before trusting any resubmission. (3) **One writer per manifest:** concurrent session fronts clobbered the deploy checksum manifest; the rule is pull-remote-first, union-merge, single writer per wave. (4) **Checkpoint everything:** the first production wave burned 94% of its hours with zero artifacts (18.7h spent, 1.1h readable) — jobs died at walls holding all state in memory. The mid-campaign checkpoint retrofit (per-stage full-state checkpoints, a graceful wall-cap exit protocol distinguishing PARTIAL from crash, bit-identical resume including the full log Z bootstrap state, gated by a 58-test suite) converted that failure mode into three-of-four readable recoveries, and is the only reason the 12-hour carousel chain and every λ -progress readout in Table 9 exist at all. Smaller pins that will save a future campaign a day each: chunked mutation sizes must divide *both* the particle count and the pilot subset; a watchdog state for “completed without artifact” doubles as the harvest reminder; BLAS thread caps matter on aarch64 login nodes; and the lensing likelihood is never run on CPU.

12 Reproducibility

Ledgers and provenance. The authoritative record is `CAMPAIGN.md`: nine locked decisions (D1–D9), the A100-hour ledger (rows appended *before* results are read, failed and zero-artifact jobs counted in full), the gate record, and a dated stage log. Every gate ever registered — including the six that ended without a run and one promoted-then-unbuilt item — has a final disposition in `research/gate_ledger_final.md`; nothing lapses silently. Among the never-run items, two of our own programmed arms are named explicitly: the S7 flow-preconditioned-MAMS arm (prerequisite unmet — the B1 $\lambda=1$ ensemble it required never came to exist) and the X3 evidence-scored Vela-ladder prototype (never mobilized). Two additional internal-reference benchmark arms were omitted under the campaign’s pre-declared external-collaboration bright line and are dispositioned in the same ledger. Every number in this report traces to an on-disk artifact via `research/synthesis_tables.md`, whose seven discrepancy flags (artifact vs. ledger prose) are binding on this text: wherever the two disagree the artifact value is quoted (e.g. the MCLMC demotion screen is 20.1σ by `data/smc_b0_report.json`, not the ledger’s “ 30σ ”; the L4 arbitration leg’s on-disk record is $\lambda = 0.4183$ at a 12.26-h resume leg, with the ledger’s $\lambda \approx 0.45/18$ -h figure including earlier checkpointed legs). The external claims register is `papers/handoff/CLAIMS.md` (§10.2). External release of this report is gated on the sign-off checklist `papers/handoff/SIGNOFF_CHECKLIST.md`, which lists the six passages characterizing unpublished upstream code state that require the team’s sign-off (internal use is unaffected; the sixth passage is the E1 epilogue’s description of the vendored experimental MCLMC driver, §1 item 3).

Pre-registration discipline. Every experimental cell was pre-registered: a design checkpoint with hypothesis, predicted direction and magnitude, falsifier, and derived thresholds, logged before submission; figures plotted before gate text was written; amendments (B1-reduced descope, the S6br budget-match, the B2’ re-registration) ledgered as dated decisions with checksummed checkpoints, never edited in place. Proposer and grader are separated throughout: every externally relevant verdict in this report is *proposed* (*UNCERTIFIED* — *external*), with certification belonging to the receiving team, and scene-substrate results are labeled “reproduced/measured, UNCERTIFIED (external)” — “validated” is reserved for the old certified stack.

Table 10: A100-hour budget by phase: caps vs. ledgered actuals (CAMPAIGN.md ledger; per-row `sacct`, $\text{Elapsed} \times 1 \text{ GPU}$, shared QOS). T1.1 was funded from the pool freed by D7’s retirement of P4; P2b by the D8 reallocation (10 h freed-P4 + 8 h stretch). The 100-h hard stop was never approached.

phase	scope	cap [h]	actual [h]
P0	scaffold, parity gates F1–F8, SMC correctness (B0)	1	0.00
P1	certification slice (T0.2/T0.3/T0.4)	10	2.21
T1.1	injection-recovery on real drizzle noise (D7 pool)	10	7.80
P2	sampler benchmark (B1–B5 waves + recovery lane)	24	22.45
P2b	carousel B1-reduced (S1r + S6br; D8/D9)	18	15.85
P3	scene-substrate payload (L0, arbitration, L0-G2)	17	5.21
P4	profile-class fork (retired at entry gate X1-G0)	10	0.00
campaign total (hard stop 100)		100	53.51

Environment freezes. The campaign venv pins python 3.13, jax/jaxlib 0.6.2, blackjax 1.3, tfp 0.25.0, numpy 2.4.6 (a declared deviation from the upstream 2.1.3 pin, covered by the gate battery), no tensorflow; all installs pass `-c constraints.txt`. The substrate is a vendored, *unpatched* snapshot of the upstream development branch (full SHA in `VENDORED_REF.txt`; the three convention reconciliations live on our side of the seam), and the old validated stack is imported by path at its certified ref. The Perlmutter-native environment re-passed the full parity battery (F8, job 55980038) with the phoenix-side F1 value bit-consistent. Production deploys are non-git rsync copies audited file-by-file against a checksum manifest (225 files at close) before every submission.

Compute. Table 10 is the phase budget against actuals: **53.51 of 100 A100-h** at program close, nothing in flight (per-row `sacct` rounding makes the component sum 53.52; flagged, not smoothed). Free-tier phoenix spend (not A100): 12.38 L4 GPU-h (B3), ~ 29 A16-h (X2), and — after program close — the E1 epilogue’s 9.56 L4-h (§1) plus the E2 epilogue’s 6.82 L4-h (§11.2). Zero-artifact and failed jobs are counted at full cost in their phases; the wave-1 post-mortem and the recovery-lane economics are part of the record, not an embarrassment to be netted out.

Rebuilding the results. The parity battery is `01_parity_scene.py` (gates frozen in the README at P0); the SMC correctness gates are `04_smc_micro_validation.py`; each benchmark cell names its runner, slurm script, seed, and artifact in its ledger row; and each gate evaluation is a standalone JSON under `data/` (Perlmutter production artifacts under `data/results-perlmutter/`). The report builds with `make -C papers pdf` against the shared `reproductions/tech-report.sty` preamble.

A Convergence Diagnostics for Every Fit

Table 11 is the complete per-fit convergence ledger: one row for every fit this campaign ever ran, including the failed, rejected, collapsed, budget-blocked, and zero-artifact arms that the headline tables summarize or omit. It extends the consolidated ledger of Table 2 (§1, item 4) from “every fit quoted in this report” to “every fit executed,” and backs each cell of the decision matrix (Table 9, §9).

Two provenance rules govern the table. First, the two inherited companion-campaign diagonal fits (the anchor $\gamma = 1.433 \pm 0.034$ and the diagonal-low $\gamma = 1.293 \pm 0.012$) appear as provenance rows

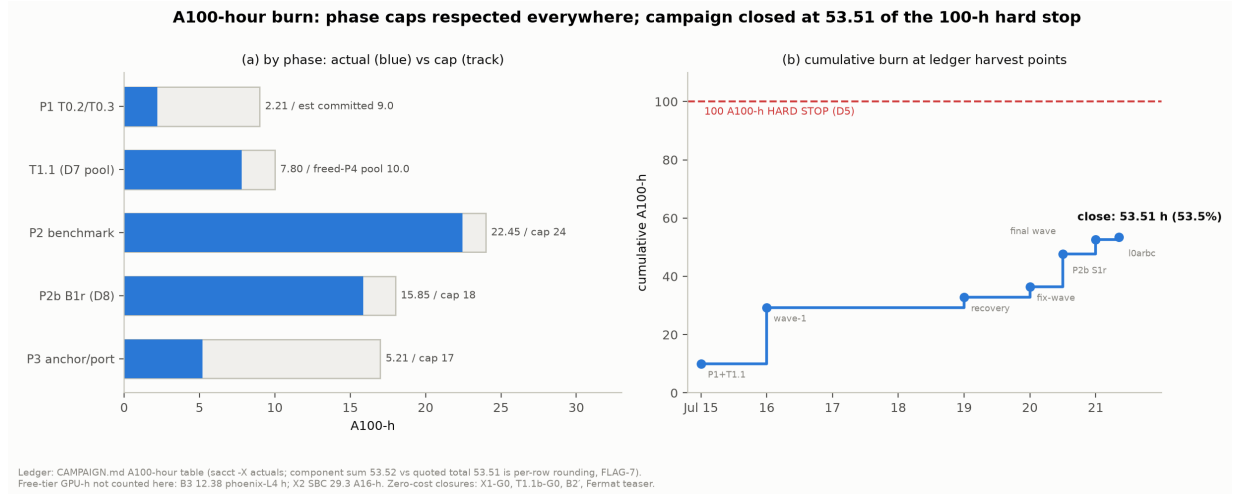


Figure 27: A100-hour burn. (a) Actual vs. cap by phase — every fence respected (P1 2.21, T1.1 7.80 of the D7-freed 10-h pool, P2 22.45/24, P2b 15.85/18, P3 5.21/17). (b) Cumulative burn at the ledger’s harvest points (events in ledger order; sub-day x -offsets are display-only), closing at **53.51** of the 100-h hard stop; the 53.52 component sum is per-row `sacct` rounding (flagged, not smoothed). Free-tier GPU-h (B3: 12.38 L4-h; X2: 29.3 A16-h) are additional and uncounted here. Zero-cost closures: X1-G0, T1.1b, B2’, the Fermat teaser. Source: CAMPAIGN.md A100-hour ledger (append-before-read protocol).

only.⁵³ Second, arms that ran the sampler but left no readable ensemble on disk (the zero-artifact wave-1 losses) are still given rows: their diagnostics cells quote the postmortem and ledger record (“no artifacts — TIMEOUT”) because that is all that exists to quote. Only the X2 free-tier SBC census ($N=32$ classic-recipe fits, summarized with the gift) and pure infrastructure failures that never reached a sampler are ledgered in the coverage notes after the table rather than rowed.

Conventions. For SMC fits the diagnostics are: tempering endpoint (λ reached; $\lambda=1$ is completion), stage count, final-stage weighted ESS (wESS; the adaptive schedule holds per-stage wESS at $0.7N$ by construction), unique particles after the final resample, MAMS acceptance against its 0.90 target, and σ_{seed} where independent-seed repeats exist. For MCMC arms: $\hat{R}_{\text{worst}}$ and ESS_{min} across parameters, against the frozen acceptance rule $\hat{R} < 1.05 \wedge \text{ESS} \geq 200$; the post-campaign E1/E2 MCLMC fits (groups M–N — lettered for the vehicle, and to avoid collision with the campaign’s L0/L1/L2 lane names) carry their own, stricter frozen gate, $\hat{R} < 1.01 \wedge \text{ESS} \geq 1000$ (the E1/E2 production fits pooled over 64 chains; the E2-O4 stage-2 arms over their 16 chains at the collaborator’s stated scale) plus a $\gamma \in [1.15, 2.0]$ containment band. Verdicts use a fixed vocabulary: **CONVERGED** (sampler completed with healthy diagnostics; any science-gate outcome is annotated separately), **PARTIAL-BY-BUDGET** (healthy sampler stopped by a pre-registered wall cap short of $\lambda=1$), **REJECTED** (completed but fails the frozen acceptance rule), **COLLAPSED** (degenerate terminal population), and **BLOCKED** (no readable ensemble — killed by a budget fence, a walltime TIMEOUT, or a since-fixed artifact-writer defect; the cause is named in each row). Every number is quoted from the artifact named in the last column; the spine `research/synthesis_tables.md` binds, and its discrepancy flags are honored here (the MCLMC bias screen is quoted at its artifact value 20.1σ , not the ledger’s “ 30σ ”; the classic-arm ESS_{min} is quoted per stage; the

⁵³Provenance footnote for the companion diagonal fits: those posteriors were produced by the companion campaign’s two-stage re-preconditioned recipe, and their per-fit $\hat{R}/\text{ESS}$ convergence diagnostics live in the companion report’s campaign ledger (Benson and Huang, 2026). They are cited here, never re-derived and never invented; this campaign consumed the fits only as inherited draws (e.g. the L0-G2 q -fit construction and the head-to-head model points of Table 1).

L4 arbitration λ is quoted from the on-disk resume-leg summary with the ledger's multi-leg figure noted).

Table 11: Convergence diagnostics for every fit the campaign ran. wESS = final-stage weighted ESS; “uniq” = unique particles after the final resample (for T0.2-family runs the stored-draw unique-row count is also given where it differs); accept = MAMS Metropolis acceptance (target 0.90; unadjusted MCLMC accepts by construction and its acceptance is not quoted). σ_{boot} = per-stage bootstrap log Z error. Verdict vocabulary as defined in the text.

fit / arm	data + likelihood		vehicle	N ch×draws	or diagnostics	verdict	artifact
A. SMC correctness gate (B0) — analytic targets with exact references; local GPU, 0 A100-h; seed 0							
B0 mix2, MAMS	analytic mixture data)	2-d bimodal (no image)	tempered MAMS mutation	SMC, 512	$\lambda=1$ @6 stages; wESS 464/512; uniq 440; accept 0.89–0.93; log Z err 0.073 ($\sigma_{\text{boot}}=0.068$); $\hat{w}_{\text{minor}}$ 0.219 vs 0.2	CONVERGED (gate PASS)	data/smc.b0.report.json
B0 mix2, HMC-mutation	same		tempered HMC mutation	SMC, 512	$\lambda=1$ @6 stages; wESS 486/512; uniq 463; accept 0.75–0.95; log Z err 0.003; $\hat{w}_{\text{minor}}$ 0.191	CONVERGED (gate PASS)	same
B0 mix2, MCLMC	same		tempered unadjusted-MCLMC mutation	SMC, 512	$\lambda=1$ @6 stages; wESS 483/512; uniq 471; log Z err 0.009; $\hat{w}_{\text{minor}}$ 0.205	CONVERGED (gate PASS)	same
B0 funnel10, MAMS	analytic nel	10-d Neal funnel	tempered MAMS	SMC, 512	$\lambda=1$ @7 stages; wESS 508/512; uniq 500; accept 0.88–0.94; log Z -0.664 vs 0 ($\sigma_{\text{boot}}=0.070$)	CONVERGED (log Z gate FAIL — sampler-class limitation, reproduced in the old validated stack)	same
B0 funnel10, HMC-mutation	same		tempered HMC	SMC, 512	$\lambda=1$ @8 stages; wESS 512/512; uniq 498; accept 0.76–0.86; log Z -0.545 vs 0	CONVERGED (log Z gate FAIL)	same
B0 funnel10, MCLMC	same		tempered MCLMC	SMC, 512	$\lambda=1$ @7 stages; wESS 461/512; uniq 446; log Z -0.383 vs 0	CONVERGED (log Z gate FAIL)	same
B0 illcond46, MAMS	analytic conditioned	46-d Gaussian	tempered MAMS	SMC, 512	$\lambda=1$ @187 stages; wESS 407/512; uniq 415; accept 0.89–1.00; worst-param $ z $ 1.83 (rule < 3); log Z -145.80 vs ref -148.20 : err 2.40 nats at $\sigma_{\text{boot}}=0.37$ ($6.6\sigma_{\text{boot}}$ — finding A)	CONVERGED (location gate PASS; log Z error noted)	same
B0 illcond46, HMC-mutation	same		tempered HMC	SMC, 512	$\lambda=1$ @195 stages; wESS 417/512; uniq 417; accept 0.71–0.98; worst- $ z $ 1.87; log Z err 21.6 nats at $\sigma_{\text{boot}}=0.41$	CONVERGED (location PASS; log Z error noted)	same
B0 illcond46, MCLMC	same		tempered MCLMC	SMC, 512	$\lambda=1$ @185 stages; wESS 445/512; uniq 425; worst- $ z $ 2.93; log Z err 13.2 nats; Δ log Z vs MAMS $+10.79 \pm 0.54 = 20.1\sigma$	CONVERGED; kernel DE-MOTED to diagnostic ($> 3\sigma$ rule)	same
B. T0.2 seed certification + T0.3 companion mask — the production correlated fit family (old validated stack; seed-2 legs inherited from the companion campaign)							

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Table 11, continued.

fit / arm	data + likelihood	vehicle	N ch $\times$ draws	or diagnostics	verdict	artifact
v3b corr low, seed 2 (money fit)	real HST/WFC3- IR v3b binned drizzle; whitened- correlated likelihood ($n_{\text{keep},w} = 9273$)	prior-seeded MAMS-SMC	128	$\lambda=1$ @28 stages; wESS 118.1/128; uniq 77 (37 stored rows); γ 1.1032; $\log Z$ -4771.08	CONVERGED	companion e2_v3b_low_smc_canary_fix.json; data/t02_t03_gate_eval.json
v3b corr low, seed 3	same	same	128	$\lambda=1$ @28; wESS 127.7/128; uniq 82 (14 stored rows — dup-cluster, reported not sick); γ 1.0967; $\log Z$ -4769.12	CONVERGED	e2_v3b_low_smc_seed3.json
v3b corr low, seed 4	same	same	128	$\lambda=1$ @28; wESS 127.5/128; uniq 79 (27 stored rows); γ 1.1005; $\log Z$ -4771.37. Across seeds 2+4: $\sigma_{\text{seed}}(\gamma) = 0.00325$ (kill $\sigma > 0.024$ not tripped), $\sigma_{\text{seed}}(\log Z) = 1.22$ nats	CONVERGED + seed-certified	e2_v3b_low_smc_seed4.json
v3b corr steep, seed 2	same	same	96	$\lambda=1$ @21; wESS 36.0/96 (final resample low); uniq 50; γ 2.6393	CONVERGED	companion e2_v3b_steep_smc_canary_p96.json; data/t02_t03_gate_eval.json
v3b corr steep, seed 3	same	same	96	$\lambda=1$ @21; wESS 96.0/96; uniq 59; γ 2.6522	CONVERGED	e2_v3b_steep_smc_seed3.json
v3b corr steep, seed 4	same	same	96	$\lambda=1$ @22; wESS 90.3/96; uniq 58; γ 2.6485. Across seeds: $\sigma_{\text{seed}}(\gamma) = 0.00664$; $\Delta \log Z$ (steep — low) = -28.9/ -32.3/ -31.2 per matched seed, $\sigma_{\text{seed}}(\Delta \log Z) = 1.79$ nats	CONVERGED ; low-basin preference seed-stable	e2_v3b_steep_smc_seed4.json
T0.3 companion-masked refit (seed 2)	same data, whiten- then-drop companion mask ($n_{\text{keep},w}$ 9273 $\rightarrow$ 8247)	same	128	$\lambda=1$ @28; wESS 127.4/128; uniq 84 (17 stored rows); γ 1.1011 (shift -0.0021 vs production, $0.26\sigma_{\text{stat}}$)	CONVERGED (companion exonerated)	e2_v3b_low_smc_compmask.json; data/t02_t03_gate_eval.json
C. T1.1 injection recovery — $\gamma_{\text{truth}} = 1.433$ arcs injected into the real v3b drizzle residual field (Perlmutter A100)						
T1.1 inj1, corr	injected scene 1; whitened-correlated ($n_{\text{keep},w} = 9273$)	prior-seeded MAMS-SMC	128	$\lambda=1$ @27 stages; wESS 125.6/128; uniq 76 (52 stored rows); $\gamma_{\text{rec}} 1.5151 \pm 0.036$ ($z = +2.25$)	CONVERGED (bias +0.082, outside both pre-registered zones)	t11_inj1_smc.json; data/t11_gate_eval.json
T1.1 inj2, corr	injected scene 2 (dup-flagged shift)	same	128	$\lambda=1$ @34; wESS 124.4/128; uniq 85 (18 stored rows — dup-cluster flag); $\gamma_{\text{rec}} 1.5719 \pm 0.043$ ($z = +3.23$)	CONVERGED (flagged; dropping it leaves the median bias +0.078, same zone)	t11_inj2_smc.json
T1.1 inj3, corr	injected scene 3	same	128	$\lambda=1$ @20; wESS 107.9/128; uniq 81; $\gamma_{\text{rec}} 1.5076 \pm 0.041$ ($z = +1.82$)	CONVERGED	t11_inj3_smc.json

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Table 11, continued.

fit / arm	data + likelihood			vehicle	N ch×draws	or diagnostics	verdict	artifact
T1.1 inj1, diag control	same	injected	scene	same	128	$\lambda=1$ @37; wESS 128.0/128 (degenerate); 1/128 unique particle rows ; $\gamma_\sigma \approx 4 \times 10^{-16}$; srcS. I_e railed $\sim 58\times$ truth; γ 1.5677 $\notin$ [1.29, 1.43]	COLLAPSED (total resample collapse; per the honesty clause the injection construction is implicated)	t11.inj1.diag.smc.json; data/t11.gate.eval.json
D. L0-G2 cross-stack reproduction — the correlated fit re-run on the scene API port, Perlmutter A100 (proposed, UNCERTIFIED — external)								
L0-G2 low leg, seed 2	v3b binned; ported CorrelatedImageData whitened-correlated likelihood			prior-seeded MAMS-SMC on the scene API	128	$\lambda=1$ @27 stages; wESS 123.5/128; uniq 81; γ 1.1005 [1.0992, 1.1065]; log Z -4770.97	CONVERGED (PASS: $\Delta\gamma = 0.0027$; log Z agrees with the old stack to 0.11 nats)	10g2.v3b.scene_seed2.json
L0-G2 steep leg, seed 2	same			same	96	$\lambda=1$ @22; wESS 47.6/96; uniq 54; log Z -4800.33 ; $\Delta \log Z(\text{steep} - \text{low}) = -29.36$, in the P1 seed family	CONVERGED (log Z agreement 0.37 nats)	same
E. B5 — T3 74-d bimodal (foundry_v3b74, diagonal likelihood, f32; Perlmutter A100)								
B5 low basin, MAMS	v3b binned, diagonal likelihood (old-stack target)			prior-seeded MAMS-SMC	128	$\lambda=1$ @31 stages; basin retention 1.000; min uniq 88; accept 0.862–0.998; log Z $38376.33 \pm 0.33_{\text{boot}}$; γ_{eqw} 1.0919	CONVERGED (G1 PASS; the 11 σ G2 FAIL indicts the frozen reference's within-basin coverage, §9.3)	b5_v3b74_low_mams_seed2.json; data/b5.gate.eval.json
B5 steep basin, same MAMS				same	96	$\lambda=1$ @22 stages; retention 1.000 (frac-low 0.0); min uniq $70 \geq N/4$; accept 0.886–0.998; log Z $38518.23 \pm 0.31_{\text{boot}}$; γ_{eqw} 2.5552	CONVERGED (G1 PASS)	b5_v3b74_steep_mams_seed2.json
B5 low basin, same MCLMC (diagnostic arm)				tempered SMC, 128 unadjusted- MCLMC mutation (demoted kernel)	128	$\lambda=1$ @36 stages; log Z $38380.36 \pm 0.36_{\text{boot}}$; γ_{eqw} 1.0935 (within-basin $\Delta\gamma = 0.0016$ vs MAMS); minor-mode weight inflated $\times 56$ [CI ₉₅ 22–146]	CONVERGED (diagnostic only; G3 ambiguous at frozen thresholds — never an evidence kernel)	b5_v3b74_low_mclmc_seed2.json; data/b5.gate.eval.json

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Table 11, continued.

fit / arm	data + likelihood		vehicle	N ch $\times$ draws	or diagnostics	verdict	artifact
B5 S1 low leg, same wave-1 attempt (job 55985449)			prior-seeded MAMS-SMC	128	no artifacts: the RC3 artifact-writer defect (TypeError after the driver returned) destroyed the write; $\lambda=1$ was reached (structural: the driver raises if $\lambda<1$), diagnostics lost; set -e killed the job before the steep leg; 1.02 A100-h. Fix landed + regression-locked; the converged rows above are the reruns	BLOCKED (writer defect — infra, not science)	research/p2-wave1.postmortem-redesign.md
B5 S2 MCLMC low same leg, wave-1 attempt (55985450)			tempered SMC, MCLMC mutation	128	no artifacts: same RC3 defect at the write (~ 36 min in); $\lambda=1$ reached (structural), diagnostics lost; 0.64 A100-h	BLOCKED (writer defect)	same
F. B2/B2' — DSPL thin-ridge cosmology (21-d, seeded mock per the team's generation recipe; Perlmutter A100)							
B2 dspl20_orig, seed 2	DSPL mock, diagonal likelihood, original (Ω_{m0}, w_0) coords		prior-seeded MAMS-SMC	512	$\lambda=1$ @64 stages; min uniq 358 ($\geq N/4$), final 468; min incremental ESS 358.4; log Z $17294.01 \pm 0.25_{\text{boot}}$; Ω_{m0} med 0.4701	CONVERGED (falsifier NOT-DECIDABLE-AS-REGISTERED: band uncalibrated for this realization, §9.4)	b2_dspl20_orig-s1_seed2.json; data/b2_gate_eval.json
B2' dspl20_ratio control, seed 2	identical data, exact ratio-coordinate reparameterization		same	512	$\lambda=1$ @64 stages; min uniq 360, final 424; accept 0.822–0.927; log Z $17290.95 \pm 0.24_{\text{boot}}$; r_2 shape PASS (width ratio $0.79 \in [0.67, 1.5]$); $\Delta \log Z(\text{orig} - \text{ratio}) = +3.06 \pm 0.35_{\text{boot}}$ where true $\Delta \approx 0$ (finding A, measured)	CONVERGED (PASS-UNDER-POWERED)	b2_dspl20_ratio-s1_seed2.json; data/b2_gate_eval.json
G. B3 — easy scene, 22-d unimodal $\times 4$ (hs2 seeded mocks; free-tier L4, 0 A100-h)							
B3 hs2_sys0 reference	classic hs2_sys0 mock, diagonal (ForwardProb-Model conventions)	diagonal	classic MAP $\rightarrow$ SVI $\rightarrow$ HMC, frozen settings	50ch $\times$ 750 (250 burn); esc. 50 $\times$ 1500 (500)	$\hat{R}_{\text{worst}}$ 7.65 (esc. 7.57) vs rule < 1.05 ; ESS $_{\text{min}}$ 62.9 (esc. 64.0) vs ≥ 200	REJECTED (blocked-by-reference)	data/b3_run_ref-sys0_l4-8.json
B3 hs2_sys1 reference	classic hs2_sys1 mock, same	same	same	same	$\hat{R}_{\text{worst}}$ 8.81 (esc. 7.87); ESS $_{\text{min}}$ 65.0 (esc. 63.7)	REJECTED	data/b3_run_ref-sys1_l4-8.json
B3 hs2_sys2 reference	classic hs2_sys2 mock, same	same	same	same	$\hat{R}_{\text{worst}}$ 4.90 (esc. 5.66); ESS $_{\text{min}}$ 74.5 (esc. 74.5)	REJECTED	data/b3_run_ref-sys2_l4-8.json
B3 hs2_sys3 reference	classic hs2_sys3 mock, same	same	same	same	$\hat{R}_{\text{worst}}$ 4.44 (esc. 4.36); ESS $_{\text{min}}$ 82.5 (esc. 83.9)	REJECTED	data/b3_run_ref-sys3_l4-8.json
B3 hs2_sys0	SMC arm same data		prior-seeded MAMS-SMC	512	none written: killed at the pre-registered 18,000-s (5-h) per-arm L4 fence before completion; no readable ensemble	BLOCKED (by budget)	data/b3_cells.json; data/b3_lane2_sys0-smc.log

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Table 11, continued.

fit / arm	data + likelihood	vehicle	N ch $\times$ draws	or diagnostics	verdict	artifact
B3 hs2_sys1 SMC arm	same	same	512	same (5-h fence)	BLOCKED (by budget)	data/b3_lane2_sys1_smc.log
B3 hs2_sys2 SMC arm	same	same	512	same (5-h fence)	BLOCKED (by budget)	data/b3_lane2_sys2_smc.log
B3 hs2_sys3 SMC arm	same	same	512	same (5-h fence)	BLOCKED (by budget)	data/b3_lane2_sys3_smc.log
H. B4 — T2 foundry_marg46 (v2d 46-d marg-ridge, cond$\sim 10^{14}$; Perlmutter A100, wave-1)						
B4 marg46 S1, seed 2 (job 55985448)	v2d native, ridge-marg likelihood (old validated stack)	marg46 prior-seeded MAMS-SMC	256	no artifacts — TIMEOUT 3:30 (3.51 A100-h): run log shows build (78 s) + warmup log-prob only; zero stage observability (no per-stage checkpointing, RC1), and the since-fixed RC3 writer defect sat on the same code path; measured cost lower bound 3.5 h at $N=256$, never re-funded	BLOCKED (TIMEOUT; ledger disposition: proposed NOT-EVALUABLE-WITHIN-BUDGET)	b4_marg46.s1_seed2.run.log; research/gate_ledger_final.md §B.1
I. B1/B1r — carousel 33-d multi-plane (Perlmutter A100): wave-1 mock arms, then B1r on REAL MUSE cutouts (sign-off-gated)						
B1 S1 cold-start SMC, seed 2 (mock; job 55985444)	carousel33 seeded-stand-in (real cutouts not yet landed), diagonal	prior-seeded MAMS-SMC	512	no artifacts — TIMEOUT 4:00 (4.01 A100-h), zero per-stage observability (RC1); postmortem estimate: the 4-h wall bought $\sim 5\text{--}8$ early stages, $\lambda \lesssim 10^{-4}$	BLOCKED (TIMEOUT; mock arms retired by the B1-REAL-DATA-LANDED amendment, superseded by B1r)	b1_carousel33.s1_seed2.run.log; research/p2_wave1_postmortem_redesign.md
B1 S1 cold-start SMC, seed 3 (mock; 55985445)	same	same	512	no artifacts — TIMEOUT 4:00 (4.01 A100-h), same failure mode	BLOCKED (TIMEOUT; retired by amendment)	b1_carousel33.s1_seed3.run.log
B1 S6b warm MAMS-alone, seed 2 (mock; 55985446)	same	MAP warm start $\rightarrow$ MAMS-alone	64 chains	no artifacts — TIMEOUT 3:00 (3.01 A100-h); run log: MAP done (302 s, 22,400 grads), burn round 0 only (~ 25 min/round incl. compile)	BLOCKED (TIMEOUT; retired by amendment; motivated the checkpoint retrofit that made the B1r rows below readable)	b1_carousel33.s6b_seed2.run.log

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Table 11, continued.

fit / arm			data + likelihood		vehicle	N ch×draws	or diagnostics	verdict	artifact
B1r S1r cold-start SMC, seed 2	real MUSE cutouts (source4-5.fits, source9.fits), diago- nal	prior-seeded MAMS-SMC, 3 chained check- point legs (11.33 A100-h)	128	$\lambda = 0.1506$ @36 stages ($\lambda=1$ structurally out of reach); uniq 92–104/128 every stage; accept 0.871–0.934 (mean 0.899 vs 0.90); stage wESS pinned at 0.7N; 0 posterior samples	PARTIAL-BY-BUDGET (sampler healthy — a cost verdict, §9.2)	b1r128_carousel133_s1_seed2.PARTIAL.json; data/b1r_decision_matrix.json			
B1r S6br warm MAMS-alone, seed 2	same	MAP (64×350) 64 chains, 0 warm start → draws MAMS-alone, budget-matched at $B^* = 3,078,912$ grads	0	4.5-h wall fence fired at burn round 11/30 — sampling never started : draws (0, 64, 32), ESS = 0, $\hat{R}$ not computable; burn healthy (accept 0.873–0.902); realized 1,277,632 grads = 41.5% of B^* ; throughput within ~5% of S1r	BLOCKED (burn-fenced; a target-cost fact, not arm pathology)	b1r128_carousel133_s6b_seed2.json; data/b1r_decision_matrix.json			
J. L0 anchor arbitration — v2d native diagonal, parity-certified scene API target (T2 class, cond~10 ¹⁴)									
L0arb MC-SMC, L4 leg	v2d native, diagonal masked-err ridge-marg	prior-seeded MAMS-SMC (phoenix L4, checkpoint- chained)	64	resume leg from stage 26: summary $\lambda_{\text{reached}} = 0.4183$ (stage-66 checkpoint; the trace’s last row is stage 67 at $\lambda = 0.4501$), 12.26-h leg; accept 0.853–0.946 (the ledger’s $\lambda \approx 0.45$ @~18 h presumably folds in the earlier leg’s wall; the on-disk summary binds)	PARTIAL-BY-BUDGET (healthy; wall-capped)	data/10_arbitration.smc.json			
L0arb MC-SMC, same A100 hbm80g (job 56168446)	same	same (Perlmutter A100 80 GB)	128	$\lambda = 0.587$ @stage 76 at the 3.5-h cap; MAMS healthy (accept ~0.88); peak 22.7 GB; third consecutive wall-capped MC-SMC attempt ⇒ vehicle amendment	PARTIAL-BY-BUDGET (vehicle budget-infeasible on this target)	CAMPAIGN.md ledger row 56168446 (ckpts stage_000–075 on CFS)			
L0arb classic warm arm, initial	same	warm MAP→ 50ch×750 SVI→HMC, B3- (250 burn) frozen settings (seeds 0/1/2)	0	$\hat{R}_{\text{worst}} = 44.3$; ESS _{min} = 53.6; all chains railed $\gamma \approx 1.987$ [1.983, 1.991] at the prior edge	REJECTED (NO-VERDICT: γ not quotable; reproduces the known T2 single-stage pathology on the scene API, §9.5)	10_arbitration_classic.json			
L0arb classic warm arm, escalated	same	same	50ch×1500 (500 burn)	$\hat{R}_{\text{worst}} = 11.7$; ESS _{min} = 63.0 (escalated stage — the initial-stage value is 53.6; quoted per stage); chains still railed	REJECTED (NO-VERDICT)	same			
K. Companion-campaign diagonal fits — provenance rows (cited, not re-derived; see the provenance footnote in the appendix text)									
anchor $\gamma = 1.433 \pm 0.034$	v2d native, diagonal	companion two-stage preconditioned recipe	see companion ledger	per-fit $\hat{R}$ /ESS ledgered in the companion report	converged there; cited	Benson and Huang (2026)			

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Table 11, continued.

fit / arm	data + likelihood		vehicle	N ch $\times$ draws	or	diagnostics	verdict	artifact
diag-low $\gamma = 1.293 \pm 0.012$	v3b	binned, diagonal	same	see companion ledger	per-fit $\hat{R}$ /ESS	ledgered in the companion report	converged there; Benson and Huang (2026)	
M. E1 post-campaign epilogue (2026-07-22) — PI-requested MCLMC diagonal fits (vendored windowed-adaptation driver; phoenix L4 free tier, 9.56 L4 GPU-h, 0 A100-h; frozen gates $\hat{R} < 1.01 \wedge \text{ESS} \geq 1000$ pooled 64 chains + containment $\gamma \in [1.15, 2.0]$)								
E1 MCLMC	v2d	native (80 ² @0.13''), certified masked-err ridge-marg (scene API)	native parity-diagonal	vendored MCLMC (windowed mass matrix, regularization ON; warm start = mapped healthy anchor chains)	2 grp $\times$ 32ch 4000 (2000 burn)	$\hat{R}_{\text{worst}} 1.0031$ (LL2 $n_{\text{sérsc}}$); ESS _{min} 30,334; per-group $\hat{R}_{\text{worst}} 1.0034/1.0031$; γ : 39,794; containment 0/64 violations (all draws in [1.324, 1.637]); kernel-nonan 1.0; 1.49 h, peak 9.1 GB. $\gamma = 1.4683$ [1.4343, 1.5048] = $0.72\sigma_{\text{comb}}$ from the anchor; seed-group medians $\Delta 0.0002$	CONVERGED (PASS G1+G2, escalated; quotable)	mclmc_diag.v2d. {npz,json}; data/e1_harvest. json
E1 MCLMC, seed 0	v3b	binned group	v3b (130 ² @0.08''), same diagonal construction	same (warm start = the 1.2941 $\gamma < 1.7$ conditional cloud, init ONLY)	16ch 4000 (2000 burn)	$\hat{R}_{\text{worst}} 1.271$ (LL2 $n_{\text{sérsc}}$), ESS _{min} 45.4; init med 1.2914 $\rightarrow$ kept med 1.1066; 16/16 chains below the band	REJECTED (E1 gate)	mclmc_diag.v3b. {npz,json}; data/e1_harvest. json
E1 group 1	v3b	MCLMC, seed	same	same	same	$\hat{R}_{\text{worst}} 1.635$ (LL3 $R_{\text{sérsc}}$), ESS _{min} 26.2; init med 1.2940 $\rightarrow$ kept med 1.0981; 16/16 below the band	REJECTED	same
E1 group 2	v3b	MCLMC, seed	same	same	same	$\hat{R}_{\text{worst}} 1.420$ (LL1 I_e), ESS _{min} 33.2; init med 1.2900 $\rightarrow$ kept med 1.1049; 16/16 below the band	REJECTED	same
E1 group 3	v3b	MCLMC, seed	same	same	same	$\hat{R}_{\text{worst}} 1.166$ (LL2 $R_{\text{sérsc}}$), ESS _{min} 67.8; init med 1.2937 $\rightarrow$ kept med 1.1074; 16/16 below the band	REJECTED	same
E1 pooled verdict (the 4 groups above)	v3b	MCLMC, same	same	same	pooled 64ch 4000	$\hat{R}_{\text{worst}} 1.379$ / ESS _{min} 139 (both LL1 I_e); 46/46 scene params over the $\hat{R}$ gate; $\gamma \hat{R} 1.261$ (ESS 181); 64/64 chains out of band (frac outside min 0.94, median 1.00; all reported, none dropped); kept-draw pooled med 1.1047 (descriptive only, 0.0015 from the corr-low 1.1032); wall 8.07 h	REJECTED — NO-QUOTE FINDING (the binned-product migration, §1); the one allowed escalation declined: structural migration, not sampling depth	same; figs/e1.v3b_ migration.png
E1 1 (PID 184374)	v3b	launch attempt	same	same (original 32ch $\times$ 2-group geometry)	2 grp $\times$ 32ch	no artifacts: OOM in the main scan (RESOURCE_EXHAUSTED, a 10.1 GB allocation atop the working set — the 130 ² lstsq-marg grad tape is $\sim 4 \times$ v2d's) $\Rightarrow$ ledgered amendment 3: 16ch $\times$ 4 groups, pooled 64 unchanged, no gate or threshold change	BLOCKED (OOM $\rightarrow$ geometry amendment)	data/mclmc_lane_v3b.log; CAMPAIGN.md E1 amendment 3

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Table 11, continued.

fit / arm	data + likelihood	vehicle	N ch $\times$ draws	or	diagnostics	verdict	artifact
E1 v3b launch attempt 2 (PID 189281)	same	same (amendment- 3 geometry)	4 grp 16ch	$\times$	no artifacts: silent kill — the relaunched lane process vanished within minutes of launch with no traceback and no log output flushed; the only trace is the watch- dog’s <code>VANISHED_NO_ARTIFACT</code> record. The subsequent relaunch is the completed run rowed above	BLOCKED (silent kill — infra, cause unattributed)	<code>data/watchdog_</code> <code>state.json</code> (<code>phoenix:189281</code>)
N. E2 post-campaign epilogue (2026-07-23) — the collaborator’s own 0.064125'' product (O2 apples-to-apples fit) + the stage-2 rescue demonstration on stored E1 chains (O4); phoenix L4 free tier, 6.63 L4 GPU-h, 0 A100-h; E1-frozen gates, §11.2							
E2 odell fit, seed group 0	collaborator’s own cutout + PSF, regis- tered 0.064125''/px (parity flip absorbed grid-side; v1 noise assumption), diagonal masked-err ridge-marg (scene API)	same vendored MCLMC as E1 (warm start burn) = transported healthy 1.433 anchor cloud, med 1.4334)	16ch 4000 (2000	$\times$	$\hat{R}_{\text{worst}}$ 2.389 (LL3 center _y), ESS_{min} 19.6; kept med 1.1153 [1.1052, 1.1307]; 16/16 chains below the band	REJECTED (E2 gate)	<code>mclmc_diag.odell.</code> <code>{npz,json}</code> ; <code>data/e2.harvest.</code> <code>json</code>
E2 odell fit, seed group 1	same	same	same		$\hat{R}_{\text{worst}}$ 2.322 (LL3 $n_{\text{sérsc}}$), ESS_{min} 19.8; kept med 1.0893; 16/16 below the band	REJECTED	same
E2 odell fit, seed group 2	same	same	same		$\hat{R}_{\text{worst}}$ 2.442 (LL2 e_1), ESS_{min} 19.4; kept med 1.1033; 16/16 below the band	REJECTED	same
E2 odell fit, seed group 3	same	same	same		$\hat{R}_{\text{worst}}$ 2.600 (EPL center _x), ESS_{min} 18.9; kept med 1.0652; 16/16 below the band	REJECTED	same
E2 odell fit, pooled verdict (the 4 groups above)	same	same	pooled 64ch 4000	$\times$	$\hat{R}_{\text{worst}}$ 5.459 / ESS_{min} 66.4 (both LL3 $n_{\text{sérsc}}$); $\gamma \hat{R}$ 3.384 (ESS 70); 64/64 chains out of band — frac — outside min 1.00, median 1.00 (<i>every</i> kept draw; + all reported, none dropped); kept-draw pooled med 1.0999 (descriptive only: 0.0033 from the corr-low 1.1032, 0.0048 from the E1 v3b shelf); health clean (nonan = kernel-nonan 1.000; final ε 0.015–0.036, the glassy-shelf signature); wall 4.49 h, peak 7.4 GB	REJECTED NO-QUOTE FINDING (pre-registered fork 2 fires: the migration extends to the collabor- ator’s product AND preparation, §11.2); the one allowed escalation declined (struc- tural migration)	same; <code>figs/e2_</code> <code>odell_migration.</code> <code>png</code>

continued on next page

Table 11, continued.

fit / arm	data + likelihood	vehicle	N ch $\times$ draws	or diagnostics	verdict	artifact
E2-O4 v3b stage-2 rescue (demonstration arm)	our stored E1 v3b mi-grated stage-1 chains (same binned diagonal target; no collaborator data touched)	stage-2 MCLMC preconditioned on the pooled stage-1 tail (init rows, tail cov as inverse mass, tail mean)	16ch $\times$ 5000 (5000 burn)	$\hat{R}_{\text{worst}}$ 1.1875 (LL0 $R_{\text{sérisc}}$), ESS_{min} 81.3 (src center $_y$); γ $\hat{R}$ 1.128; vs stage-1: $\hat{R}_{\text{worst}}$ 1.379 $\rightarrow$ 1.188, γ $\hat{R}$ 1.261 $\rightarrow$ 1.128; split-half $\hat{R}_{\text{worst}}$ 1.033 (ch 0–7) / 1.321 (8–15); 16/16 chains 100% out of band; γ med 1.0849 (telemetry; -0.019 vs the stage-1 tail — F2 not fired); 1.65 h	REJECTED at the frozen gates — falsifier F1 FIRE D (no pristine- $\hat{R}$ -at-the-shelf; the rescue improves but does not certify; reported as loudly as a pass, §11.2)	o4_v3b_stage2. {npz,json}; research/checkpoint_ e2_o4.md; figs/e2_ o4_rescue.png
E2-O4 v2d stage-2 rescue (control arm)	our stored E1 v2d healthy stage-1 chains (native diagonal target)	identical rescue	16ch $\times$ 5000 (5000 burn)	$\hat{R}_{\text{worst}}$ 1.0029, ESS_{min} 8305; γ : $\hat{R}$ 1.0007, ESS 12,058; γ 1.4683 [1.4344, 1.5040] (telemetry) = $0.0008\sigma_{\text{comb}}$ from the E1 quotable 1.4683; split-half 1.0033/1.0027; 0/16 chains out of band; 0.49 h	CONVERGED (control pre-diction O4-D2 MET exactly as pre-registered; F3 not fired)	o4_v2d_stage2. {npz,json}; data/e2_harvest. json

Coverage notes (what is summarized rather than rowed). For completeness of the “every fit ever ran” claim: (i) the **X2** SBC lane ran $N=32$ classic-recipe fits on the free tier; they are gift deliverables scored as a set, so their health census (2/32 healthy, 30/32 unhealthy at that budget) is reported with the SBC gift rather than as 32 rows (§10.1, `data/x2_sbc.json`) — consistent with the B3 and arbitration classic-arm rows here. (ii) The q -fit proposal constructions (SVI-cov canaries, e.g. `t11_inj*_canary_svicov.json`) are prep stages of the prior-seeded-SMC vehicle, not posterior fits, and are subsumed in their fit’s row. (iii) Pure infrastructure failures that never yielded a sampler state — the T1.1 inj3 and l0arb OOMs, the vendor-guard path-identity trip, the B5-steep snapshot-ordering reruns, and the classic-arm attempt #1 `IndexError` at MAP — are ledgered per-job in `CAMPAIGN.md`; their successful reruns are the corresponding rows above. (iv) After the two dead E1 v3b launches (group M BLOCKED rows), a second concurrent launch of the *same* production lane — identical config and seeds (`data/mclmc_lane_v3b_r3.log`) — ran alongside the lane that completed and was killed unread at harvest open; it wrote no artifact and none of its draws were read or used, so the completed lane’s `npz/json` is the sole v3b E1 record. It is noted here rather than rowed because it is a duplicate of a rowed fit, not a distinct one.

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